



# Exploring Earth's Interface with Space

The Scientific Case for a Satellite Mission to the  
Lower Thermosphere-Ionosphere Transition Region



**A report by the ESA-NASA Lower Thermosphere-Ionosphere Science Working Group**

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## EXECUTIVE SUMMARY

This report describes the scientific case for a mission concept to study the Earth's Lower Thermosphere-Ionosphere – the LTI – which is the interface region between our atmosphere and space. The LTI is situated at altitudes between 100-200 km and is uniquely characterized by complex interactions between the co-existing neutral and ionized gases. The LTI receives and dissipates hundreds of gigajoules of energy every second from above due to solar wind interactions with our planet's magnetosphere. Simultaneously, this region continuously receives energy and momentum input from below in the form of tides and gravity waves originating in the lower and middle atmosphere as well as impulsive input from thunderstorms, volcanoes, and earthquakes, whose energy content rivals that provided by the magnetosphere. The LTI responds dramatically to all of these driving forces producing large variations in many key properties, such as neutral and plasma number densities, motions, temperatures, and composition, as well as electric fields, conductivity, and currents. Importantly, the LTI also acts as a critical interface that subsequently regulates this energy transfer from the magnetosphere above and from the neutral atmosphere below.

The study of the Lower Thermosphere-Ionosphere has three fundamental motivating factors:

1. The LTI is a boundary layer and transition region between the Earth's atmosphere and space making it a critical link in understanding the holistic nature of our geospace system. Significant gaps exist in our current knowledge of the LTI with the lack of comprehensive observations preventing a causal understanding of its behavior and limiting the advancement of geospace modeling capabilities.
2. The LTI is where numerous phenomena play a major role in generating space weather and may hold important information about climate change. Many of these phenomena are associated with the complex chemistry of the upper atmosphere as well as the unique electrodynamics of the region, both of which remain poorly understood. Direct measurements in the LTI will lead to substantial progress in space weather modeling with explicit benefit to society.
3. Fundamental physical processes in the terrestrial LTI are also present in other magnetized planetary environments, such as Jupiter and Saturn, and these processes bear similarities to un-magnetized atmosphere-space transition regions of Mars and Venus. Understanding the terrestrial LTI will thus significantly advance our deciphering of analogous phenomena in other planetary systems.

Understanding the fundamental LTI energetics and dynamics, including a host of space weather effects, represent long-held goals of the scientific community. Simultaneous measurements of the neutral and ionized gas properties, local electric and magnetic fields, and energetic particles must be gathered directly in the location where the requisite physical effects occur in order to understand the LTI. Orbiting platforms with *in situ* measurements are the established way to gather the required simultaneous, multi-parameter measurements at critical altitudes with the necessary range of spatial and temporal scales, while sampling across all local times, longitudes, and latitudes.

Unfortunately, progress has been hindered by the inability to collect multi-parameter measurements simultaneously in the LTI. Despite its proximity, the LTI resides at altitudes that are too high for direct measurements by instruments on balloons and research aircraft, and yet are too low for repetitive and longer duration direct measurements by traditional satellite-borne instruments. Suborbital sounding rocket experiments and ground-based instruments have provided important observations but with very limited duration and restricted coverage.





NASA's pioneering Atmosphere Explorers (AE) of the late 1970s have been the only spacecraft that gathered, albeit limited, sampling of this region, journeying to altitudes as low as 128 km. The Pioneer Venus and the more recent MAVEN mission to Mars have demonstrated very successful direct measurements in the Venusian and Martian thermosphere-ionosphere with excursions to similar low altitudes as Earth's LTI. The AE satellites and previous, successful planetary missions have demonstrated that a satellite mission dedicated to the exploration of Earth's LTI will provide an abundance of accurate, unique measurements that will transform our knowledge of this critical transition region.

This report provides the scientific motivation, measurement needs, and preliminary design concepts to carry out an unprecedented study of the interaction between the Earth's atmosphere and the ionized gases of space at LTI altitudes. ***The overarching goal is to understand how the transition from Earth's atmosphere to space is governed by the fundamental neutral-plasma interactions that are intrinsic to the lower thermosphere-ionosphere.*** Such a mission would provide the first detailed, systematic investigation of this important unexplored region, vastly improve and constrain models of the near-Earth environment, and significantly advance our understanding of its coupling with the magnetosphere above and the middle atmosphere (stratosphere and mesosphere) below.

To address the overarching goal, three high-level science objectives are identified that together will determine how collisions between neutral and charged species affect the electrodynamics, energetics, and neutral atmospheric dynamics, composition and structure within the LTI. These objectives are targeted towards improving our understanding of the critical processes and properties that determine how electric currents flow and close in the LTI, and thereby couple this transition region both to the magnetosphere and to the lower atmosphere.

The scientific goal and objectives described in this report lead to a set of next steps to be undertaken as soon as possible within the existing ENLoTIS working group framework:

1. Flow-down of the scientific needs expressed in this report into firm mission requirements supported by analyses, tools and capabilities to justify and verify these requirements;
2. Assessment of instrumental capabilities needed to meet measurement requirements;
3. Study of orbits, propulsion, as well as mission and science operational concepts capable of meeting sampling requirements;
4. Further progress on the mitigation approaches to the special environmental challenges that affect the measurement techniques, data processing, spacecraft design choices and concepts of operations.

Indeed, while there are precedents for successful space-based *in situ* measurements of the LTI environment, there are significant challenges to operating for extended periods in very low Earth orbit (i.e., below 200 km altitude). Further work is necessary to mitigate these challenges and progress toward fully developed design requirements for any new LTI mission, and should include detailed technical studies of the following topics: effects of high ram pressure and plasma wakes on measurements; atmospheric drag and aerothermal heat flux effects on spacecraft performance at low altitudes; and effects of atomic oxygen influence on spacecraft materials.

This report represents the first steps toward potential ESA-NASA cooperation on developing a new *in situ* LTI mission, which would result in significant scientific advancements in our understanding of plasma-neutral interactions that connect our upper atmosphere with solar and geomagnetic drivers, and apply more generally to transition regions of other planetary atmospheres.



## **Table of Contents**

|                                                                                 |           |
|---------------------------------------------------------------------------------|-----------|
| <b>Executive summary.....</b>                                                   | <b>4</b>  |
| <b>Preface.....</b>                                                             | <b>8</b>  |
| <b>1. Introduction.....</b>                                                     | <b>10</b> |
| <b>2. Background, scientific justification and needs .....</b>                  | <b>15</b> |
| 2.1. Defining characteristics of the Lower Thermosphere-Ionosphere .....        | 15        |
| 2.2. Collisional coupling between ionized and neutral gases within the LTI..... | 21        |
| 2.3. Societal needs addressed by studying the LTI region.....                   | 43        |
| 2.4. Critical LTI modeling needs .....                                          | 45        |
| 2.5. Summary .....                                                              | 49        |
| <b>3. Scientific goals and objectives .....</b>                                 | <b>50</b> |
| 3.1. SO1: Collisional Electrodynamics .....                                     | 53        |
| 3.2. SO2: Collisional Energetics .....                                          | 56        |
| 3.3. SO3: Collisional Dynamics .....                                            | 58        |
| <b>4. Scientific strategy and needs.....</b>                                    | <b>61</b> |
| 4.1. Scientific needs .....                                                     | 63        |
| 4.2. Observational needs .....                                                  | 71        |
| 4.3. Supporting models and observations .....                                   | 83        |
| 4.4. Science Strategy Matrix.....                                               | 84        |
| <b>5. Notional mission elements and expected challenges .....</b>               | <b>85</b> |
| 5.1. Mission elements.....                                                      | 85        |
| 5.2. Mission design features and challenges .....                               | 91        |
| 5.3. Towards a notional observatory .....                                       | 96        |
| <b>6. Synergies and international context.....</b>                              | <b>98</b> |
| 6.1. International programs and collaborations .....                            | 98        |
| 6.2. Current and planned synergistic observations .....                         | 99        |
| 6.3. Uniqueness and complementarity .....                                       | 102       |



|                                                   |            |
|---------------------------------------------------|------------|
| <b>7. Summary and direction forward .....</b>     | <b>104</b> |
| <b>References .....</b>                           | <b>107</b> |
| <b>Acronyms .....</b>                             | <b>124</b> |
| <b>Appendix A    Definition of Terms.....</b>     | <b>126</b> |
| <b>Appendix B    Science Strategy Matrix.....</b> | <b>128</b> |



## PREFACE

The ESA-NASA Lower Thermosphere-Ionosphere Science (ENLoTIS) Working Group was formed in May 2022 to cooperatively explore future lower thermosphere-ionosphere (LTI) satellite mission concepts, targeting very low altitudes (100-200 km) with *in situ* sampling of relevant geophysical parameters associated with the neutral atmosphere, the ionosphere's plasma, electromagnetic fields, and energetic particles, which, together with modeling, would enable significant advancements in the understanding of neutral-ion interactions and other related science and space weather topics in this critical region of Geospace.

The LTI region has been identified as one of considerable interest to both NASA and ESA. Most recently, the Daedalus mission study was carried out under the remit of ESA's Earth Observation Programmes (EOP) Directorate competitive Earth Explorer 10 pre-feasibility (Phase 0) activities. Furthermore, many NASA studies have also focused on the LTI region, including both directed missions with dipping spacecraft, such as the initial TIMED dual-satellites and the GEC constellation, as well as numerous highly-rated Explorer proposals targeting the LTI.

Although the Daedalus mission was not selected, the ESA Advisory Committee on Earth Observation (ACEO) ranked it highly on scientific grounds and encouraged further study activities to mature the concept, exploring potential international collaboration. Subsequent bilateral discussions with NASA's Science Mission Directorate (SMD) noted that such a concept was in alignment with the 2020 SMD science plan – Science 2020-2024: A Vision for Scientific Excellence – along with other complimentary activities within the NASA Heliophysics Division.

Building on NASA's and ESA's long history of very successful collaborations, this mutual interest in LTI science led to the establishment of a new inter-agency and cross-discipline science connection, linking the ESA EOP Climate Action, Sustainability and Science Department and the NASA Heliophysics Division. Initial exploratory discussions led to the formation of the ENLoTIS Working Group, which was directed to explore the science case behind a potential joint LTI mission.

Members of the ENLoTIS Working Group are listed below, consisting of 7 scientists from ESA Member and Cooperating States and 7 scientists from the United States. The working group held 3 "in person" meetings over the course of 18 months, interspersed with regular virtual meetings on a more frequent basis. This report constitutes their chief findings and recommendations.

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## 1. INTRODUCTION

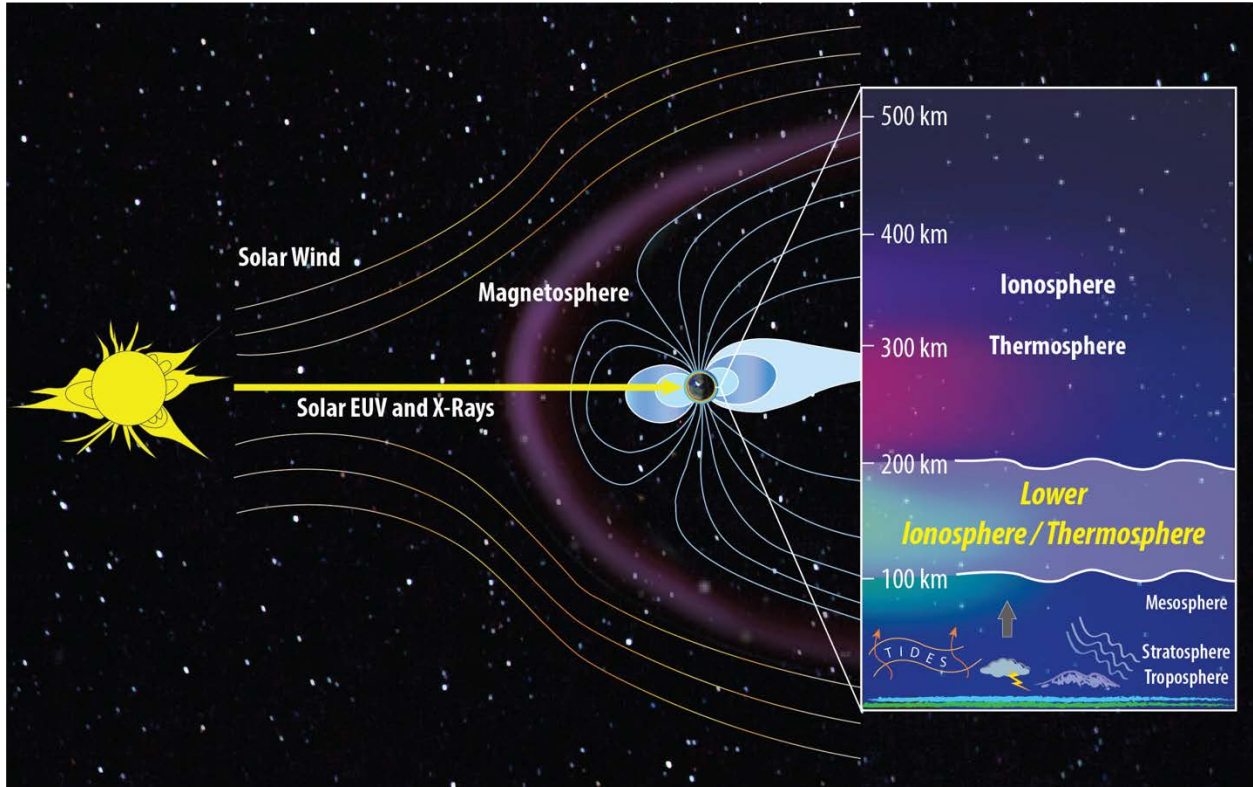


Figure 1.1: Schematic representation of the LTI within the coupled Sun-Earth system (credit: NASA/GSFC – R. Jennings).

Fundamental to any extended planetary atmosphere is its interaction with its star, manifested through both the absorption of stellar radiation and the impact of the stellar wind. A planet's upper atmosphere directly absorbs the incident stellar radiation at high energy wavelengths ranging from extreme ultraviolet (EUV) to X-rays, becoming partially ionized to form an ionosphere. For planets with an extended atmosphere and a strong magnetic field, the role of the high-speed stellar wind is amplified: its interaction with the planet's magnetic field creates a powerful electric generator which gives rise to a complex 3D system of electric currents and plasma flows throughout the magnetosphere and ionosphere. Furthermore, disturbances in the lower atmosphere propagate vertically upwards and deposit their momentum and energy into the upper atmosphere.

At Earth, the lower thermosphere-ionosphere – the LTI – extends between approximately 100 and 200 km of altitude (Figure 1.1), and is a region where collisional coupling between neutral and ionized gases plays a dominant role. While the LTI covers a relatively small portion of the combined upper atmospheric and near space environments – together known as geospace – and acts as the transition region between atmosphere and space, it plays a crucial role in determining the Earth system's response to solar variability and related geomagnetic storms. Here, intense radiation, high energy charged particles, and electromagnetic fields energized by solar wind-magnetosphere interactions all converge and transform into heat and momentum that can dramatically alter the circulation and composition of the neutral atmosphere via plasma-neutral interactions. These geomagnetic storms can be episodic due to sudden disturbances on the sun, such as solar flares or coronal mass ejections, or more periodic, such as active regions with 27-day rotation periods and corotating interaction regions in the solar wind due to persistent coronal holes. Notably, the different types of geomagnetic disturbances manifest differently



in how the LTI responds. The lower and middle atmosphere (i.e., troposphere, stratosphere, and mesosphere in Fig. 1.1) also influence LTI properties through highly variable upward propagating fluxes of energy and momentum that converge in the LTI. Sources of LTI forcing from the lower atmosphere are wide ranging and can be short-lived, including episodic wave events generated by such disturbances as thunderstorms, hurricanes, earthquakes, and volcanic eruptions, as well as more periodic disturbances associated with land-sea distributions around the globe, orography, and day-night solar illumination.

Importantly, these varied, dynamic sources of energy dramatically alter the nature of the LTI itself, including its vertical and horizontal structure, through physical processes that are often irregularly structured and turbulent, difficult to predict, and challenging to measure. The recent Tonga volcanic eruption registered significant disturbances in the upper thermosphere and ionosphere, transferring energy through a chain of processes involving the LTI that are poorly understood. As a transition region between atmosphere and space, the LTI may be considered both a “crucible” where momentum and energy from terrestrial and extraterrestrial sources are transformed, as well as a “turnstile” where magnetospheric currents close and neutral winds circulate within a crescendo of tides and gravity waves. Figure 1.1 depicts the LTI region of geospace that is the focus of this report within the context of the broader Sun-Earth system. The importance of the LTI region from scientific, climate, and societal perspectives is summarized below.

**Fundamental science.** The LTI embodies intense electrical phenomena with long-range connections to Earth’s surface and regions further out in space. Within the LTI, ionized species, moving under the action of electric fields, transfer energy and momentum to neutral particles via collisions giving rise to neutral winds. The inertia of these winds can redistribute ionospheric plasma and modulate electrical current flow both within the LTI region and throughout the entire ionosphere. This perpetual but unsteady exchange of energy and momentum spawns a feature-rich LTI environment fueled by plasma-neutral interactions in the region. As described in Chapter 2, steep vertical gradients of almost every property exists in the LTI. To sustain such gradients, a persistent flux of momentum and energy must be deposited into the LTI – tens of gigajoules enter the LTI every second from terrestrial and extraterrestrial sources. The inner workings of the LTI process this energy, often revealing emergent behavior unique to the region. For example, narrowly confined, enhanced currents, called electrojets, emerge only within the bounds of the auroral oval and equatorial regions of the LTI. This emergent behavior is a consequence of plasma-neutral collisions confined by altitude and magnetic field topologies and occurs because of complex and highly non-linear two-way interactions taking place between the neutrals and the plasma in these regions. Cross-scale coupling, feedback, persistent behavior or memory, instabilities, and emergent behavior all occur in the LTI, and are general attributes of a nonlinear dynamical system. Consequently, sources of energy input, internal processes, and responses of physical properties must be studied together in the highly non-linear LTI – a quintessential example of system science – to unravel the physics of this complex domain. How to sample, interpret, and predict such a dynamic system within the Earth’s upper atmosphere is a major challenge, impacting instrument and satellite design as well as operations concepts. Devising a notional plan to gather such knowledge via direct measurements is a major motivation of this report.

**The need for observations.** Although the LTI is of fundamental importance to understanding many key physical processes within the Earth’s upper atmosphere and ionosphere, there are very few simultaneous observations of both ionized and neutral properties within this critical region of geospace. Most direct LTI observations come from sounding rocket experiments that offer instantaneous views of a specific region but lack a sustained, systematic representation of evolving processes over time. Ground-based radar and similar facilities can offer much-needed long duration observations, but are restricted in their geographic coverage. A series of spacecraft in the 1970s provided important, though limited, global LTI observations using low altitude excursions and (at the time) innovative



instrumentation. To date, our description of the LTI relies mainly on theoretical models driven by idealized inputs that are poorly constrained by very sparse data sets. To understand the plasma-atmosphere interactions within the LTI, it is necessary to perform a sustained study by *in situ* spacecraft that probe directly this critical region of geospace. The 100-200 km altitude range has never been systematically explored with a complete instrumentation suite to investigate simultaneously both the electrodynamics and the neutral atmosphere response, an objective which cannot be effectively fulfilled by spacecraft that lack *in situ* instruments and on-board propulsion to counteract the high orbital drag environment within the LTI region.

**Climate connections and long-term effects.** The Earth's climate is changing on decadal time scales. This long-term evolution is not isolated to the lower atmosphere but has also been observed in the middle and upper atmosphere in response to anthropogenic production of greenhouse gases such as methane (CH<sub>4</sub>) and carbon dioxide (CO<sub>2</sub>). Carbon dioxide in the LTI causes extensive radiative cooling, and observed increasing concentrations are associated with decreasing temperature trends of 1–2 K per decade, as inferred from measurements covering several decades (Mlynczak et al., 2022). As the amount of CO<sub>2</sub> and radiative cooling increases, analysis of observations and state-of-the-art climate models indicate that the neutral gas density of the upper atmosphere at a fixed altitude (400 km) will decrease at a rate between 2-7% per decade (Brown et al., 2021), with serious implications for orbital debris mitigation. This is a clear example of the intimate links between various components of the Earth climate system. Exploration of the LTI provides an opportunity to track markers of climate change and their trends, bringing additional data to complement those obtained from current and planned future Earth observing systems. An additional effect of climate change that alters the forcing of the LTI is the increased occurrence and amplitude of severe tropospheric storm activity. This translates to additional energy content within the atmospheric waves that propagate vertically from these storms, and which ultimately is transferred to the LTI. Furthermore, as the LTI base state changes due to climate evolution, its response to both solar and terrestrial drivers may also change. One example is how the production of ozone-depleting chemical compounds in the LTI during geomagnetic storms by solar energetic particles may be impacted by long-term trends in temperature and density in the LTI. Predictive models based on decades-old surveys of global LTI properties will likely be insufficient to determine how the LTI's role in coupling different regions of geospace could evolve in the future.

**Technological impacts to society.** Fundamental processes in the LTI not only have a profound effect on the geospace environment, but also include a significant societal impact through the ever-increasing use of space-based technology. For example, operations of near-space assets in low-Earth orbit (LEO) are directly affected by rapid changes in the LTI related to both solar and terrestrial drivers. As more nations become spacefaring and as space assets are used increasingly for communications, navigation, and Earth observations, the hazards associated with space debris and satellite conjunctions increase. Fleets of thousands of LEO satellites are being deployed (or planned to be deployed) to provide critical services on Earth, and there is growing interest in satellite operations at very-low-Earth orbit (VLEO), below 300 km altitude. In such conditions, space debris path prediction, conjunction avoidance and the need for accurate re-entry predictions also increase and become a central issue in the space economy with key risks to be managed by the space insurance industry. Uncertainty in LTI system properties, particularly neutral gas density and motions, is the foremost source of error in predicting the future position of LEO spacecraft and debris. A recent (and costly) example is the loss of a major part of the fleet of newly deployed Starlink satellites during their initial low altitude orbits in early 2022. Moreover, variability in plasma properties of the LTI translates to disturbances of many operationally critical infrastructure systems ranging from ground-induced currents that can disrupt power distribution to global navigation satellite signal transmission and long-distance communications. Overall, acquiring accurate data bases of the properties of the LTI system and understanding its fundamental physical processes and variability is imperative to elucidate the long-term evolution of the LTI system and maintain a sustainable future for all impacted activities in space as well as on Earth.





**Relation to other LTI missions and concept studies.** The only systematic global *in situ* observations within the Earth's LTI region were those gathered by instruments on the three NASA Atmosphere Explorer (AE) satellites, AE-C, AE-D, and AE-E, between 1973 and 1981 (Burgess and Torr, 1987; Spencer et al., 1973). These satellites carried instrumentation that successfully measured plasma density, ion drifts, neutral and ion composition and temperatures, and one component of the neutral wind to perigees between 130 and 150 km altitude. Together the AE spacecraft served as important scientific and technological pathfinders by demonstrating the feasibility of low-perigee operations of both the satellites and instruments.

Since the AE missions, a number of low perigee mission concepts have been studied over the past several decades. These include: the initial Thermosphere-Ionosphere-Mesosphere Energetics and Dynamics (TIMED) concept that included two spacecraft capable of journeying as low as 120 km altitude (NASA, 1991); the Geospace Electrodynamics Connections (GEC) mission with four identical satellites that would perform *in situ* measurements of the high-latitude LTI region with perigees as low as 130 km (NASA, 2001); and the Atmosphere-Space Transition Region Explorer (ASTRE) mission (Pfaff et al., 2022). More recently, the Daedalus mission was proposed to ESA's Earth Explorer program (ESA, 2020; Sarris et al., 2020) that included an elliptical orbit of 150 km perigee by a 2000-3000 km apogee with deep dipping campaigns as low as 120 km.

Investigations of the plasma-neutral interactions present in the LTI can extend beyond Earth to other planets in our solar system. For example, NASA's MAVEN mission has successfully performed measurements in the Martian thermosphere-ionosphere since 2014 that include several successful low perigee "dips" to altitudes as low as 125 km as part of its high apogee elliptical orbits. Data from these low orbits have indicated the presence of complex ion-neutral collision processes in the Martian upper atmosphere (Collinson et al., 2020), even in the absence of a planetary magnetic field, and demonstrate the viability of this approach for carrying out repeated sampling of the Martian lower thermosphere-ionosphere.

The last several decades have underscored that the Earth's space environment should not be studied in isolation but rather, the geospace transition region should be viewed as our most accessible opportunity to understand the universal physics of how atmospheres and space interact for all planetary bodies with atmospheres, both in our solar system and beyond. Studying the Earth's LTI region thus provides new and essential information which can be expanded to other planetary systems with dipole magnetic fields, such as Saturn and Jupiter, where strong upper atmospheric dynamos similar to the Earth's are conjectured to exist. In a larger sense, the Earth's geospace region may also be viewed as a relatively accessible laboratory for comparative studies of exoplanets outside of our solar system.

**Notional mission concept principles.** The notional mission concept described in this report focuses on one or more instrumented spacecraft that can *journey directly* into the LTI on repeated orbits under a variety of solar driving conditions. The measurements obtained from this type of mission will be crucial for understanding how the undersampled 100-200 km altitude transition region responds to the solar forcing and magnetospheric drivers from above as well as to the atmospheric forcing from below. This can be accomplished using a comprehensive suite of well-proven, high heritage scientific instruments that will gather the first systematic, direct measurements of plasma-atmosphere coupling processes within the LTI region. The concept embraces the complete specification of the drivers, response, and coupling within the LTI needed to address key science objectives described herein, and thus gathers the necessary knowledge to fully understand, and ultimately be able to predict, the fundamental processes that link the coupled Sun-Earth system.

**Outline of this report.** Chapter 1 briefly introduces the LTI including both the imperatives and challenges to exploring this region in detail. Chapter 2 provides a detailed description of the physical





properties and processes within the LTI, including what is known and, importantly, what remains to be explored via a dedicated, *in situ* mission. This chapter also further describes how targeted LTI measurements can improve modeling capabilities vital for future space weather prediction and climate change studies. Chapter 3 defines the overarching scientific goals and presents the fundamental science objectives of an *in situ* LTI mission concept. Chapter 4 formulates high-level observational needs associated with the science objectives. Chapter 5 presents preliminary design concepts for a notional low-perigee mission that would perform simultaneous *in situ* multi-parameter observations. Chapter 6 identifies synergies with ongoing and planned future agency activities and outlines the framework for possible future international collaborations in support of a dedicated LTI mission concept. Chapter 7 provides a summary and recommendations for a direction forward.

## 2. BACKGROUND, SCIENTIFIC JUSTIFICATION AND NEEDS

Earth's Lower Thermosphere-Ionosphere (LTI) is a critically important region of geospace, representing the transition region between Earth's neutral atmosphere and the ionized gases of space. The collisions between neutral and charged particles unique to this region drive a wide range of highly complex non-linear physical processes. Understanding the collective interplay of these processes as a system is required to advance our understanding of how the Earth's atmosphere-space transition region works. Indeed, knowledge of the LTI physical processes and space weather effects within the LTI represent long-held goals of the international scientific community, particularly to advance the models that describe and predict the physical behavior of the LTI.

### 2.1. Defining characteristics of the Lower Thermosphere-Ionosphere

The LTI may be considered an envelope that surrounds the Earth containing two commingled fluids – a neutral gas representing a mixture of atomic and molecular species and an ionized gas consisting of several different ion species and electrons. Characterized as a thin “shell” extending from 100 – 200 km above the Earth's surface, the physical properties and processes of the LTI differ markedly from that of the neutral gas below, comprising the middle and lower atmosphere, and that of the collisionless plasma above, consisting of the upper ionosphere and magnetosphere. These include, *at all latitudes*:

- Creation of the ionosphere by photoionization during the day;
- Creation of the thermosphere through heating of the ambient neutral gas by impinging solar EUV radiation;
- Coupling of the neutral and charged components by collisional processes, sustaining global systems of electric currents;
- Deposition of energy and momentum by planetary waves, tides, and gravity waves propagating up from the troposphere and through the stratosphere and mesosphere below.

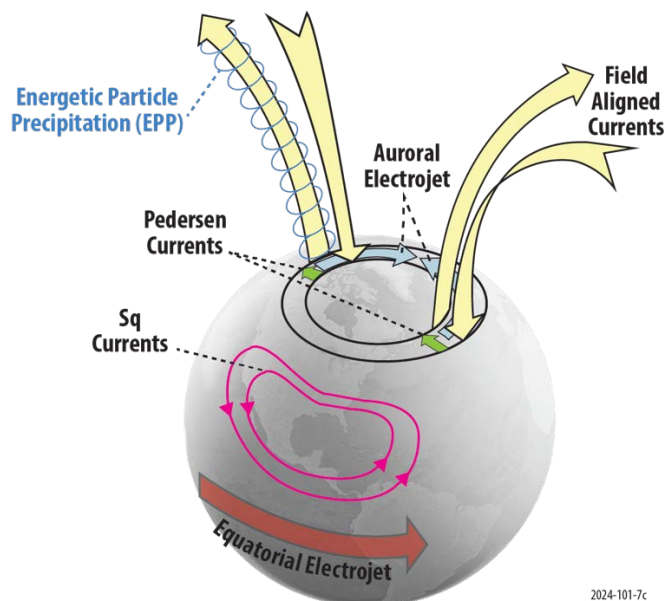


Figure 2.1: Overview schematic of the lower thermosphere-ionosphere as the seat of global current systems at high, middle, and low latitudes (credit: NASA/GSFC – R. Jennings).

The global system of electric currents that flow horizontally at the base of the ionosphere is depicted in Figure 2.1. Importantly, the currents are enabled in the Earth's LTI by the different collision rates between the neutral species and the ambient electrons and ions, as we discuss below.

Furthermore, *at high latitudes*, the LTI is further characterized by these intrinsic processes:

- Closure of field-aligned currents of magnetospheric origin;
- “Braking” of precipitating auroral energetic particles by collisional interactions with the neutral gas, contributing to ionization and heating in the region;
- Joule heating resulting from the conversion of impinging electromagnetic energy into heat.

In the following, we provide a brief review of the fundamental properties and features of the lower thermosphere and of the ionosphere. This is followed by a discussion of the physical processes within the *combined* lower thermosphere-ionosphere for which our understanding is particularly bereft.

**Fundamental properties of the lower thermosphere.** The neutral density in the LTI region decreases by over three orders of magnitude from 100 km to 200 km altitude. Short wavelength solar UV radiation is directly absorbed here by upper atmospheric constituents. In particular, molecular oxygen ( $O_2$ ) undergoes photodissociation as it absorbs photons with wavelengths below 242 nanometers (nm):



where  $h\nu$  is the symbol for photon energy. The excess energy of the UV absorption is converted to heat. Since the major atmospheric constituents at these altitudes are very ineffective at radiating away this heat into space through emission of infrared radiation, the neutral atmospheric temperatures increase significantly with altitude above ~100 km, forming the thermosphere. The energetic solar EUV radiation varies by a factor of 5 or more within the 11-year solar cycle, leading to substantial variability in the LTI, as shown for the temperature profiles in Figure 2.2 (Doornbos, 2012). In addition to heating the upper atmosphere, the EUV radiation also ionizes a small fraction of the gas, creating the ionosphere, as discussed below.

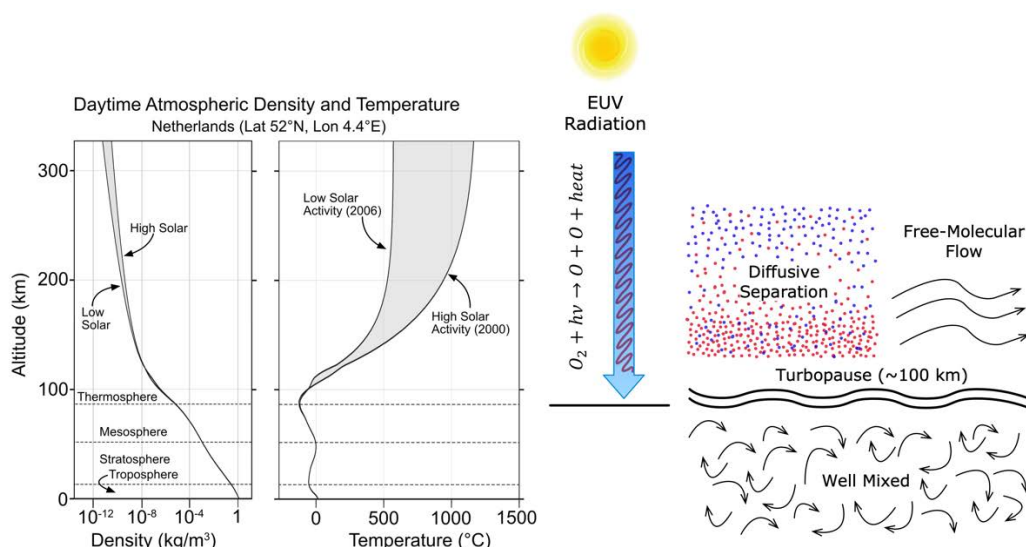


Figure 2.2: Left, altitude profiles of neutral density and temperature during low and high solar activity (adapted from: Doornbos, 2012). Right, schematic illustrating the transition from well-mixed gases below 100 km to free-molecular flow and diffusive separation in the LTI (credit: NASA).

At altitudes above about ~100 km, the atmosphere undergoes a transition at the turbopause, where the gases cease to be well-mixed and turbulent, eventually converting to free-molecular flow at higher altitudes with a large mean-free path, as depicted in Figure 2.2 (right). Here, the different gases attain different altitude distributions via diffusive processes, with the lighter species prevailing within the higher altitudes.

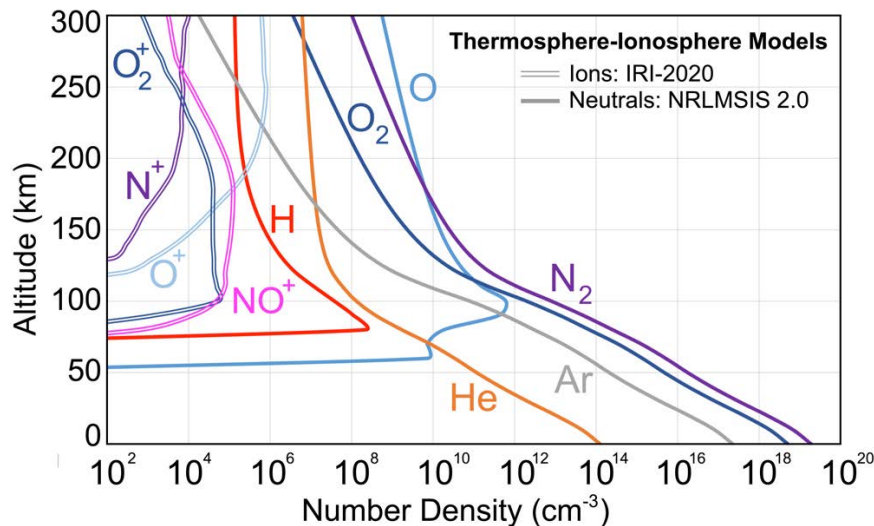


Figure 2.3: Altitude profiles of ion and neutral species, based on IRI-2020 and NRLMSIS 2.0 empirical models (credit: DUTH – T. Sarris).

The LTI processes are highly sensitive to the altitude distribution of the different species, which vary considerably with latitude, local time, and solar activity. The concentrations of atmospheric constituents and their altitude distribution controls the penetration of the ionizing solar radiation, heating and ionization in the LTI. Notional altitude profiles of various species of atoms and molecules that make up the atmosphere are shown in Figure 2.3, for mid-latitude, daytime conditions with moderate solar activity, as computed from the empirical NRLMSIS 2.0 model (Emmert et al., 2020). Empirical models specify the average atmospheric behavior based on disparate sets of observations; they have limited applicability for understanding the fundamental physical processes that produce variability in the LTI over a wide range of spatial and temporal scales. Advancing our understanding of LTI processes beyond state specification to skillful predictive capability requires knowledge based on accurate, simultaneous, and co-located observations of the neutral gas density, composition and temperature, along with other plasma parameters as discussed below.

**Fundamental properties of the lower ionosphere.** There are three main mechanisms by which ambient or “thermal” plasma is created in the upper atmosphere, each depending on an interaction of an external energy source with the Earth’s extended atmosphere: (1) photoionization of the neutral gas via impinging ionizing solar radiation, (2) impact ionization of the neutral gas via energetic (auroral) particle precipitation, and (3) meteor ablation.

Photoionization of the ionosphere occurs when solar photons have sufficient energy to detach an electron from a neutral atom or molecule to create an ion and electron pair. For atomic oxygen:



The photoelectrons produced in this process can have sufficient energy to further contribute to ionization. In the lower part of the LTI, below about 150 km, molecular constituents  $N_2$  and  $O_2$  are photoionized as well.

Importantly, photoionization is only the first step in a chain of processes by which various ion species are formed in the LTI. A range of reactions including charge exchange readily occurs. For example, atomic oxygen ions react with the ambient nitrogen and oxygen molecules ( $N_2$  and  $O_2$ ), creating  $NO^+$  and  $O_2^+$ :



At the lower altitudes, the molecular ions are lost quickly (with a time scale of seconds) through dissociative recombination. Since the molecular ions dominate below about 160 km, when the direct illumination of the sun vanishes at dusk, the lower ionosphere, which rotates with the planet, becomes largely depleted of any significant plasma until local sunrise when the main photoionization process starts over again. In other words, at the lower altitudes, the ionosphere is essentially “re-plenished” every day. At higher altitudes, the slower process of radiative recombination and transport of plasma imply longer ionization lifetimes.

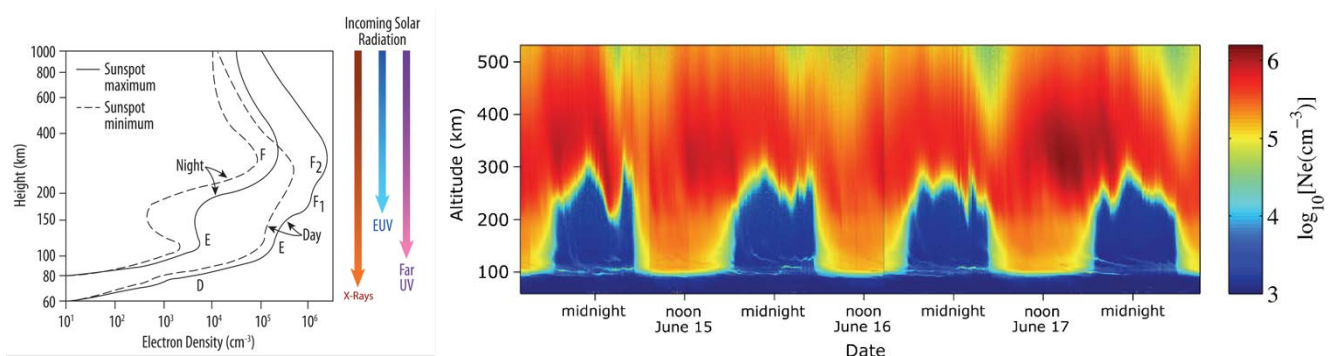


Figure 2.4: Left, model altitude profile of electron density (after Hargreaves, 1992). Right, Arecibo incoherent scatter radar measurements of altitude profiles of electron density for 4 consecutive days in 2004 (courtesy N. Aponte, in: Pfaff, 2012).

The interplay of the ionizing radiation penetrating to different altitudes based on wavelength and the different recombination and transport processes results in a “layered” structure of the plasma concentration in the ionosphere. Historically the ionization peak at 100–130 km is referred to as the E region. At lower altitudes, the D region is produced by soft X-rays. The peak at 300 km is part of the F region, where the night-time ionosphere is preserved due to the longer lifetime of the ions.

The average number density distributions of the different ion species that constitute the lower ionosphere at different altitudes are shown in Figure 2.3, taken from the International Reference Ionosphere (IRI) empirical model (Bilitza et al., 2022). In general, the actual number density profiles of these species at any given time can be highly variable, which the empirical model cannot fully capture because extensive measurements are missing.

*Modelled* day and night ionization profiles versus altitude are shown in Figure 2.4 (left), for a mid-latitude location (Hargreaves, 1992). Note the strong dependence of the resulting plasma density on solar activity. *Measured* plasma density data from 50–500 km, illustrating the diurnal variations of the ionosphere at a single location, are plotted for 4 continuous days in Figure 2.4 (right) as observed by the mid-latitude Incoherent Scatter Radar (ISR) at the Arecibo Observatory in Puerto Rico (Pfaff, 2012).



The strong day-night difference is especially apparent below about 250 km altitude. The lower altitude ionosphere builds up very quickly after sunrise and then disappears abruptly at sunset through recombination processes described above. During the night, the ionospheric density may be reduced over two orders of magnitude at altitudes up to about 250 km, where the lower edge of the F-region can be very “ragged”, influenced by transport effects and perhaps local instabilities. The very low night-time E region density produces large spatial gradients at the dawn and dusk terminators. The relationships between these gradients and neutral wind systems have an important impact on the electrodynamics of the LTI, that is poorly sampled and hence poorly understood.



Figure 2.5: A view of auroral “curtains”, photographed from the International Space Station at an altitude near 400 km (credit: [NASA](#)).

In addition to photoionization from solar EUV, at high latitudes the upper atmosphere is also ionized by precipitating energetic electrons with typical energies of 1 – 30 keV, which also create the optical aurora as shown in Figure 2.5. Typically produced at altitudes between 80–250 km, displays of discrete auroral forms are striking, visible manifestations of the interactions of space plasma with the upper atmospheric neutral gases. The auroral emissions show vertical striations that correspond to the direction of the ambient magnetic field lines along which the precipitating particles are guided while gyrating around these lines. The various colors of the aurora are due to emissions from different neutral species and altitudes.

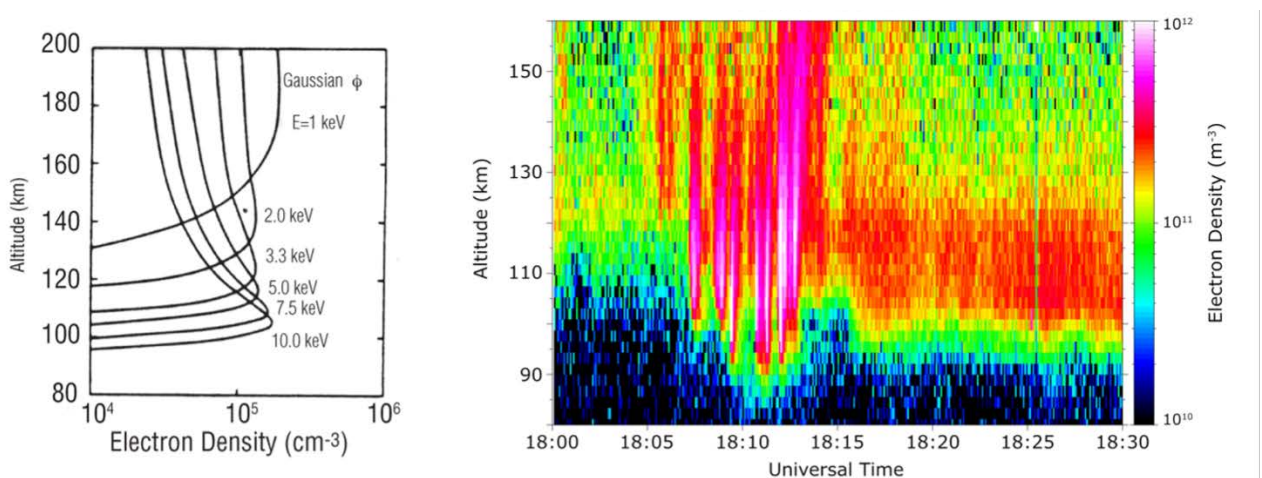


Figure 2.6: Left, effective penetration altitude of precipitating electrons as a function of energy (from Strickland, 1983, in: Pfaff, 2012). Right, EISCAT incoherent scatter radar observations from 1<sup>st</sup> December 1986, showing abrupt thermal plasma enhancements within the lower ionosphere, created by impinging keV electrons (courtesy I. Häggström, in: Pfaff, 2012).

The interaction altitude of the energetic electrons with the atmosphere depends on both the energy of the precipitating electron and the neutral density and constituency of the atmosphere. The effective penetration altitude of precipitating electrons as a function of energy is shown in Figure 2.6 (left) for a model atmosphere (Strickland et al., 1983) showing the resulting thermal plasma density. Incoherent scatter radar observations reveal how broad spectra of impinging 1-10 keV electrons create distinct thermal plasma enhancements within the lower ionosphere, as shown in Figure 2.6 (right) (Pfaff, 2012).

High latitude observations of these precipitating energetic electrons measured by the FAST satellite at 437 km altitude are shown in the upper panel of Figure 2.7 (Pfaff, 2012). The inverted-V shapes represent spatial variations of the electron energy that occur across auroral arcs. The lower panel shows the expected thermal plasma in the lower thermosphere associated with these measured electron energies according to established theory (Strickland et al., 1993). Note how the altitude of the modelled thermal plasma in the lower panel varies in accordance with the measured energy of the precipitating electrons, following the general penetration depths shown in Figure 2.6 (left).

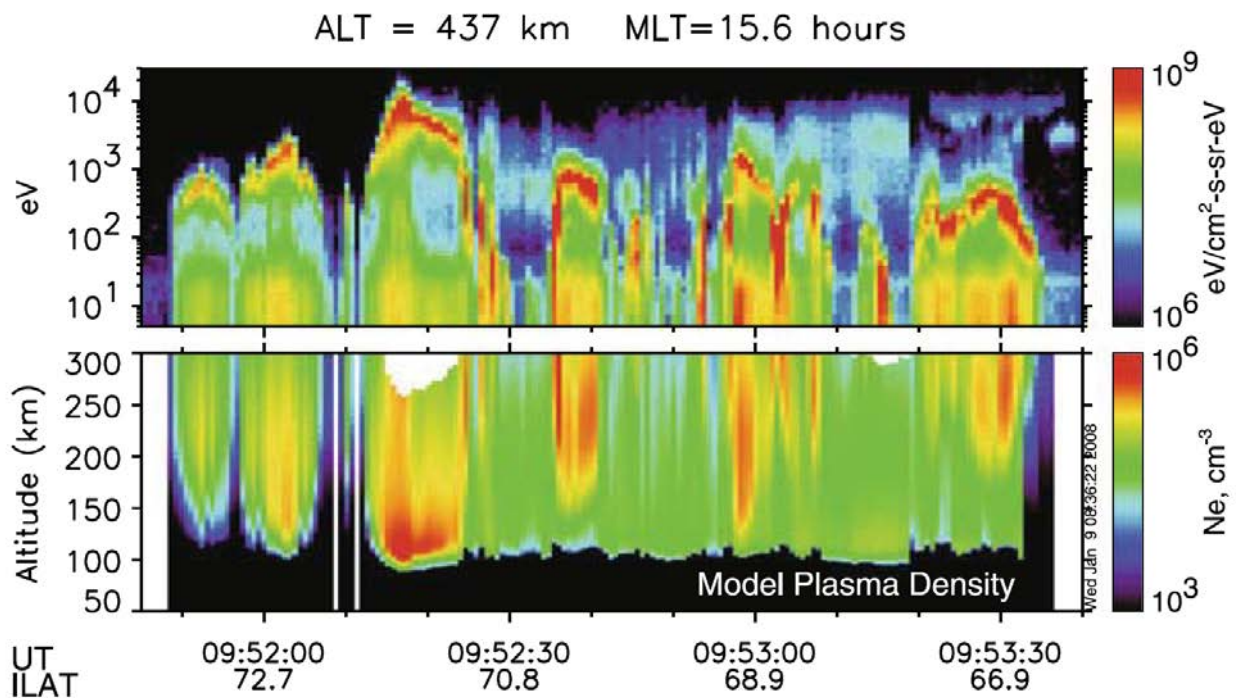


Figure 2.7: FAST satellite high-latitude observations on 5<sup>th</sup> December 1998, of down-going energetic electrons (upper panel) with model plasma density profiles created by the measured precipitating electrons (lower panel) (from Pfaff, 2012).

Contrary to discrete aurora, diffuse aurora, which are characterized by electron precipitation with a broad energy spectrum (100 eV to 50 keV) over a wide latitudinal and longitudinal extent, can persist for hours, building up substantial ionization over the 100-300 km altitude range (Nishimura et al., 2020). At higher altitudes, ionized constituents have a long lifetime due to the much longer recombination rates of the  $O^+$  ions, as discussed above.

Another source of ions in the near-Earth plasma results from the influx of meteors, which occur due to their ablation as they strike the upper atmosphere. This mechanism directly produces ionized gases when the kinetic energy of the incoming meteor is sufficient to remove electrons from the neutrals (Vondrak et al., 2008). As many meteors contain metals, this process produces metallic ions, not otherwise present in the LTI. The glowing gas associated with the ablation of larger meteors produces the familiar “shooting stars” that are observed in the night sky at all latitudes. The metallic ions in the



Earth's ionosphere have recombination lifetimes that are very long; when these ions settle in the lower thermosphere, they can accumulate in narrow altitude regions to form long-lasting “sporadic-E” layers, as discussed further on below.

## 2.2. Collisional coupling between ionized and neutral gases within the LTI

Having described the main characteristics of the LTI, we now discuss several key physical processes within this region that establish why the LTI is so *critical* to geospace. This discussion provides the foundation for the main science objectives of a dedicated LTI mission, which are discussed in Chapter 3. In particular, we show below how the collisional interactions between the plasma and neutral gases inherent to the LTI represent fundamental characteristics of the Earth-space transition region, which enable electric currents to flow mainly perpendicular to the Earth's magnetic field, and therefore mainly *horizontally*. In the LTI, these currents serve as an integral part of the larger-scale global electrical circuit, in particular linking the field-aligned currents that connect the magnetosphere and ionosphere (Figure 2.11 below). These major magnetospheric currents flow nearly vertically along magnetic field lines through the ionosphere, closing in the LTI at high latitudes while merging at mid- and low latitudes with wind driven currents, also in the LTI, forming the Earth's global electric circuit. To fully appreciate the importance of the LTI, we underscore that neither the neutral gas nor the plasma alone can support electric currents perpendicular to the magnetic field. Rather, it is a consequence of the *combined* neutral and charged particle behavior that enables such complex and highly variable global scale cross-field currents to exist.

Below, we discuss physical processes that are organized among three different themes which each describe critical aspects of the collision-dominated behavior that characterizes the LTI:

**Collisional electrodynamics:** The processes governing the motion of charged species in the presence of neutrals, an ambient magnetic field, and the behavior of the resulting LTI electric currents. Special reference is made in this section to ionospheric Hall and Pedersen conductivity, electrodynamic parameters that characterize the response of the LTI to imposed field-aligned currents and electric fields.

**Collisional energetics:** The collisional interactions between neutrals and charged species that lead to heating and energy exchange between species in the LTI. Key processes include heating by precipitating energetic particles and the dissipation of electromagnetic energy.

**Collisional dynamics:** The processes related to the motion of neutral gases within the LTI system. This includes the effects of collisional forces from plasmas (ion drag) and the largely unknown effects of atmospheric waves (gravity waves, planetary waves and tides) that propagate into the LTI from the atmosphere below, resulting in intense lower thermosphere winds and shears.

### 2.2.1. Collisional electrodynamics

**Coupling of the Lower Ionosphere and Thermosphere.** Having shown how the ionized and atmospheric gases co-exist within the same volume in the LTI, we now discuss how the ionized and neutral gases are coupled. Although only ion and electron motions are influenced by the Earth's magnetic field, both the ionized and neutral gas motions are subject to collisional interactions that vary with altitude. A profusion of behaviors results from these interactions.

Charged particles gyrate about the ambient magnetic field at their gyro frequencies,  $\Omega_j$ , given by  $\Omega_j = e|B|/m_j$  where  $e$  is the electric charge,  $|B|$  is the magnetic field strength, and  $m_j$  is the mass of the species which is designated by  $j$ . For reference, in a magnetic field of 50  $\mu\text{T}$  which is typical in the lower



ionosphere at high latitudes, the gyro frequency for  $O^+$  ions is 48 Hz, whereas the gyro frequency is 1.4 MHz for electrons.

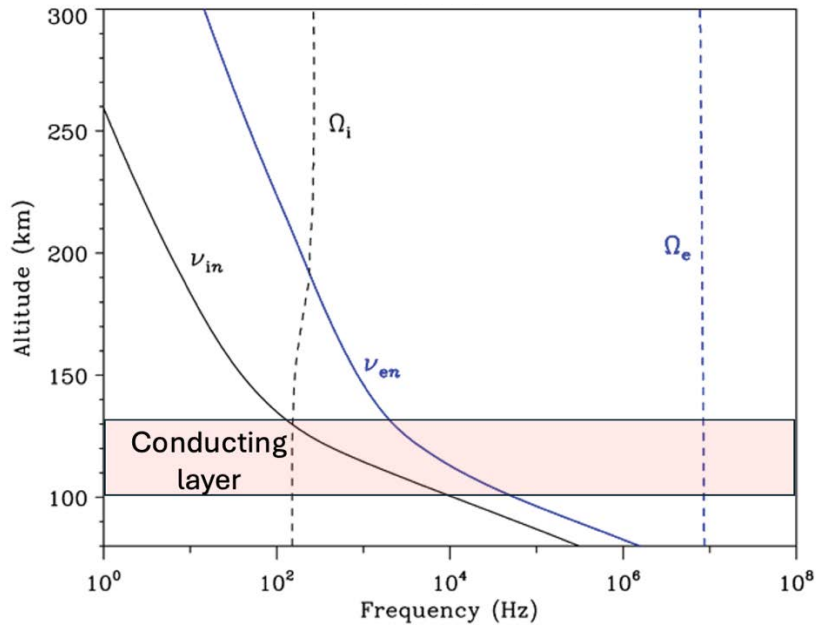


Figure 2.8: Altitude profile of model ion and electron gyrofrequencies and ion-neutral and ion-electron collision frequencies for a mid-latitude location (Wallops Island, Virginia, USA) and mid-day (after Pfaff, 2012).

Since the upper atmosphere neutral gases are dense compared to the ionized species, collisions between charged and neutral species significantly influence the individual and bulk motions of the plasma. Various expressions have been put forward for the ion-neutral collision frequency,  $\nu_{in}$ , and the electron-neutral collision frequency,  $\nu_{en}$ , (Banks, P. M., 1973) that depend on the neutral density, neutral and ion masses, and their temperatures. For the lower altitude regions of the LTI, mean *modelled* ion-neutral and electron-neutral collision frequency profiles are shown in Figure 2.8 for a mid-latitude location at mid-day. While empirical models of the collision frequencies are reasonable to capture some LTI features, they have limited applicability since they depend on the ambient gas constituents and temperatures which are not known. Collocated, simultaneous *in situ measurements* of the neutral and ionized atmospheric gases are critical to quantify collision frequencies to better understand and model many LTI processes.

Perhaps the most prominent role of the LTI is to provide a path where currents flowing in the Earth's electrical circuit can connect, or close, due to different motions of the charge carriers (ion and electrons) of the ionospheric plasma. This behavior is different from that of non-collisional cold plasmas in the presence of electric and magnetic fields where both ions and electrons gyrate under the influence of the magnetic field, as shown in Figure 2.9 (a), but drift together across field lines in the same  $\mathbf{E} \times \mathbf{B}$  direction. On the other hand, in the LTI, collisions with the neutral gas interrupt the ion gyromotion so frequently that the ion motions deviate from the  $\mathbf{E} \times \mathbf{B}$  direction, as shown in Figure 2.9 (b). The difference in charge motion between ions and electrons due to their differential response to collisions creates a cross-field current, or one that is perpendicular to the ambient magnetic field. Below about 130 km, the ion motion is dominated by collisions with the neutrals ( $\nu_{in} > \Omega_i$ ) whereas the electrons are able to execute their gyro motions about the magnetic fields ( $\nu_{en} \ll \Omega_e$ ) and consequently undergo  $\mathbf{E} \times \mathbf{B}$  drifts across magnetic field lines (the electrons are said to be “magnetized”). The narrow altitude region of 100–130 km where the electron  $\mathbf{E} \times \mathbf{B}$  drifts are unimpeded by neutral collisions constitutes the atmospheric dynamo region where strong currents flow, to be discussed below. Figure 2.9 (c)

illustrates this process at high latitudes by showing the cross-field motion of the ionosphere plasma as a function of altitude. The downward magnetic field of the northern hemisphere is shown in this figure together with a horizontal electric field. Another illustration of these effects is given in Figure 2.9 (right), where modelled ion and electron velocities are shown versus altitude for a 50 mV/m poleward (meridional) DC electric field at high latitudes ( $|B| = 54 \mu\text{T}$ ). Ions and electrons both undergo an  $\mathbf{E} \times \mathbf{B}$  drift of approximately 950 m/s at 220 km. At lower altitudes however, the ion drift is slower and changes its orientation towards the direction of the DC electric field. At about 125 km altitude, where the Pedersen mobility peaks (as discussed below), the ion drift is one half the value given by  $\mathbf{E} \times \mathbf{B}$  and is equally divided between flow along and perpendicular to the electric field direction. At lower altitudes, the ion drift decreases significantly such that the electrons are the only charge carriers with appreciable motion. As the ion-neutral collisions slow the ion drifts, momentum is transferred to the neutral gas, as will be explained below.

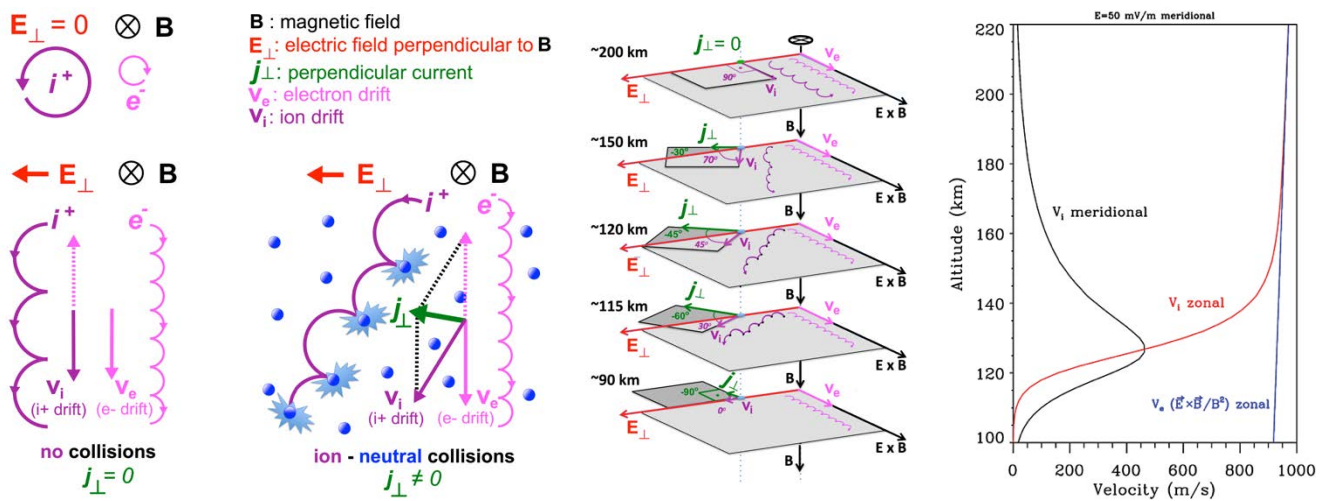


Figure 2.9: Ion and electron ( $\mathbf{E} \times \mathbf{B}$ ) drifts in the frame of the neutral gas, (a) in the absence of collisions, (b) in the presence of ion-neutral collisions, and (c) as a function of altitude (i.e. neutral density and collision frequency) (from ESA, 2020). Right, ion and electron ( $\mathbf{E} \times \mathbf{B}$ ) drifts in the presence of an applied 50 mV/m electric field in the meridional (or northward) direction at Poker Flat, Alaska (from Pfaff, 2012).

The fundamental interactions of the ions and electrons with the neutral gas are specified by the mobility and conductivity tensors. The mobility  $\mu$  relates the velocity of mean motion of a species to the applied electric field. The presence of a magnetic field makes the mobility anisotropic, it is high along the magnetic field, while lower across the magnetic field. Moreover, a perpendicular electric field induces the motion both along its direction (Pedersen mobility) and in a direction perpendicular to both electric and magnetic fields (Hall mobility). Due to different altitude profiles of the collision frequencies and gyrofrequencies for ions and electrons, Hall and Pedersen mobilities exhibit different altitude profiles. The conductivity  $\sigma$  relates the electric current to the electric field. Pedersen and Hall conductivities can be defined through the charge concentration and the corresponding mobility. Pedersen and Hall conductivities are enhanced in the daytime LTI, and represent distinguishing electrodynamic characteristics of the lower ionosphere as they enable largely horizontal currents to flow, magnetospheric field-aligned currents to close, and significant Joule heating to occur.

The current density ( $\mathbf{J}$ ) [ $\text{A/m}^2$ ], is expressed in terms of the plasma density ( $N_e$ ), the relative motion between ions and electrons, and the electric charge ( $e$ ):

$$\mathbf{J} = e \cdot N_e \cdot (\mathbf{V}_i - \mathbf{V}_e) \quad (2.5)$$



Clearly, where the ion velocities may be considered very small, at least with respect to the neutral gas, ( $V_i \sim 0$ ), for example below 110 km altitude (see Figure 2.9), the current is essentially carried by the electron drift which corresponds to the  $\mathbf{E} \times \mathbf{B}$  velocity. Below about 100 km, where the ion-electron vector difference is near its maximum, the current magnitude decreases drastically because of the reduction in plasma concentration at the base of the ionosphere.

Another way of expressing electric current  $\mathbf{J}$  is through Ohm's law, as the product of the electric field  $\mathbf{E}$  and the conductivity tensor,  $\underline{\sigma}$ :

$$\mathbf{J} = \underline{\sigma} \cdot \mathbf{E} \quad (2.6)$$

The conductivity may be expressed in terms of the mobilities such that  $\underline{\sigma} = N_e \cdot e^2 \cdot (\underline{\mu}_i + \underline{\mu}_e)$ . The components of the conductivity are thus (Rishbeth and Garriott, 1969):

$$\sigma_{\parallel} = N_e \cdot e^2 \cdot (\mu_{\parallel e} + \mu_{\parallel i}) \quad \text{parallel conductivity} \quad (2.7)$$

$$\sigma_P = N_e \cdot e^2 \cdot (\mu_{Pe} + \mu_{Pi}) \quad \text{Pedersen conductivity} \quad (2.8)$$

$$\sigma_H = N_e \cdot e^2 \cdot (\mu_{He} - \mu_{Hi}) \quad \text{Hall conductivity} \quad (2.9)$$

The mobility tensor is defined in a geometry where the z-axis is parallel to the magnetic field,  $\mathbf{B}$ , and the x and y axes are in the plane perpendicular to  $\mathbf{B}$ . In the reference frame moving with the neutral flow the mobilities are expressed as follows with  $j$  corresponding to the species (Rishbeth and Garriott, 1969):

$$\mu_{\parallel j} = \frac{1}{m_j v_{jn}} \quad \text{parallel mobility} \quad (2.10)$$

$$\mu_{Pj} = \frac{1}{m_j v_{jn}} \cdot \left( \frac{v_{jn}^2}{v_{jn}^2 + \Omega_j^2} \right) \quad \text{Pedersen mobility} \quad (2.11)$$

$$\mu_{Hj} = \frac{1}{m_j v_{jn}} \cdot \left( \frac{v_{jn} \Omega_j}{v_{jn}^2 + \Omega_j^2} \right) \quad \text{Hall mobility} \quad (2.12)$$

We emphasize here that the collision frequencies, represented by  $v_{jn}$ , are key for quantifying the mobility and hence the conductivity. For parallel mobilities electron-ion collisions, represented by  $v_{ic}$ , also become important. Model profiles of the Hall and Pedersen mobility components (combined for ions and electrons) are plotted in Figure 2.10 (left) for three different locations, one at the magnetic equator (Jicamarca, Peru), one at high latitudes (Poker Flat, USA), and one at mid-latitudes (Wallops Island, USA). In the LTI, the Pedersen mobility peaks at around 125 km and the Hall mobility peaks near 105 km. The mobilities are higher at lower latitudes due to the weaker magnetic field strength since they are roughly proportional to  $\sim 1/\Omega_i^2$ , and the altitude of the peak is somewhat higher. The Pedersen mobility peak in the mesosphere results from electron mobility contributions.

The ionospheric conductivity depends linearly on the charged particle density and the anisotropic mobilities controlled by the magnetic field. Figure 2.10 (right) shows typical vertical profiles of the Pedersen, Hall, and parallel conductivities at noon at mid-latitudes for medium solar activity (Richmond, 2016). The Pedersen conductivity is directed along the electric field vector perpendicular to the ambient magnetic field. The Hall conductivity is in the  $-\mathbf{E} \times \mathbf{B}$  direction. These conductivities enable an ionospheric current system perpendicular to the Earth's magnetic field, which provides the closure path of the currents in the solar wind magnetosphere circuit that flow along magnetic field lines. Note that the parallel conductivity is orders of magnitude higher than the corresponding Hall and Pedersen

conductivity, though it falls off in the lower ionosphere due to the higher neutral density and collision frequencies there.

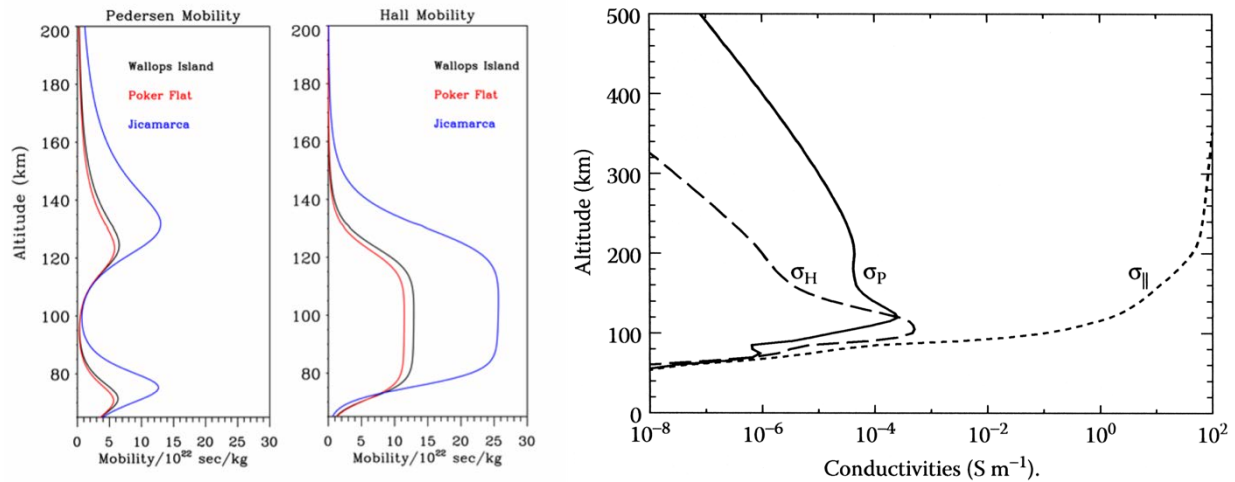


Figure 2.10: Left, Pedersen and Hall mobility components (combined for ions and electrons) for three different locations, at the magnetic equator (Jicamarca, Peru), at high latitudes (Poker Flat, USA), and at mid-latitudes (Wallops Island, USA) (from Pfaff, 2012). Right, noontime, mid-latitude conductivity profiles for Hall, Pedersen, and parallel conductivities for medium solar activity on March 21 (from Richmond, 2016).

**Key processes in the LTI involving electrodynamics.** The preceding description of LTI electrodynamics is at the core of four different, fundamental processes involving currents intrinsic to the LTI region:

1. Closure of field-aligned currents of magnetospheric origin;
2. Auroral electrojet currents driven by magnetospheric electric fields;
3. Global dynamo daytime currents;
4. Equatorial Electrojet currents.

Below, we first describe the two sets of currents associated with the high latitude LTI and magnetospheric forcing. These are followed by descriptions of the daytime dynamo and equatorial electrojet currents driven by neutral winds.

**Closure of field-aligned currents.** Numerous satellite magnetometer measurements have shown how field-aligned currents flow along magnetic field lines between the magnetosphere and the ionosphere. These currents are associated with precipitating energetic particles that create the aurora as well as return currents. *in situ* magnetometer data from polar-orbiting satellites show that such field-aligned currents are typically observed to first order as a pair of sheets that extend in longitude along roughly equal magnetic latitudes. Statistical studies (Iijima and Potemra, 1976) show that the field-aligned currents form a characteristic pattern, with currents on the poleward edge of the polar cap flowing into the ionosphere on the dawn side and out on the dusk side, with opposite flows on the equatorward edge of the oval.

Closure of the field-aligned currents occurs in the lower ionosphere where the conductivity is largest, as sketched in Figure 2.11. This conductivity is controlled both by the altitude profile of the local thermal plasma, primarily determined by the energy spectra of energetic electrons, and by the mobilities that depend on the collision frequencies between the neutral and the charged particles and thus on the

properties of the atmospheric gas (as discussed above). Although the *average current closure diagram* shown in Figure 2.11 is generally “accepted”, there are no observations demonstrating that the 3-D currents in the high latitude ionosphere actually connect in this way. The diagram is particularly oversimplified for current closure during geomagnetically disturbed conditions. Furthermore, the smaller scale and instantaneous current flows are certainly more complex as is the connection to mid- and low-latitude currents. Understanding these currents and their closure await detailed observations in the LTI which then support associated, detailed modeling.

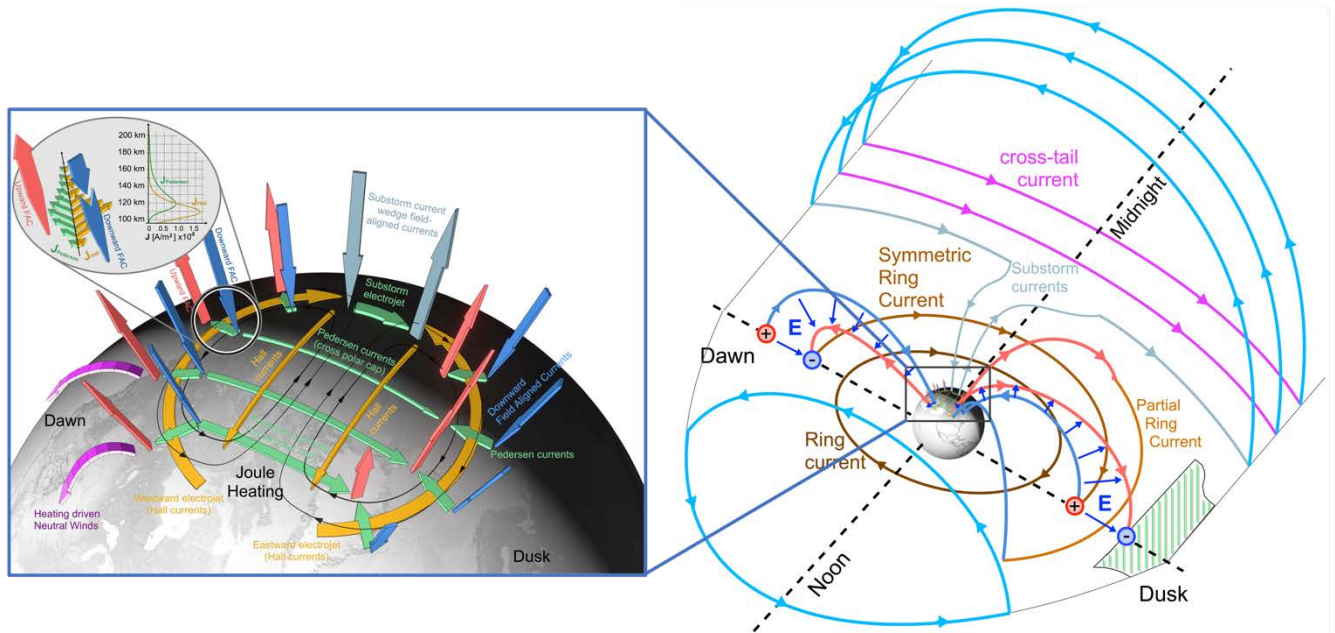


Figure 2.11: The current system and closure of field-aligned currents in the ionosphere by horizontal currents (after Palmroth et al., 2021). The insert on the right shows how the field-aligned currents connect to magnetospheric currents.

**Auroral electrojets.** At the lowest altitudes in the high-latitude LTI where the ion drifts are almost entirely impeded by collisions with the neutral gas, the magnetospheric electric fields impose  $\mathbf{E} \times \mathbf{B}$  motion on the electrons which drive Hall currents referred to as “auroral electrojets”, shown as wide yellow arrows in Figure 2.11. The pre-midnight poleward DC electric field drives the electrons westward in the auroral oval, resulting in an eastward electrojet, while the opposite is true during post-midnight conditions where the equatorward DC electric field drives a westward electrojet.

The auroral electrojets are considerably stronger than the Pedersen currents that close the field-aligned currents. Their magnitudes depend on both the DC electric field and the plasma density which is largely dependent on auroral precipitation. Auroral electrojet signatures observed in ground-based magnetometers can be 500-1000 nT or even larger, particularly during significant magnetic storms. Importantly, sudden large variations of electrojets may induce currents in long terrestrial conductors on the ground, such as power grids and pipelines, impacting power systems and infrastructure.

The behavior of the auroral electrojet currents, particularly where they converge near midnight and in the dayside cusp, is not understood. Figure 2.11 depicts the westward current at midnight which may surge towards earlier local times during substorms, creating a pronounced “Harang discontinuity” between the opposing electrojet current directions. How such intensified electrojets are closed is an open question. Weaker Hall currents may provide closure by forming across the polar cap, as shown in Figure 2.11, although how and whether these currents are set up is not presently known. The characteristics and dynamics of these fundamental, high-latitude auroral electrojets are not known,

particularly their spatial and temporal variability, and thus require systematic measurements to understand their behavior.

**Atmospheric dynamo.** Atmospheric currents are set up by neutral gas motions or winds which, via ion-neutral collisions, drag the ionosphere plasma across the ambient magnetic field. The current density,  $\mathbf{J}$ , is mainly produced within the lower ionosphere (90–140 km) as the ambient ions are collision dominated yet the electrons are magnetized, as described above. As the winds,  $\mathbf{U}$ , push ions across the magnetic field,  $\mathbf{B}$ , electromotive fields are set up equivalent to  $\mathbf{U} \times \mathbf{B}$ . In order to restrict the currents to be non-divergent (i.e.,  $\nabla \cdot \mathbf{J} = 0$ ), local polarization electric fields,  $\mathbf{E}$ , are set up. The total current is thus given by the dynamo equation:

$$\mathbf{J} = \underline{\sigma} \cdot (\mathbf{E} + \mathbf{U} \times \mathbf{B}) \quad (2.13)$$

where  $\underline{\sigma}$  is the conductivity tensor. Figure 2.12 illustrates how the neutral wind drives currents in the LTI at mid-latitudes (Jin et al., 2008). Here, tidal winds drive the currents within the narrow altitude region of 100–140 km where conductivities are enhanced. The complex height varying current density can be simplified through height integration to a global current system which is called the “solar quiet” (or  $S_q$ ). In both the northern and southern hemispheres, these dynamo currents form global current loops on the dayside in the LTI from the upper mid-latitudes to the magnetic equator, as shown in Figure 2.13 (Yamazaki and Maute, 2017). To maintain a divergence free current system to satisfy  $\nabla \cdot \mathbf{J} = 0$ , polarization electric fields perpendicular to the magnetic field are set up and, field-aligned currents flow between conjugate points at mid and low latitudes (Heelis and Maute, 2020). These polarization electric fields map along the magnetic field and influence plasma flows at higher altitudes, affecting the entire dayside ionosphere.

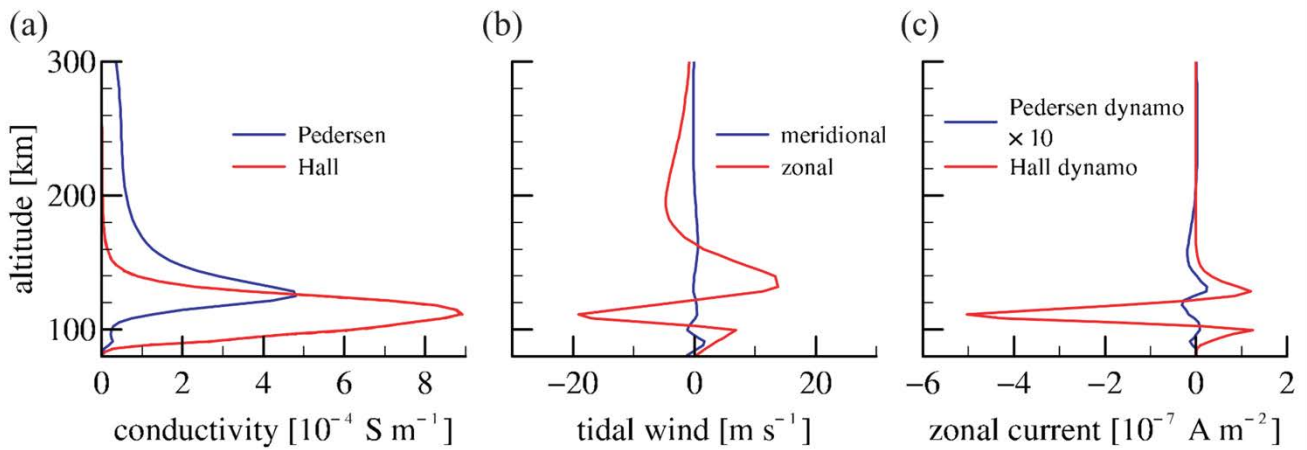


Figure 2.12: Model results which illustrate how the neutral wind drives currents in the LTI at mid-latitudes (from Jin et al., 2008). Here, tidal winds in (b) drive the currents (c) within the narrow altitude region of 100–140 km based on the enhanced conductivities (a).

Although it is generally accepted based on sparse observations and modelling that global, daytime dynamo currents are driven by neutral winds between 100–140 km altitudes, their physical properties remain largely unknown and poorly understood because of a lack of measurements. Hence, this fundamental geospace phenomenon still eludes our understanding, particularly the extent in which the dynamo drives local polarization electric fields and field-aligned currents which, in turn, influence ionospheric dynamics at higher altitudes.



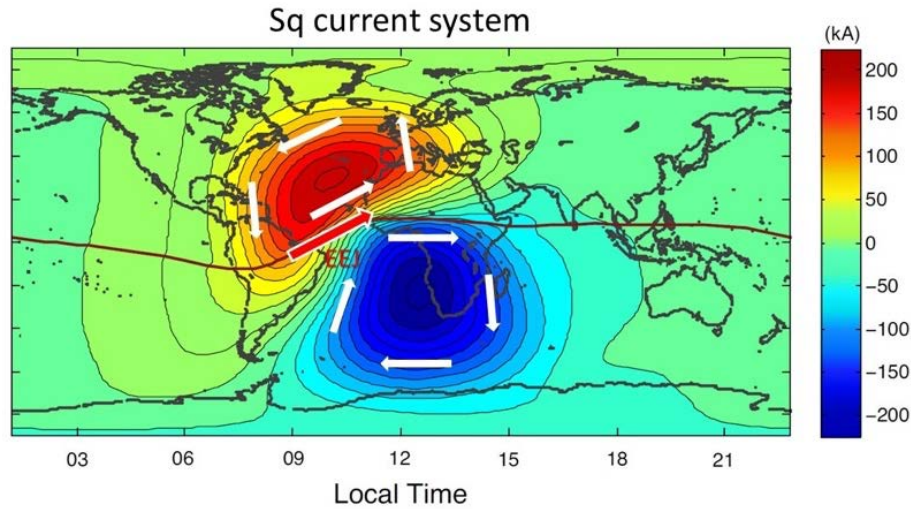


Figure 2.13: The daytime Sq dynamo currents including the equatorial electrojet (after Yamazaki and Maute, 2017).

**The equatorial electrojet.** The creation of enhanced current flow at the magnetic equator, the equatorial electrojet, is an example of emergent behavior resulting from the constraints imposed on the plasma and electrodynamics. Here, an enhanced Cowling conductivity is set up by the horizontal magnetic field geometry and the fact that below 90 km no current can flow creates a significantly stronger current in this region, the equatorial electrojet. At this location, in the altitude range of roughly 95–115 km, significant zonal (east-west) electron drifts are set up in the following way. Vertical ion  $\mathbf{E} \times \mathbf{B}$  drifts are inhibited due to collisions. The electrons on the other hand are magnetized and drift vertically. The resulting vertical charge separation establishes a vertical polarization field,  $\mathbf{E}_p$ , which, in turn, creates a horizontal  $\mathbf{E}_p \times \mathbf{B}$  drift,  $\mathbf{V}_D$ , which reinforces the small, existing dynamo current and thus creates the strong electrojet current. Sounding rocket data show the existence of the vertical, polarization electric field and enhanced current density associated with the daytime electrojet at 105 km in Figure 2.14 (Pfaff et al., 1997). The equatorial electrojet is strongest within a degree of the magnetic equator, where the magnetic field vector is precisely horizontal, thus enabling the vertical, polarization electric field described above to be set up.

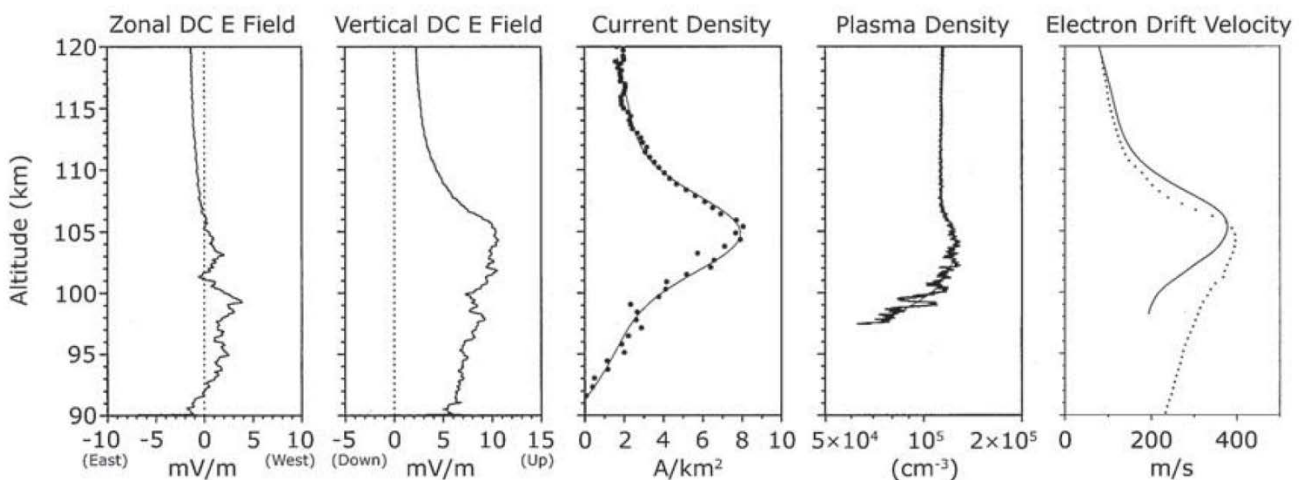


Figure 2.14: Sounding rocket observations on 21<sup>st</sup> September 1994 (10:56) at Alcantara, Brazil, of the DC electric field, current density, and plasma density in the equatorial electrojet. The panel on the right shows consistent calculations of the plasma drift velocity from the independently measured current and plasma density and the  $\mathbf{E} \times \mathbf{B}$  velocity (after Pfaff et al., 1997).



Despite our general understanding of the daytime eastward electrojet, evidence of "counter" or westward electrojet which can occur, particularly in the afternoon as well as in the early morning sector, are not understood. Furthermore, strong electric fields (as evidenced in Jicamarca radar data), including large-scale structure, particularly at night, and their possible association with strong wind shears represent major aspects of the electrojet system that are poorly explored.

### 2.2.2. Collisional energetics

Energy within the LTI can take several forms: kinetic, chemical, potential, and thermal. Many processes occurring in the LTI that change the energy of the system result in the transformation from one energy form to another. Here, we examine "collisional energetics" which concerns how collisional processes in the LTI result in a significant input and transformation of energy.

**Transformation/dissipation of electromagnetic energy and auroral energetic particles.** Heating of the high latitude LTI region, besides EUV solar insolation, results from two major processes – Joule heating due to electromagnetic energy dissipation and heating from the direct impact of precipitating energetic particles. These two high-latitude heating processes have global impact on the LTI and their combined energy deposition rates can rival that of the sun's EUV energy deposition during geomagnetic storms, often concentrated in and around the auroral oval. One of the consequences of such energy input and the collisional exchange of energy between ions and neutrals is a change in the chemical makeup of the ionosphere and thermosphere. Below, we describe these two important energy sources followed by a discussion of the chemical consequences that tie to collisional energetics.

**Joule heating in the LTI.** When an electrical current flows through a conductor with a finite conductivity, part of the electromagnetic energy is converted to thermal energy in the Joule heating process. Within the LTI, the electromagnetic power converted to heat is significant, on the order of several GW globally, at times rivalling the heating caused by solar radiation absorption. The process arises from the same basic plasma-neutral collisions that are at the origin of electrical currents circulating in the LTI. The energy deposition process peaks in the LTI and has a global impact on the structure of the upper atmosphere. Joule heating in the LTI is highly variable in space and time, and difficult to estimate, as the electromagnetic energy source can change rapidly and the LTI properties contributing to Joule heating have different response rates.

A major source of energy to the LTI system is electromagnetic energy imposed by the solar wind-magnetosphere-ionosphere interaction. Energy flow to the LTI can be considered in a quasi-steady state where the convergence of the Poynting vector is balanced by the conversion of electromagnetic energy quantified by  $\mathbf{J} \cdot \mathbf{E}$  (Richmond, 2010; Thayer and Semeter, 2004). In the magnetosphere generator region,  $\mathbf{J} \cdot \mathbf{E} < 0$  indicates flow kinetic energy from the solar wind is converted to electrical energy. In the dissipative region of the LTI,  $\mathbf{J} \cdot \mathbf{E} > 0$  indicates electromagnetic energy is converted to thermal and flow kinetic energy. This electromagnetic conversion rate in the LTI primarily results from processes that occur perpendicular to the magnetic field and can be expressed by the corresponding perpendicular component of electric fields, currents, and velocities,

$$\mathbf{J}_{\perp} \cdot \mathbf{E}_{\perp} = \mathbf{J}_{\perp} \cdot (\mathbf{E}_{\perp} + \mathbf{U} \times \mathbf{B}) + \mathbf{U} \cdot \mathbf{J}_{\perp} \times \mathbf{B} \quad (2.14)$$

The first term is the Joule heating rate of the gas and is positive indicating a conversion of electromagnetic energy to thermal energy. The second term represents the mechanical power term, related to the bulk momentum transfer primarily to the neutral gas. This term can be of either sign depending on whether the LTI medium, primarily through neutral wind motion, is acting to generate or dissipate electromagnetic energy. Joule dissipation, frictional heating, and the related energy exchange processes in the ionosphere-thermosphere system represent an important means by which the



atmosphere acts as a load on the magnetosphere (Strangeway, 2012; Thayer, 1998, 2000; Thayer et al., 1995). It is clear from the above relation that the vector orientation of the neutral wind and electric field to the net current is critical for determining the partitioning of energy from electromagnetic to thermal and bulk flows energies.

The relative motions of the ion and neutral populations have a profound effect on their respective temperatures, which results from frictional heating occurring between the ion and neutral gases when they are not flowing at the same speed and direction. The Joule heating rate per unit volume,  $q_j$ , accounts for heating of the neutral gas via currents flowing through the gas in a manner similar to Ohmic heating of a wire. However, the medium through which current flows in a wire is not moving. Joule heating accounts for the more general situation where the medium is moving and, for the LTI, that motion is at the neutral wind speed. The Joule heating rate may be represented as:

$$q_j = \sigma_p \mathbf{E}'^2 \quad (2.15)$$

where  $\mathbf{E}' = (\mathbf{E} + \mathbf{U} \times \mathbf{B})$  is the electric field in the neutral reference frame, and  $\sigma_p$  is the Pedersen conductivity. Joule heating depends linearly on the plasma density, which varies with solar insolation and auroral precipitation, and on the square of the electric field when determined in the reference frame of the neutral wind. The wind reference frame is introduced because the current is passing through a moving gas, such that, the electric field of consequence to heating will depend on its vector relation with that moving media. Joule heating has a strong altitude dependency due to the variation of the Pedersen conductivity and winds with altitude.

Estimates of Joule heating profiles (per unit volume) are shown in Figure 2.15 (left) using European Incoherent Scatter (EISCAT) radar *remote sensing* data for different levels of magnetic activity and local time (Baloukidis et al., 2023). The peak Joule heating altitude for these cases corresponds roughly to the location of the maximum of the Pedersen conductivity which is determined using the *model* values of collision frequency since these are not measured. Because this conductivity depends on the plasma density, the peak Joule heating altitude also varies according to the plasma density profile. Joule heating profiles using measured plasma density, electric field, and neutral winds from 6 different sounding rockets flown from Poker Flat, Alaska, are shown in Figure 2.15 (right). While there is a qualitative understanding that Joule heating rates peak in the LTI region, there is considerable variation and structure in the profiles, affecting neutral gas dynamics and current systems in ways not incorporated into present descriptions of the region.

To further illustrate how Joule heating per unit volume is distributed in latitude and how particularly prominent it is in the LTI region, Figure 2.16 (from Figure 4.3) shows calculations of Joule heating rate, given in  $\text{mW/m}^3$ , as a function of altitude and latitude at a fixed longitude from the TIE-GCM physics-based model. These type of model calculations and predictions have never been comprehensively validated using global observations of all the needed plasma and neutral parameters and at the resolutions that are needed, with the exception of some near-instantaneous altitude profiles that are provided by sounding rockets.

Since Joule heating is related to the square of the electric fields, structured and time variable electric fields produce significantly higher Joule heating rate in reality, than the smoothed fields that typically are used in *model* calculations (Matsuo and Richmond, 2008). Indeed, various studies (Emery et al., 1999; Thayer, 1998) have invoked small scale electric fields to account for a “missing energy” in circulation models and other related calculations. Furthermore, travelling ionospheric and atmospheric disturbances are known to be launched during times of sudden enhanced Joule heating associated with geomagnetic storms, which would also contribute additional heating over a large range of scale sizes.

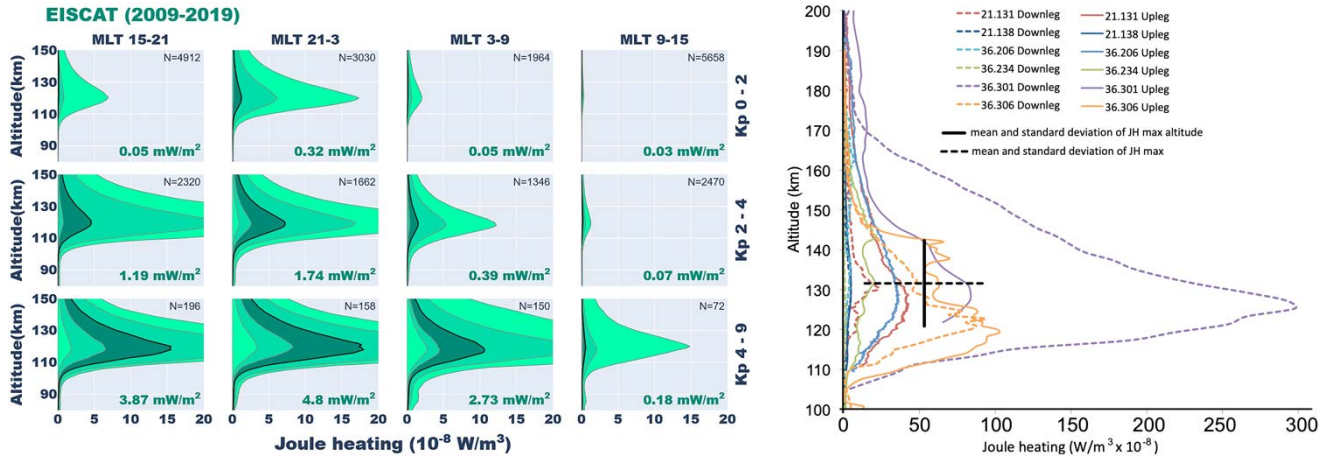


Figure 2.15: Left, diurnal and geomagnetic activity variations of Joule heating based on EISCAT data (from Baloukidis et al., 2023). Right, Joule heating calculated from high-latitude sounding rocket measurements (from ESA, 2020).

Dedicated and accurate observations of all relevant parameters of the LTI are necessary to properly quantify Joule heating in order to characterize and predict the global behavior of the upper atmosphere, especially during geomagnetically active times.

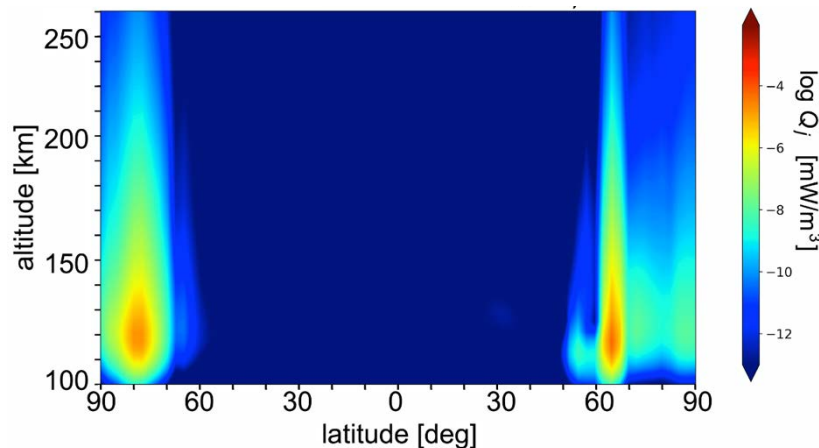


Figure 2.16: Altitude-latitude plot of Joule heating rate along a meridional cut at fixed longitude from the TIE-GCM model (from Sarris et al., 2023b).

**Poynting flux and heating associated with Alfvén waves and two-stream waves.** The Joule heating discussed here represents electromagnetic energy conversion which can also be expressed in terms of the Poynting Flux,  $\mathbf{S}$ , where  $\mathbf{S} = 1/\mu_0 (\mathbf{E} \times \mathbf{B})$  as discussed by (Thayer and Semeter, 2004). An important source of Poynting flux in the high-latitude ionosphere is that associated with electromagnetic Alfvén waves which typically accompany auroral precipitation and field-aligned currents, entering the high-latitude ionosphere from the magnetosphere. Indeed, Alfvén waves are very common in the high-latitude ionosphere at higher altitudes (Chaston et al., 2007) and have been shown to be an effective source of ionospheric heating, particularly in the cusp (Lotko and Zhang, 2018). Alfvén waves have also been observed in the lower ionosphere on auroral zone sounding rockets (Akbari et al., 2022; Cohen et al., 2013) exhibiting large amplitude electric and magnetic fields of tens of mV/m and 100's of nT, respectively. The waves have been interpreted as standing waves associated with the Alfvén “resonators” (Akbari et al., 2022), which have been speculated to exist within regions bounded by large gradients in the plasma density, such as in the LTI between the base of the F-region and the



lower E-region in the nighttime auroral ionosphere. Furthermore, low latitude C/NOFS satellite data suggest that resonators need not be confined to the high latitude region (Simões et al., 2012).

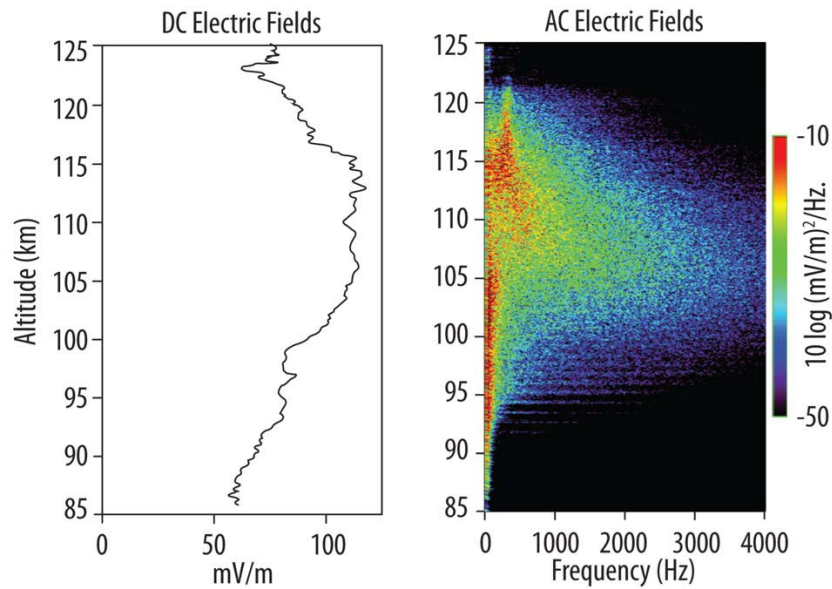


Figure 2.17: Right, wave electric field observations of two-stream waves vs. altitude observed on 9<sup>th</sup> April 1988 on a sounding rocket from Esrange, Sweden. Left, simultaneous measurements of the DC electric field (from Pfaff, 2012).

In addition to energy deposition and heating associated with Alfvén waves, two-stream waves and associated plasmas turbulence are another importance source of frictional heating in the LTI, particularly at high latitudes, where the presence of such waves may have profound effects on conductivity and the polar cap potential. The high-latitude E-region where Hall currents are driven by large DC electric fields is frequently unstable to collisional two-stream waves (Buneman, 1963; Farley, 1963), a condition which occurs when the relative ion-electron drift exceeds roughly the local acoustic velocity. In the high-latitude LTI, this condition is met for DC electric fields of  $\sim 20$  mV/m or greater, where the ambient  $\mathbf{E} \times \mathbf{B}$  velocities exceed the local acoustic speed which is  $\sim 400$ - $500$  m/s near 110 km altitude. Although two-stream waves are commonly associated with the auroral electrojet where the driving magnetospheric DC electric fields may be particularly strong, this instability and the resulting waves and turbulence may commonly exist throughout the polar cap wherever this driving condition is met.

Sounding rocket observations of two-stream waves are shown in Figure 2.17 (right), where the DC electric field (left) reached values of over 100 mV/m (Pfaff et al., 1992). The data clearly show that the altitude region associated with the high Hall mobility is where the two-stream waves are unstable. The unstable region expands to higher altitudes as the driving DC electric field increases (Pfaff, 2012). The instability is cut off below 95 km due to increased collisional effects. Two-stream waves are also present in the equatorial electrojet (Pfaff et al., 1987, 1997) although the unstable altitude region is narrower, limited to altitudes of 103-109 km where the electrojet current is strongest and hence the ambient, differential electron-ion drift is supersonic.

Importantly, two-stream waves have been shown to be a local source of frictional heating. Using global modelling, (Dimant and Oppenheim, 2011) show that during storms, the anomalous conductance and heating due to two-stream waves may increase the polar cap conductance by a factor of two, resulting in an overestimate of the cross polar cap potential.

**Heating from precipitating auroral energetic particles.** Another important energy source to the upper atmosphere that also occurs in the LTI at high latitudes is the conversion of the energy of precipitating particles to heat. Electrons are accelerated in the magnetosphere and precipitate along largely vertical magnetic field lines into the LTI. The collision cross-section with neutrals is energy dependent, such that the higher the particle energy, the smaller the neutral collision cross-section. Thus, higher energy particles, which are constantly losing energy as they descend, are able to penetrate deeper into the atmosphere before experiencing significant collisions to stop their progression. At this level, inelastic collisions become dominant and kinetic energy is more efficiently converted to thermal and chemical energy. Thermal efficiency, often referred to as representative of the amount of kinetic energy that is converted to thermal energy, is highly dependent on the particle kinetic energy spectrum and, therefore, is strongly altitude dependent in the LTI. The thermal efficiency is poorly known and detailed measurements of the neutral response in the presence of precipitating particles are required to better estimate this parameter.

Heating of the neutral gas through direct absorption of kinetic energy of incoming particles represents a heat source that was estimated to be about one fourth of typical Joule heating on average (Lu et al., 1995), although it can be more significant during large auroral events (Thayer and Semeter, 2004). The heating caused by the precipitating particles depositing their energy is important for the high latitude energy budget as well for the creation of localized effects. For example, (Clemmons et al., 2008) have invoked heating by soft particle precipitation in the cusp to suggest that upwelling might explain the depletions in the neutral density measured by the Streak satellite at low altitudes.

A comparison of Joule heating and particle heating is provided in Figure 2.18 during storm conditions as well as quiet conditions, from a model (Richmond and Lu, 2000) that assimilates a variety of ground-based and space-based measurements. For the two processes, the energy input to high-latitude regions during storm conditions is significantly increased compared to quiet conditions. As noted by these authors, the Joule heating has a greater effect on the thermospheric circulation not only because of the higher heating rate, but also because the particle energy is deposited at lower altitudes extending as low as 90 km, where the higher density of the neutral gas reduces the amplitude of its disturbances.

The contribution of the direct heating by the precipitating particles to the energy balance in the LTI needs a more careful characterization, requiring direct *in situ* measurements of the precipitating particle energy spectra as well as the local properties of the atmosphere.

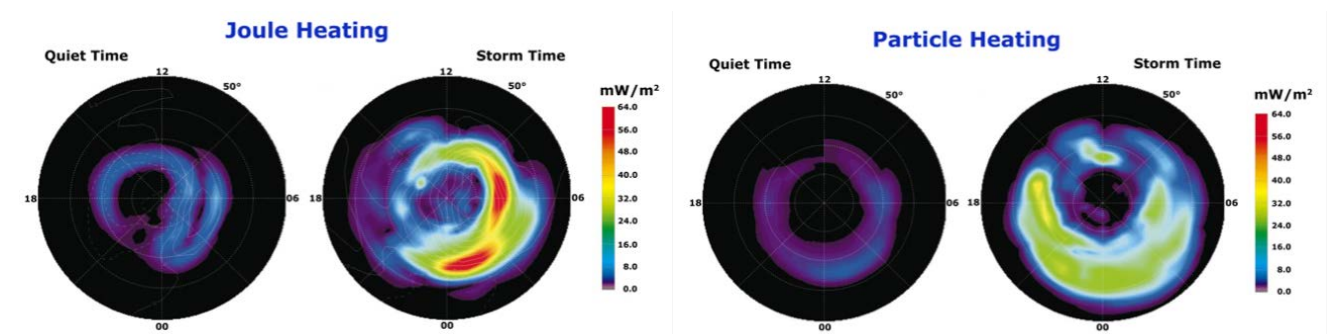


Figure 2.18: Joule heating and particle heating, during quiet time and storm-time conditions (from Richmond and Lu, 2000).

**Chemical changes in the LTI due to Energetic Particle Precipitation (EPP).** Precipitating energetic electrons at higher energies (at and above 10 keV) provide important kinetic energy input to the LTI and the mesosphere below, with significant chemical consequences. The high energies (above 50 keV) of precipitating radiation belt particle populations and solar energetic particles ionize the lower thermosphere, the lower mesosphere and even the upper stratosphere (solar particles). Ionization is a

key process that produces an abundance of charged particles and increases in electron density in the LTI. The formation of the ionospheric constituents within the LTI involves several multi-step, chemical processes producing predominantly  $\text{NO}^+$  and  $\text{O}_2^+$  ions. The ion and electron concentrations shape the E-region conductivities. The closure of the currents, the resultant neutral gas heating, and the transfer of momentum to the neutral gas associated with EPP are thus highly dependent on chemical production and loss processes forming the E-region. Chemical time constants to produce these ion species are much faster than transport, thus the effects of EPP on the LTI electrodynamics are localized but strongly altitude dependent.

As a result of strong EPP ionization and subsequent chemical reactions, trace species important to ozone depletion, such as odd hydrogen ( $\text{HO}_x$  defined as the sum of H, OH, and  $\text{HO}_2$ ) and odd nitrogen ( $\text{NO}_x$  defined as the sum of N, NO,  $\text{NO}_2$ ), are produced. While most  $\text{HO}_x$  is produced just below the LTI in the mesosphere where the related ozone depletion also takes place,  $\text{NO}_x$  forms throughout the thermosphere. At polar latitudes, in the darkness of winter,  $\text{NO}_x$  species have long lifetimes, which allow them to be transported deeper into the atmosphere and contribute more significantly to polar ozone loss in the stratosphere (Sinnhuber et al., 2012).

Kinetic energy sources and inelastic collisions are common LTI processes that can lead to the generation of excited neutral species. A portion of these excited species de-excite and radiate a photon before experiencing another collision. This acts as a cooling mechanism to the LTI energy budget and is an important process to offset the heating mechanisms described previously. The infrared emissions of 5.3- $\mu\text{m}$  by NO, 15- $\mu\text{m}$  by  $\text{CO}_2$ , and 63- $\mu\text{m}$  by O are particularly important cooling sources in the LTI. During geomagnetic activity, NO emissions increase significantly, followed by  $\text{CO}_2$ , and less to little change in the 63- $\mu\text{m}$  emission (Mlynarczyk et al., 2005). The appreciable increase in NO emission during geomagnetic activity makes NO a very important trace species to measure and understand. The chemical pathways for NO production are local to the LTI where excited states of atomic nitrogen interacting with molecular oxygen undergo exothermic reactions to produce NO. The primary excitation mechanism of NO vibrations under quiescent conditions is via inelastic collisions with atomic oxygen O. These collisions convert translational thermal energy of particles into internal vibrational energy of the NO molecule, which is radiated by spontaneous emission from NO (primarily 5.3- $\mu\text{m}$ ), resulting in a cooling of the atmosphere. This process is often referred to as nature's thermostat for the LTI region where the heating created in the auroral region is offset by cooling produced by NO.

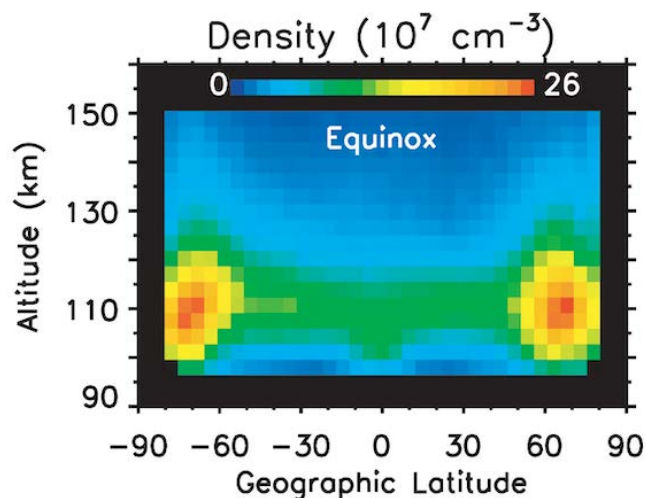


Figure 2.19: Distribution of NO in the LTI region during equinox, determined from measurements of solar radiation fluorescent scattering with a limb-scanning ultraviolet spectrometer on the Student Nitric Oxide Explorer satellite (from Barth et al., 2003).

Factors that contribute to NO enhanced emissions during geomagnetic activity include (1) increased NO abundance associated with auroral electrons, (2) increased kinetic temperature, (3) increase in atomic oxygen, and (4) increased rates of reaction. Inferences from other measurements suggest the two primary factors are an increase in NO abundance and increase in kinetic temperature throughout the LTI. Figure 2.19 presents observations of NO abundance from (Barth et al., 2003) for the equinox seasons near solar maximum. Notice the enhancement in NO predominantly at the high latitudes. The NO enhancement around the auroral oval is due to ionization by 1-10 keV EPP. This leads to greater NO emission at high latitudes, but the overall emission is also affected by the increase in temperature produced by EPP and Joule heating in the auroral region.

EPP has a direct influence on the LTI by ionizing and heating the region. However, a cascading of chemical processes ensue that can alter the overall LTI response. EPP and chemical processes affect conductivities, currents, heating and momentum in the region. It remains a challenge to attribute NO emissions to specific factors. *in situ*, multi-parameter observations of temperature, composition, and EPP are needed to resolve those factors and understand how the LTI responds to an EPP event.

### 2.2.3. Collisional dynamics

Dynamics in the LTI refer to the overall momentum budget associated with the motion of the neutral gas subject to various internal and external forces. To first order, hydrodynamic forces including gravity, pressure gradients, Coriolis, and viscous forces control the dynamics of the neutral upper atmosphere in the LTI, acting on the neutral fluid within the rotating reference frame of Earth's upper atmosphere. However, a supplementary complexity arises since neutrals in the upper atmosphere are also subjected to collisional forces from the local plasma, referred to as the ion drag force. Furthermore, waves excited in the lower atmosphere by the absorption of solar radiation and latent heat release in the tropics propagate upward and add to the complex wave spectrum of tides, planetary waves and gravity waves impressed upon the neutral gas. While some waves dissipate before reaching the LTI region, others break in the LTI region, deposit their momentum, and alter the LTI mean circulation. Furthermore, upward-propagating waves can produce large wind shears in the LTI region. A schematic in Figure 2.20 presents the complex picture of the LTI dynamics, which have been summarized in a comprehensive review (Vincent, 2015). In the following, some of the key processes related to LTI neutral dynamics are reviewed, including plasma-neutral momentum transfer processes and the effects of planetary, tidal and gravity waves.

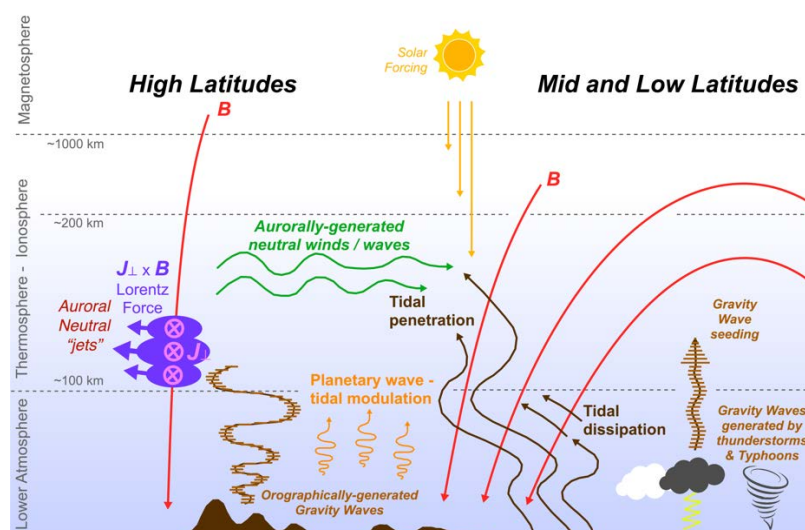


Figure 2.20: Schematic of processes related to LTI neutral dynamics (credit: DUTH – T. Sarris).



**Momentum transfer via ion drag.** As discussed in sections 2.2.1 and 2.2.2, the LTI is a region where electromagnetic energy is converted to mechanical motion and thermal energy of the neutral gas. In this process, currents flowing in the LTI can be a sink or source of momentum to the neutral gas via the  $\mathbf{J}_\perp \times \mathbf{B}$  force, also called Ampere's force (e.g., Richmond and Thayer, 2000). This force due to currents is commonly referred to as the ion drag force because it results primarily from collisional drag between differentially drifting ions and neutrals. Thus, Ampere's force accounts for electrical forces on the neutral gas while the Lorentz force accounts for electrical forces on the plasma. In terms of energy processes related to Ampere's force, the transfer of electromagnetic energy to bulk kinetic energy of the neutral gas is captured by the work rate expression in equation (2.14), i.e.:  $\mathbf{U} \cdot (\mathbf{J}_\perp \times \mathbf{B})$  where Ampere's force is directed along or opposing the neutral wind. Similar to Joule heating, the work rate is a single-fluid representation of bulk kinetic energy transfer to the neutrals due to currents flowing perpendicular to the magnetic field within the LTI. Although the Joule heating rate is a positive definite quantity leading to heating of the neutral gas, the work rate term can be of either sign depending on the wind direction relative to  $\mathbf{J}_\perp \times \mathbf{B}$ . This indicates that Ampere's force can decelerate or accelerate the neutral wind via momentum exchange between the neutral gas and the currents.

The physical pathway by which neutral winds exchange momentum with currents is much more complex. At the core of the process is collisions between the ions and the neutral gas. Ampere's force on the neutral gas can be written in component form as  $\mathbf{J}_\perp \times \mathbf{B} = N_e \cdot m_i \cdot v_{in}(\mathbf{V}_{i\perp} - \mathbf{U}_\perp)$ . In this form, the collision process and the differential motion between ions and neutrals are explicitly described. The complexity arises because ion and neutral motions are subject to quite different constraints; ions are influenced by electromagnetic forces while neutrals are influenced by hydrodynamic forces. In general, the ions are considered a resistive force to the neutral dynamics because of their constraint to remain tied to geomagnetic fields – a valid condition in the upper thermosphere. However, in the LTI the overabundance of ion-neutral collisions overcomes the Lorentz force and the ions move across magnetic field lines. This “demagnetization” of the ions is the cause of the perpendicular electric current flow in the LTI.

A prediction of how high-latitude neutral circulation may be accelerated at LTI altitudes by a sustained electric field is shown in Figure 2.21 (Kwak and Richmond, 2007). This shows TIE-GCM model neutral circulation patterns at 142 km from an imposed electric field model that forces a two-cell convection pattern. The impressed pattern on the neutral gas illustrates the impact of ion drag on the neutrals but also demonstrates a lag in neutral wind response. This lag is strongly dependent on altitude due to the changing collision frequency, persistence of the ion-drag forcing, and the neutral gas inertia. In the LTI, it can take several hours for the neutral winds to reach a steady flow. A model predicting winds driven by steady ion motion during post-midnight, diffuse aurora is shown in Figure 2.22 (left) (Brinkman et al., 1995) which displays the formation of “jet streams” peaking near 150 km where the ion-drag force per unit mass maximizes. Neutral wind jets at 130 km with speeds over 300 m/s have been observed in the auroral oval, as shown in Figure 2.22 (right) (Larsen et al., 2022).

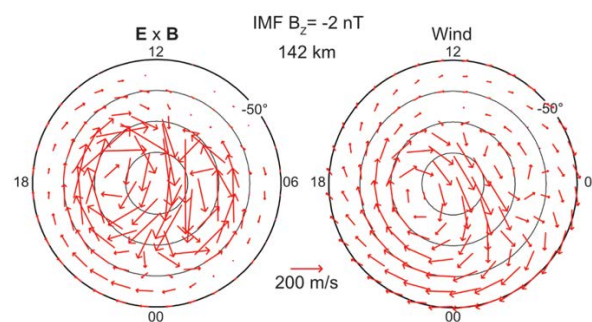


Figure 2.21: TIE-GCM model results of  $\mathbf{E} \times \mathbf{B}$  drifts (left) and neutral winds (right) at 142 km (after Kwak and Richmond, 2007).

Thus, ion drag in the LTI is a significant influence on neutral motion but also impactful on electric currents. Lagging response and altitude dependencies in neutral gas motion makes it challenging to resolve the full influence of ion drag on the LTI system. To investigate this important momentum transfer process, *in situ* and multi-parameter measurements of wind, ion motion, plasma density, and neutral density in the LTI are needed, similar to those required for estimating the Joule heating rate process.

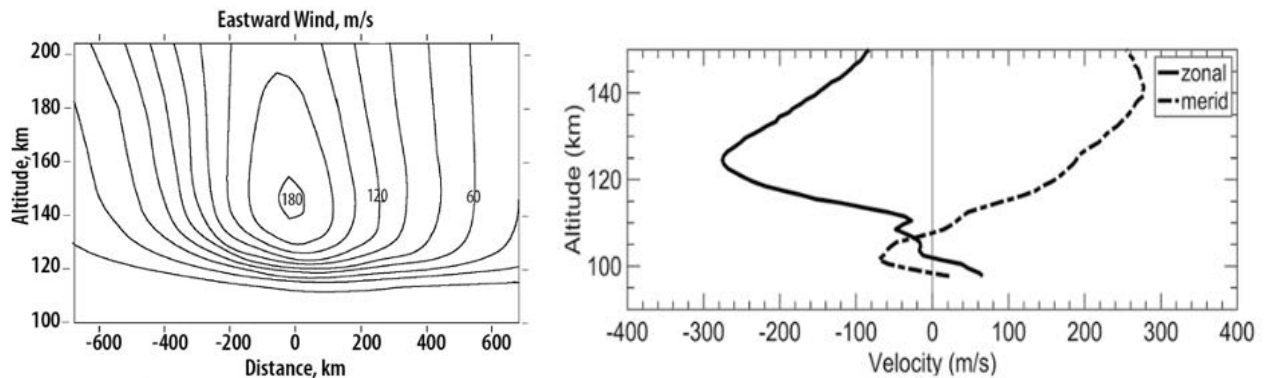


Figure 2.22: Left, model prediction of neutral wind jets during diffuse aurora (from Brinkman et al., 1995). Right, observations of auroral neutral wind jets from vapor trail measurements on a sounding rocket at Poker Flat, Alaska, on 2<sup>nd</sup> March 2017 (from Larsen et al., 2022).

**Planetary waves, tides, and gravity wave input to the LTI from below.** The neutral dynamics of the LTI is largely dominated by atmospheric motions originating from below, including planetary waves, tides, and gravity waves. The generation and propagation of waves and their dissipation in the LTI is considered the primary mechanism through which energy and momentum are transferred from the lower to the upper atmosphere and ionosphere. Waves extending over a broad spectrum are excited in the lower atmosphere. Planetary or Rossby waves, which occur naturally in rotating fluids, and atmospheric tides are global-scale modulations of the upper and middle atmosphere that include thermal and gravitational tides. The thermal tides are excited by regular solar differential heating with periods that are harmonics of a solar day e.g., with diurnal, semidiurnal, terdiurnal periods of 24, 12, and 8h respectively. Smaller-scale gravity waves generally form in the troposphere and can be of meteorological (convective, shear, geostrophic) or topographic (mountain waves) origin (Fritts and Alexander, 2003). They have also been linked to strong disturbances in the troposphere, such as thunderstorms and typhoons.

The upwards propagation of tides, planetary waves, and gravity waves are influenced by the background atmosphere and by nonlinear interaction between waves. Their amplitudes grow exponentially with altitude due to the decrease in atmospheric density (Andrews et al., 1987). These large wave amplitudes lead to wave breaking and hence momentum deposition, often occurring in the LTI. This, in turn, produces forces that drive large scale residual circulations. Below, we provide examples of these phenomena which each have a significant influence on the LTI.

In the last 20 years, clear evidence in the ionosphere of non-migrating, DE3 tides (eastward propagating 24h tide with 3 peaks in longitude) has been observed (Immel et al., 2006). The DE3 tide, generated by latent heat release in the tropics over the three continents, is a prominent example of the global atmosphere coupling from the troposphere to the upper atmosphere, but so far data to quantify the complex spectrum of waves from the lower atmosphere has been sparse. Such tidal waves propagate from below at all latitudes and have some of their most profound effects in the LTI region, creating highly variable and strong neutral winds below 150 km altitude. These winds drive the daytime dynamo system of currents, discussed above.

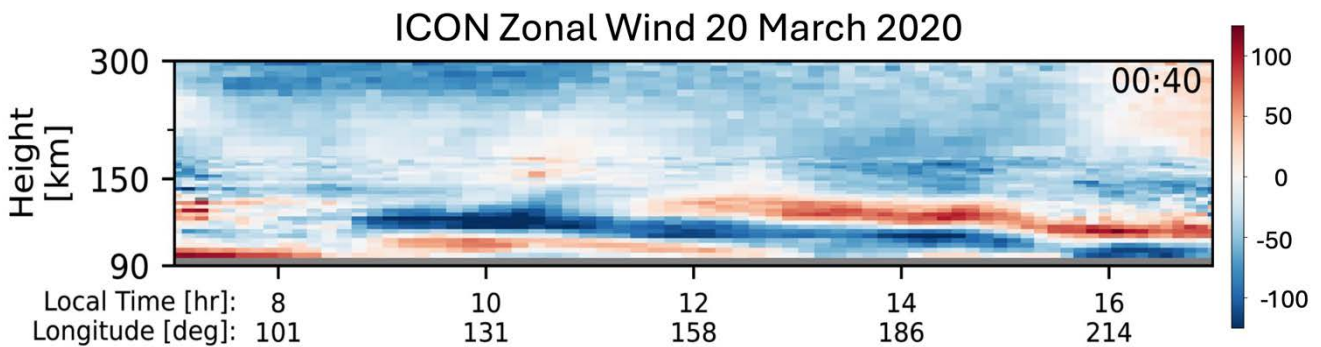


Figure 2.23: ICON observations of large amplitude tidal winds and vertical shears in the LTI (from England et al., 2022).

*Remote measurements* from the MIGHTI instrument on ICON confirm the existence of such tidal layers (England et al., 2022), as illustrated in Figure 2.23, showing winds of significantly larger magnitude than those predicted by simple tidal theory (Jin et al., 2008). Such larger neutral winds have also been observed routinely using vapor trails from sounding rockets, particularly at night, as reported by (Larsen, 2002). Recent studies have found that tidal waves can play a central role in the day-to-day variability of equatorial and mid-latitude physical processes (Zhou et al., 2020). Furthermore, tidal and planetary waves, which operate at global scales, are also associated with daily variations not only in the winds, but temperature and composition (Liu, 2017). Both the winds and local conductivities must be known to quantify their role in driving local current systems.

In addition to the global neutral winds driven by tidal forcing in the LTI, this region is also influenced by continuous streams of gravity waves propagating from below. Indeed, observational and theoretical studies have shown that gravity waves contribute to vertical coupling, energy and momentum transport and deposition, travelling atmospheric and ionospheric disturbances, and irregularities and turbulence (Fritts and Lund, 2011).

Gravity waves are dissipated in the thermosphere at altitudes between 100 and 200 km through molecular damping, modifying thermospheric temperatures (Liu, 2000; Walterscheid, 1981). The forcing by gravity waves is believed to be a crucial aspect of upper atmospheric dynamics, however due to a lack of measurements it remains poorly quantified. As a result of this forcing, the LTI transport is known to be affected, for instance with the formation of vertical wind shears above the mesopause (Liu, 2007), and by the triggering of plasma instabilities in the equatorial ionosphere (Hysell and Kudeki, 2004; Kelley et al., 1991). Often, atmospheric gravity waves are associated with ionospheric signatures known as a travelling ionospheric disturbance consisting of a horizontal modulation of the ionospheric electron density propagating over large distances (Jonah et al., 2016).

Notable advances in the understanding of gravity waves and their effects come from very-high resolution runs of whole atmosphere general circulation models, which can capture individual gravity waves in the upper atmosphere (Liu et al., 2014a), providing a global *theoretical* view of gravity wave sources and propagation effects as shown in Figure 2.24. Most importantly, such models show the impact of gravity waves on the momentum balance of the mesosphere and lower thermosphere. The gravity waves are readily visualized by modulations of airglow layer emissions in the mesosphere, providing a 2D view of the wavefield. However, comprehensive *in situ* multi-parameter measurements in the LTI are needed to validate the gravity wave perturbations and their effect in the LTI region.



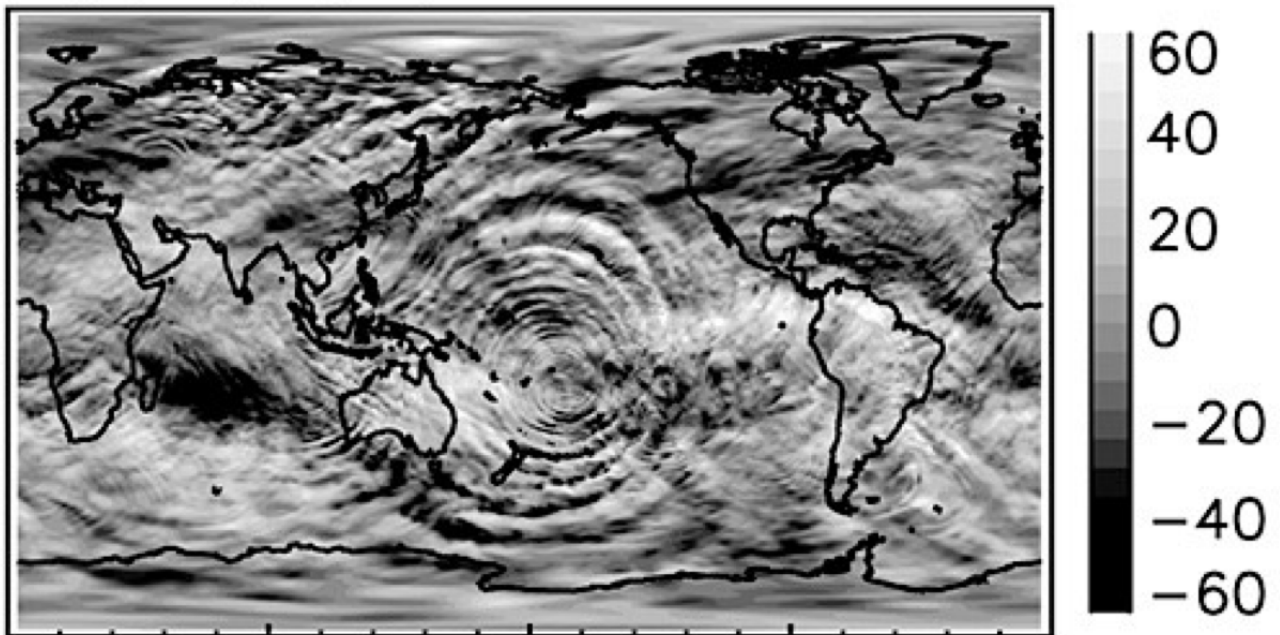


Figure 2.24: Simulation results based on a mesoscale-resolving whole atmosphere general circulation model, showing meridional winds associated with gravity waves propagating from below to the  $\sim 100$  km altitude cross-section of the plot (from Liu et al., 2014a).

The role of gravity waves in the momentum budget of the LTI depends on the quantification of the power of the waves at different scales. For example, currently, spatial scales of 200 km and larger (termed meso- $\alpha$  scales) are included in a few high-resolution models such as WACCM (Liu et al., 2019), WACCM-X (Huba and Liu, 2020). However, scales at 20–200 km (meso- $\beta$  scales) are not resolved in models (Liu et al., 2019) and play an important role in the global momentum budget (Liu, 2022). Furthermore, gravity waves at even smaller scales, in the 2–20 km range (meso- $\gamma$ ), may still be prominent according to high-resolution regional simulations. It thus remains to be seen whether the momentum deposition comes from a few sporadic large amplitude wave events or through deposition by many small amplitude waves. Importantly, gravity wave characteristics evolve as they extend to higher altitudes within the LTI, eventually developing additional spatial structure as shown in Figure 2.25 (Fritts et al., 2020).

Whereas some gravity waves continue to propagate through the LTI to higher altitudes, forming travelling atmospheric disturbances (TADs), and, via direct ion-neutral coupling, travelling ionospheric disturbances (TIDs), other gravity and tidal waves actually “break” in the LTI, depending on their intensity, wavelengths, and velocity. In such cases, the collapsing waves are expected to generate small scale structures and turbulence, while depositing energy. The effects of these breaking waves can also trigger instabilities that produce secondary and higher order gravity waves originating in the LTI region (Vadas et al., 2018). Figure 2.25 shows high resolution model simulations of gravity waves breaking near 100 km and the subsequent emergence of secondary waves propagating from that level to altitudes up to 200 km (Fritts et al., 2020). The effects of these primary and higher order gravity waves in the LTI region have a major influence on the large-scale mean meridional circulation, which is important for understanding the impacts of compositional effects and neutral density changes on the ionosphere at higher altitudes (i.e., above 200 km).



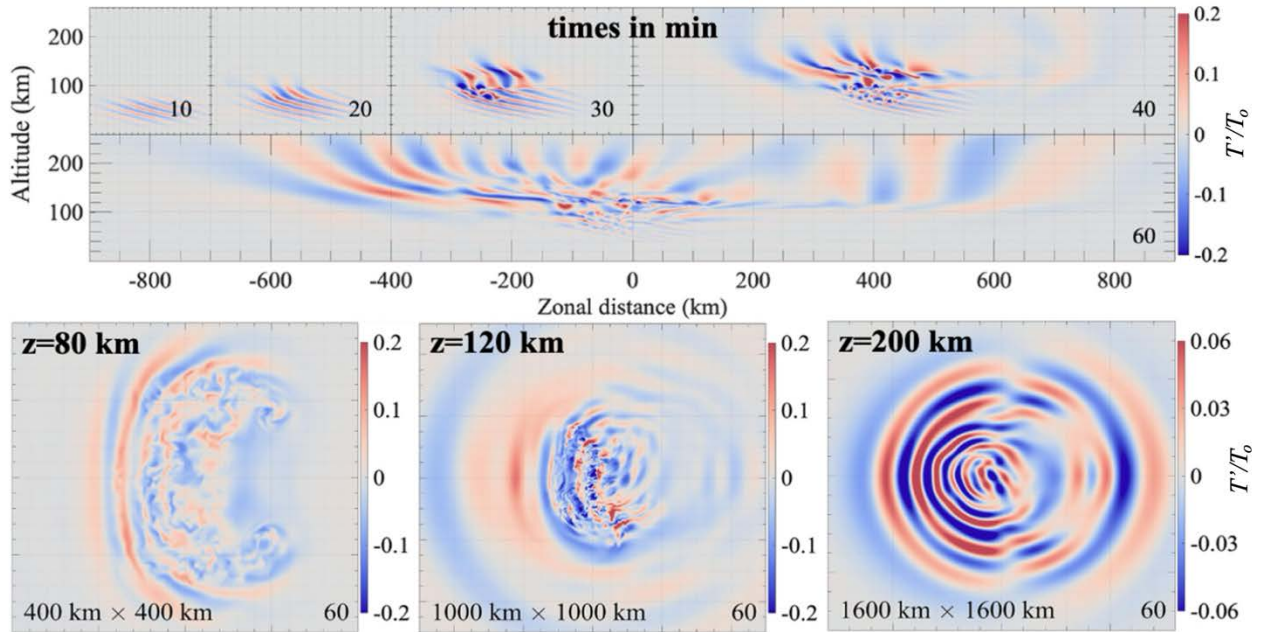


Figure 2.25: High resolution model simulations showing how gravity-wave induced temperature perturbations evolve within the LTI (after Fritts et al., 2020). Top: 2D (vertical-zonal) cross-sections of temperature perturbations at 10, 20, 30, 40 and 60 min from initial wave perturbation. Bottom: Horizontal (zonal-meridional) cross-sections of temperature perturbations at 60 min for 80, 120, and 200 km altitude.

There are only sparse observations of gravity waves by *in situ* probes. For example, observations of oscillations within neutral gas composition measurements gathered on the Dynamics Explorer-2 satellite have been attributed to gravity waves (Innis and Conde, 2002). Oscillations of both neutral and plasma density and motions were found using probes from the Dynamics Explorer-2 satellite (Earle et al., 2008), as shown in Figure 2.26. These data were gathered at 261 km, near the end of the DE-2 mission. These authors interpret the oscillations to correspond to gravity waves with wavelengths of ~225 km along the line satellite path and to be consistent with propagation from below. It is highly likely that a mission that executes horizontal orbits at lower altitudes in the LTI will encounter copious observations of gravity waves, from which detailed measurements should reveal important information regarding their wave characteristics including non-linear evolution and associated structure.

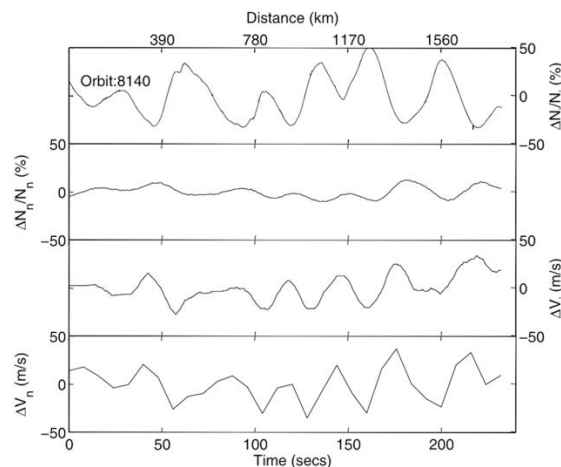


Figure 2.26: Gravity wave signatures in both neutral and ionized gas density and velocity data observed on the Dynamics Explorer-2 satellite at 261 km, on 22<sup>nd</sup> January 1983 (from Earle et al., 2008).

**Neutral wind shears and Sporadic-E layers.** The large variability of the neutral winds also leads to large vertical shears of the neutral winds, which play an important role in the upper atmosphere dynamics (England et al., 2022; Larsen, 2002; Liu, 2017). The strong shears with altitude of the winds are responsible for the convergence of long-lasting metallic ions that form distinct sporadic-E layers, as advanced by (Whitehead, 1970). Arecibo ISR observations show how sporadic-E layers exist in both the day and night, driven by wind shears associated with descending tidal motions (Haldoupis, 2012), as shown in Figure 2.27.

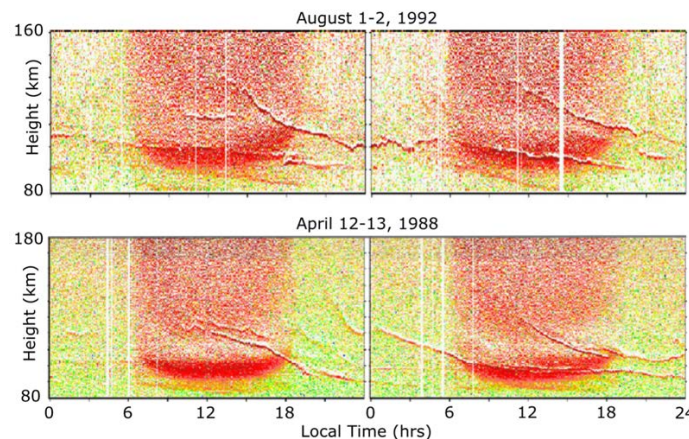


Figure 2.27: Arecibo ISR observations of enhanced plasma density showing sporadic-E layers during both day and night, characterizing the descending tidal motions (after Haldoupis, 2012).

The wind shears also drive Kelvin-Helmholtz instabilities, depicted in Figure 2.28 (lower left), producing distinct quasi-periodic (QP) coherent radar echoes (Larsen, 2000). Such Kelvin-Helmholtz instabilities generate both neutral and plasma structures within the mid-latitude LTI region. Figure 2.28 (right) shows electric fields associated with plasma structures produced by such waves (Pfaff et al., 2005) during a strong sporadic-E event in the presence of QP-echoes that “map” upwards along magnetic field lines to higher altitudes.

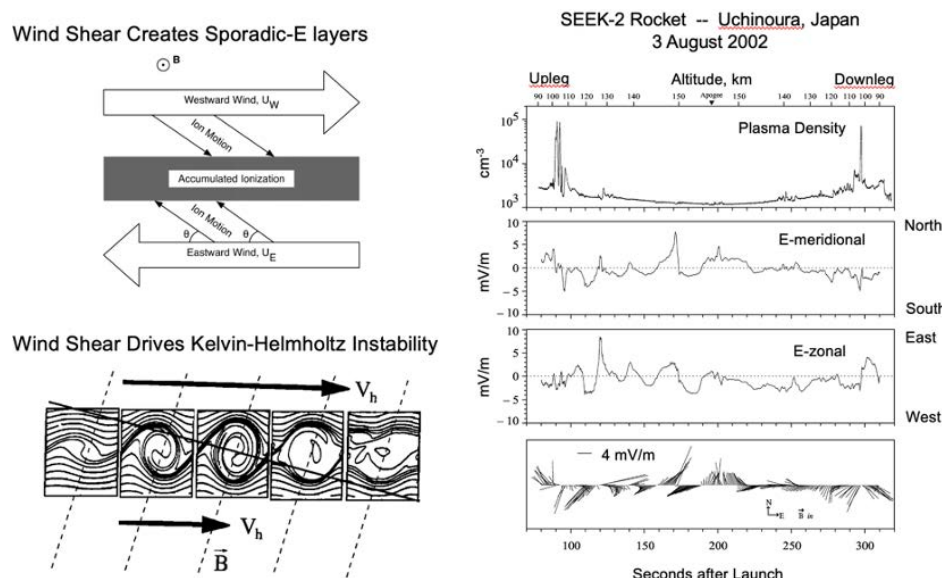


Figure 2.28: Upper left, depiction showing how wind shears create sporadic-E layers (from Pfaff, 2012). Lower left, depiction showing how wind shear also drives Kelvin-Helmholtz instabilities (from Larsen, 2000). Right, sounding rocket observations of electric field structures associated with sporadic-E and K-H waves (from Pfaff et al., 2005).

Significant, larger-scale neutral structures, possibly resulting from gravity waves or other processes, have also been shown to be common in the lower thermosphere. Direct evidence of neutral structuring of the LTI was provided by measurements on a low-altitude U.S. DoD satellite. Figure 2.29 shows high time resolution neutral density structuring in data from the Streak instrument (Clemmons et al., 2009) gathered near 150 km. These surprising and unexplained results underscore that in the LTI, the balance of momentum forcing and solar heating produces an upper atmosphere that behaves in ways that are presently not understood and for which we have essentially no systematic, direct, and comprehensive measurements. Such large, abrupt gradients in the upper atmospheric density may induce significant disturbances in the attitude and drag of very low Earth orbit satellites, anticipating which is currently a top-level space weather priority. Understanding how large shears and sharp gradients develop in the LTI in response to driving from below, as well as how their morphology and relevant scales drive plasma irregularities, are key open questions in LTI dynamics.

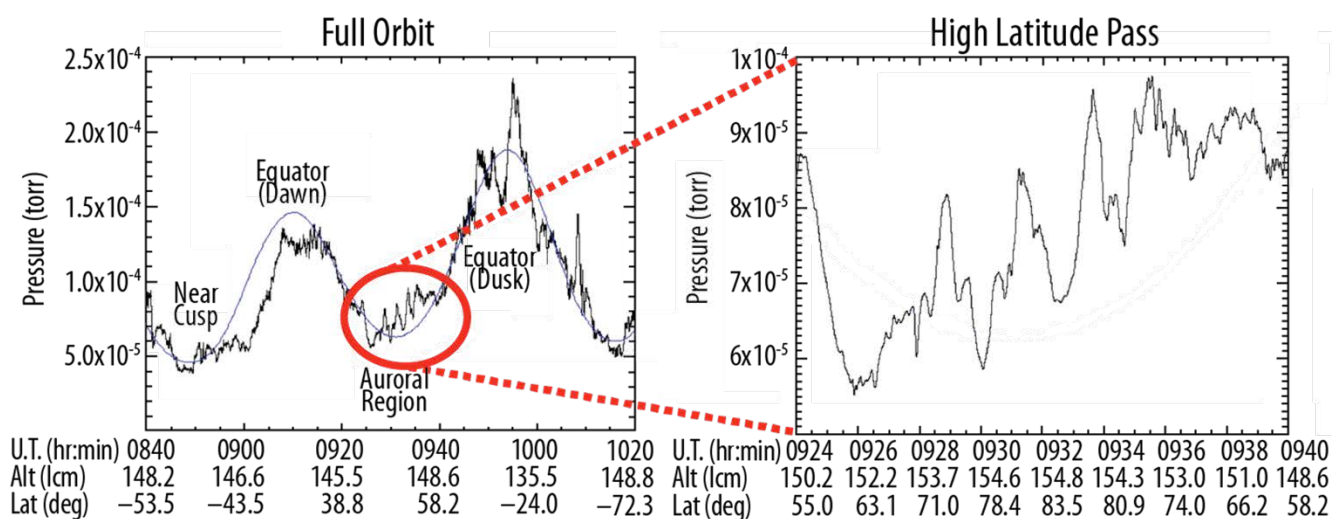


Figure 2.29: Neutral density measurements gathered on the Streak satellite on 28<sup>th</sup> June 2006 near 150 km reveal both large- and small-scale structures (as short as 1 km) (from Pfaff et al., 2022).

**Coupling of thunderstorm energy to the LTI.** In addition to planetary waves, tides, and gravity waves, the LTI also receives upwards-directed tropospheric energy associated with thunderstorms and hurricanes, although the coupling mechanisms at the LTI interface are not well understood. Such energy may be in the form of neutral gas dynamics, such as gravity waves discussed above, and/or as electrical energy, which we discuss here.

At low and mid latitudes, thunderstorms generate the Earth's fair-weather electric field in the troposphere, stratosphere, and mesosphere. The extent by which these electric fields also couple to the LTI and thereby contribute to the ionosphere's electrical environment is currently not known, although there exists strong experimental evidence for upwards penetration of thunderstorm-related electric fields into the ionosphere.

Observations on the low-latitude San Marco satellite provide evidence of upwards-directed, quasi-DC electric fields above tropospheric storm systems without corresponding variations in the plasma density (Farrell et al., 1994). Importantly, although San Marco's apogee was 610 km, such electric fields were only observed when the satellite was below 300 km and only at night, when the ionospheric conductivity was low. In fact, the most intense storm-related event occurred when the satellite was at an altitude of 188 km, two days before it re-entered.



These observed electric fields were clearly associated with tropospheric weather systems, as evidenced in concurrent microwave images from the DMSP F8 satellite (Farrell et al., 1994). Although the electric field signatures were localized, they were somewhat extended in horizontal extent, existing for tens to several hundred km along the satellite trajectory. Other reports of electrical signatures associated with thunderstorms displayed a more transient nature, such as the DC electric field “pulses” observed with sounding rocket probes launched over an active thunderstorm (Kelley et al., 1985). Another example of a transient electric field signature associated with a tropospheric-driven energy source is shown in Figure 2.30. Here, probes on the Dynamics Explorer-2 satellite observed a 40 mV/m electric field pulse as well as upwards accelerated energetic (keV) electrons, precisely as the satellite traversed Hurricane Debby (Burke et al., 1992).

Although ionospheric models show that DC and low frequency electric fields should not penetrate efficiently upwards through the ionosphere barrier, satellite and sounding rocket data, though sparse, show that this is not the case. Whereas the enhanced conductivity of the ionosphere is thought to shield much of the quasi-DC electric fields, the observations suggest that the ionosphere shielding may not be “perfect”. Thus, this fundamental energy source has significant implication for our understanding of the energy balance of the mid and low latitude ionosphere not only for the earth, but for other planets as well. Measurements in the LTI will help elucidate the nature and amplitudes of these tropospheric energy sources in the ionosphere and significantly advance our understanding regarding how they couple into the LTI region.

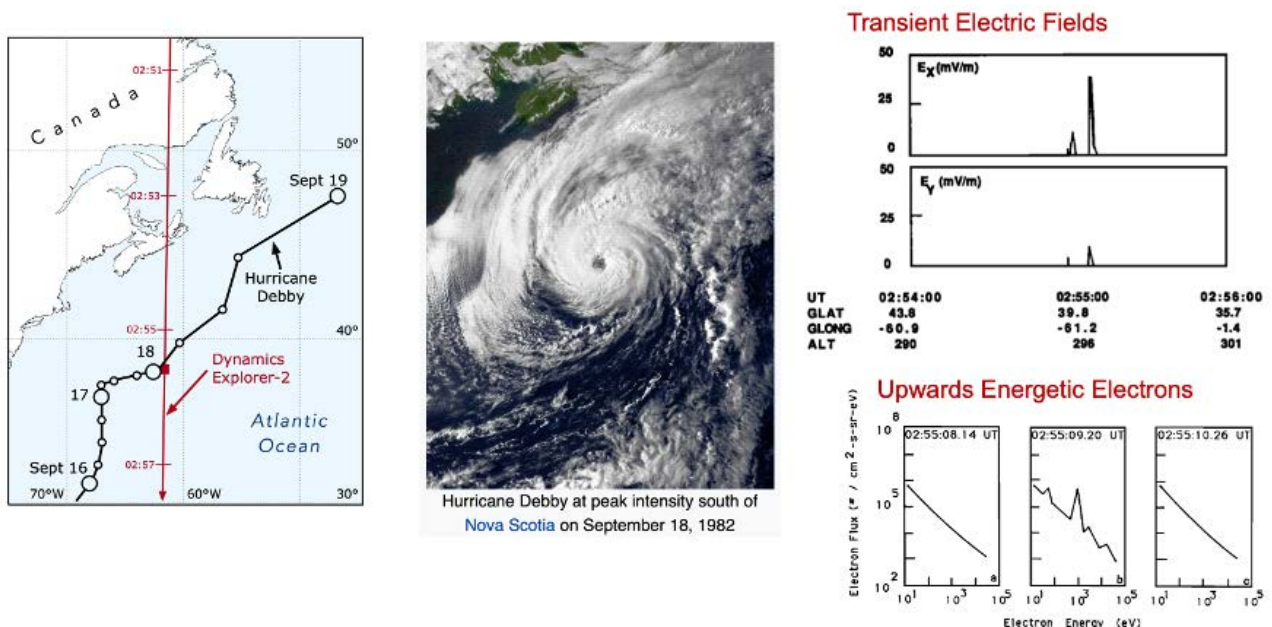


Figure 2.30: Dynamic Explorer-2 satellite observations over Hurricane Debby on September 18, 1982. The illustration on the left shows the path of the hurricane over 3 days and the DE-2 satellite trajectory. The middle panel shows a photograph of the hurricane off the coast of Nova Scotia when it was most intense (NOAA). The panels on the right show evidence of a transient electric field pulse and upward directed keV electrons when the satellite traversed the hurricane path (after Burke et al., 1992).

## 2.3. Societal needs addressed by studying the LTI region

The fundamental processes of the lower thermosphere-ionosphere not only affect the state of the near-Earth geospace environment but also carry a significant societal impact. The importance of the LTI for society is underscored by the growing relevance of the region and its influence on low-Earth orbit (LEO)





satellites, communication and navigation systems, and power grids and pipelines, in addition to its role in environmental and climate impacts. Some of the key aspects of these societal impacts are discussed below.

**Atmospheric drag and spacecraft operations.** The current space age is characterized by an increasing number of commercial satellites in LEO, providing significant economic and societal benefits. Knowledge of atmospheric density and its variability within the low altitudes of geospace, which is strongly influenced by processes occurring within the LTI, is crucial to satellite operators because it determines the orbit characteristics and evolution of any object in the near-Earth regions of space. Indeed, this is the region where satellites experience substantial drag ultimately resulting in atmosphere re-entry for near-Earth circular orbits. The February 2022 Space-X loss of 38 satellites is believed to have been caused by short-term thermospheric density enhancements due to a moderate geomagnetic storm and the resulting enhanced Joule heating in the LTI (Dang et al., 2022; Hapgood et al., 2022; Zhang et al., 2022). Furthermore, the increasing abundance of spacecraft and spacecraft debris within LEO makes predicting an object's path for collision avoidance an increasing problem which is compounded by the lack of accurate models of the upper atmospheric density.

Developing a mission that can withstand the challenging conditions within the LTI while also gathering relevant *in situ* measurements would significantly advance satellite technology and lead to future improvements in understanding satellite drag and gas-surface interactions. This improved understanding would result in more accurate modelling of satellite dynamics and aerothermodynamics, which are important for mission analysis and satellite operations, including when exploring options of operating in Very Low Earth Orbit (VLEO), that is, at altitudes below 300 km.

**Protecting critical societal infrastructure – communication/navigation/radar systems.** Our society relies on satellites and radiowave transmission within the ionosphere to support communication and navigation infrastructure. Knowledge of the state of the ionosphere is crucial for reliable nowcasts and forecasts of potential disruption or loss of vital radio linkages necessary for position determination from global navigation satellite systems (such as GPS and Galileo) as well as over-the-horizon (OTH) radar target detection and coordinate registration. Radiowaves propagating through the ionosphere are affected by gradients in the plasma density and its associated spatial variations. A broad spectrum of plasma density irregularities occurs in the polar regions as well as in vicinity of the auroral and equatorial electrojets, which may induce scintillations with profound effects on radiowaves due to their transition through the LTI. In addition, plasma density enhancements such as sporadic-E layers in the LTI reflect lower-frequency radiowaves and block their propagation to higher altitudes in the ionosphere.

**Protecting critical societal infrastructure – power grids and pipelines.** Strong variations in ionospheric currents generate Ground-Induced Currents (GIC) in long, ground-based conductors, such as power grids and pipelines, which can be damaged or lead to infrastructure failure. The strongest and highly variable horizontal ionospheric currents flow in the LTI region where conductivities are large during geomagnetic storms. Detailed knowledge of the current flow and conductivity variations will advance our understanding of how sudden current surges affect power infrastructure during geomagnetic activity, such as storms and substorms, and would enable better preparedness to mitigate space weather impacts on power grids and pipelines.

**Climate and environmental impacts.** Not surprisingly, variations in the Earth's climate in the lower atmosphere also have important impacts in the upper atmosphere, although these are less studied due to the lack of measurements. For example, a slow decrease in average thermospheric density has been inferred from over 4 decades of satellite drag data (Emmert et al., 2008) that is believed to result from increases of CO<sub>2</sub>, which traps heat at low altitudes but acts as a radiative cooling agent in the upper atmosphere. Lower temperatures lead to a lower air density at LTI altitudes, resulting in longer orbital



lifetimes of satellites and space debris. Globally distributed observations within the LTI will provide the necessary, unprecedented knowledge to verify such hypotheses.

Furthermore, SABER data from NASA's TIMED satellite show that CO<sub>2</sub> increases at a rate of 5-6% per decade in the upper Mesosphere-Lower Thermosphere (MLT) around 80 km altitude and above, similar to the rate measured on the ground at the Mauna Loa Observatory in Hawaii (Mlynczak et al., 2022). This study concludes that "radiative cooling effects of CO<sub>2</sub> on the temperature of the MLT are now larger than at any time since the beginning of the Industrial Age" and provides detailed estimates of cooling rates and radiative relaxation times based on the SABER observations.

Anthropogenic changes over the last 100 years have been invoked to account for increased visible noctilucent cloud sightings (Lübken et al., 2018), which result from increased CO<sub>2</sub> and water vapor caused by increases in methane. Although noctilucent clouds result from ice formation on aerosols which occur at the extremely low temperatures around the mesopause near 83 km altitude, associated chemical effects at higher altitudes in the LTI would also be expected.

Other long-term climate effects in the LTI may occur in the polar regions due to energetic particle precipitation (EPP) producing odd nitrogen. Odd nitrogen is long-lived in the absence of solar radiation and therefore is likely to be transported down to the winter stratosphere where it can catalytically destroy ozone. The downward transport is dynamically controlled by the polar vortex which may extend up to the LTI region (Harvey et al., 2022). To accurately predict ozone levels with climate models, realistic odd nitrogen distributions and dynamics are critical. Climate models are believed to underestimate the odd nitrogen produced by EPP (Funke et al., 2017). Furthermore, the EPP impact is estimated to increase in the future due to strengthening of the meridional circulation by increasing greenhouse gases (Maliniemi et al., 2020) and allowing for additional odd nitrogen to access the stratospheric ozone than before. Such climate changes and unknown environmental impacts illustrate how poorly we understand the long-term effects of our planet's near-Earth space environment and its potential unknown consequences.

**Summary.** As demonstrated above, the LTI plays a critical role in many space weather effects and therefore it is imperative to understand the processes in the LTI to protect space-related societal infrastructure. A dedicated LTI mission with its unique measurements would gather the necessary data to enable a better understanding of how this region participates in the space weather events discussed herein, including, in particular, their extreme effects during magnetic storms. Such new LTI observations would, in turn, lead to improved thermospheric density and composition specifications in empirical models of the upper atmosphere, such as NRLMSIS-2.0 (Emmert et al., 2020), ultimately leading to a better representation of the global atmosphere-ionosphere system.

## 2.4. Critical LTI modeling needs

Together with the need to fundamentally advance our understanding of the physics of the atmosphere-space transition region (Karlsson et al., 2020; Palmroth et al., 2021; Pedatella and Liu, 2018; Zettergren and Semeter, 2012), there is also an urgent need to significantly improve the climatological specification and empirical models of the LTI at all latitudes and local times by incorporating new LTI datasets (Emmert et al., 2020) and increasing resolution to accommodate the rapid variations with altitude of both state variables and dynamic forcing that characterize the LTI. Furthermore, as discussed in the last section, improvements in this modeling will also significantly advance our space weather forecasting capabilities used for a variety of operational needs and applications, as emphasized by numerous researchers (Crisp et al., 2020).



Although fully coupled whole atmosphere models including the ionosphere, thermosphere, and, in some cases, the magnetosphere, are currently being developed (Gombosi et al., 2021; Liu et al., 2018a; Pham et al., 2022), many fundamental characteristics of the LTI – for example of the Hall and Pedersen mobilities and their effects on the subsequent conducting layer, the thermal plasma and currents associated with auroral particle precipitations, and the persistent neutral wind shears at all latitudes – cannot be captured realistically by models, especially at horizontal scales less than ~100 km. Despite these shortcomings, modeling of the geospace system has progressed considerably in recent years to include coupling of the thermosphere-ionosphere system to the lower and middle atmosphere below and the magnetosphere above. Such models are capable of better simulating and accounting for processes in the LTI which affect variations on scales of ~100 km and larger of surrounding regions. Such advanced modeling systems are well-suited for advancing our understanding of the system science as well as improving forecasting space weather effects in geospace.

Despite these improvements, the present lack of sufficient observations of the LTI transition region between the atmosphere and space presents a challenge to our attempts to fully quantify the large number of different physical processes within the LTI in models, to incorporate more realistic and accurate descriptions of the LTI state variables in the models, and to assess the impact of these LTI processes on the surrounding regions and thus evaluate the models. We next describe how specific modeling needs can be met by LTI measurements obtained by an *in situ* space flight mission. These needs can be broadly categorized into the following topical areas: ionospheric conductivities at all latitudes including feedback to the magnetosphere at high latitudes; energy and momentum distributions and exchanges; and the connection between neutral dynamics and compositional changes.

**Ionospheric conductivities and feedback to the magnetosphere.** Ionospheric conductivities are essential parameters in the modeling of several key processes such as plasma flow and electric fields, Joule heating, ionospheric current flow and closure, and coupling to the magnetosphere on meso- and large-scales. Auroral precipitation can account for 25–50% of total precipitating energy during active times with the creation of thermal plasma in the LTI modifying conductivities on a wide range of scales (Nishimura et al., 2020). In magnetosphere-ionosphere coupling, the ionospheric conductivities and the wind dynamo impose restrictions on the relation between electric fields and field-aligned currents (FAC), in addition to the magnetospheric plasma pressure and plasma acceleration which constrain the FAC flowing into and out of the ionosphere. As a result, the LTI region is a significant component of ionosphere-magnetosphere coupling. In many models the effect of the wind dynamo is neglected in this magnetosphere-ionosphere coupling even though it can locally drive FACs larger than the average 10% change predicted by the wind dynamo (Zhu et al., 2019). Modeling studies (Wiltberger et al., 2017) have shown that including plasma kinetic processes lead to increased ionospheric conductivities and associated changes in the strength of the plasma velocity which feeds back to the magnetosphere.

While the ionosphere is often simplified as a boundary condition for magnetospheric simulations, it plays a crucial role in predicting the geospace environment and associated space weather effects especially during geomagnetically disturbed time periods. The ionosphere is coupled to magnetospheric models via dynamic, complex field-aligned current distributions and electron precipitation, which determine the 3D ionospheric conductivity variation and the electric potential patterns. The electric potential patterns are mapped to the magnetosphere as a boundary condition (Janhunen et al., 2012). However, the ionosphere is not a passive component that simply responds to magnetospheric forcing but is one which can modify the magnetosphere as well. The LTI region plays a critical role in this coupling, particularly at high latitudes since particle precipitation significantly enhances the conductivity within the region. Indeed, the ionospheric conductivity distribution has been shown to influence magnetospheric dynamics by coupling via the magnetic field lines and the closure of field-aligned currents (Ridley et al., 2004; Wiltberger et al., 2017). Other examples of the ionospheric feedback to



magnetospheric dynamics are manifest in dawn-dusk asymmetries (Walsh et al., 2014) and the active role of the ionosphere, and particularly the LTI, in magnetospheric tail dynamics (Runov et al., 2021).

Meso-scale features in the ionosphere, such as sub-auroral polarization streams (SAPS) and Strong Thermal Emission Velocity Enhancement (STEVE) events, are associated with extreme ion drift and/or LTI heating which can influence the ionospheric conductivities, dynamics, and composition. While numerical models are at the brink of simulating these phenomena, comprehensive observations including neutral winds, electric fields, composition, etc., are needed to understand the resulting processes in the LTI region and their effects.

**Ionospheric conductivities at low and mid latitudes.** At low latitudes, very little detailed information is available with respect to the altitude variation in conductivity across the solar terminator and nighttime, which significantly impacts the strength of the evening pre-reversal enhancement (PRE). The PRE is a strong upward ionospheric plasma motion at low latitudes at sunset where its variability and magnitude is not well captured by models but has been shown to be associated with the occurrence of plasma depletions and associated irregularities in the post-sunset low latitude F-region (Bhattacharyya, 2022; Huang, 2018; Stolle et al., 2008). Despite progress in understanding the effects of the LTI conductivity on the PRE (Liu et al., 2018b; Richmond and Fang, 2015), the PRE magnitude alone has been shown to be insufficient to account for post-sunset plasma depletions and irregularities and therefore, some additional process might be at play, such as a seeding mechanism associated with the upward propagation of gravity waves (Huba et al., 2023; Huba and Liu, 2020) or simply differences in the wind and ion drift vectors (Kudeki et al., 2007). These mechanisms have been suggested as means to instigate large scale equatorial plasma depletions. Enhanced zonal electric fields (upwards plasma drifts) have also been reported at the dawn terminator at low latitudes (Kelley et al., 2014), for which the conductivity in the LTI is believed to play a significant role. Clearly, enhanced, detailed models of the LTI conductivity informed by *in situ* observations at all local times, but particularly at dusk and dawn, are required to augment the simplified parametrizations currently used to calculate the nighttime conductivities at low and mid-latitudes (Lin and Chu, 2017; Ren et al., 2012).

Local processes which can occur at all latitudes and local times are important features of the LTI which are generally not included in detailed global models. One such process is the ablation of meteors. This process increases the metallic ion content and which, in concert with neutral wind shears, forms layers of high density and conductivity, called sporadic E layers as discussed above. The evolution of such layers and related polarization electric fields may describe how such layers couple to the F-region above and may also be important factor in the development of medium-scale TIDs (Cosgrove and Tsunoda, 2003; Yokoyama et al., 2011). Although some numerical models include these processes (Baumann et al., 2015; Shinagawa et al., 2021; Verronen et al., 2016; Wu et al., 2021), understanding how sporadic E layers evolve and influence the LTI and lower F-region electrodynamics requires collocated, high resolution observations of winds, electric fields, and ion density and composition.

**Quantification of energy distribution in the LTI.** One of the most significant unknowns in numerical modeling of the ionosphere-thermosphere system is the quantification of the electromagnetic energy and EPP associated with the solar wind and magnetospheric coupling and its effect on the upper atmosphere. Not knowing the amount and location of the energy deposited in the upper atmosphere on different spatial-temporal scales has important local and global implications for capturing geomagnetic storm effects, such as missing the magnitude and variability of neutral density enhancements important for space traffic coordination and accurately determining effects on navigation and communication systems.

Energetic Particle Precipitation (EPP) changes the chemical composition in the lower LTI and the layers below. Odd nitrogen ( $\text{NO}_x$ ) production which among others is important for understanding ozone





depletion and long-term changes in the lower atmosphere, introduces large uncertainties in models particularly because its consequences are not known. The EPP-driven  $\text{NO}_x$  production due to solar protons and radiation belt energetic electrons is relatively well described. However, models still underestimate  $\text{NO}_x$  production (Randall et al., 2015), indicating that major  $\text{NO}_x$  production from auroral electrons plays a significant role (Sinnhuber et al., 2012). Climate models lack a description of auroral electrons and use oversimplified parameterization for auroral  $\text{NO}_x$  production (Schmidt et al., 2006). Currently, there are efforts to describe the auroral particle energy flux and spectrum with the Vlasior magnetospheric model (Alho et al., 2022; Palmroth et al., 2018), with promising results comparing proton precipitation with satellite data (Grandin et al., 2023). This will eventually improve simulations of the auroral impact on  $\text{NO}_x$  production. To better quantify the effects of  $\text{NO}_x$  and evaluate the upper atmosphere response on different scales, co-located measurements at high spatial and temporal resolution of both neutral and plasma parameters are needed.

Processes related to energy distribution within the LTI are not well quantified by observations and are subsequently poorly parametrized in LTI models. An example is the contribution of small-scale electric fields on Joule heating which is crudely parameterized in models due to missing information about its magnitude and variability beyond sparse statistical knowledge (Codrescu et al., 1995; Deng et al., 2009; Krcelic et al., 2023; Zhu et al., 2018). A fixed partitioning of Joule heating into heating ions and neutrals in the LTI is assumed in most models which feeds back to the chemistry, dynamics and electrodynamics of the system. Modeling studies have highlighted that knowledge of the height variation of Joule heating is essential for quantifying its effect on neutral density and other LTI parameters (Huang et al., 2012). However, comprehensive data sets of derived Joule heating rate profiles with high temporal and spatial resolution are very rare.

In addition to heating, the energetics of the LTI region also includes processes such as radiative cooling through species such as  $\text{CO}_2$ ,  $\text{NO}$  and  $\text{O}$ . It has been shown, observationally and via modeling studies, that the thermosphere can cool significantly after strong geomagnetic storms due to energetic particle precipitation producing  $\text{NO}$  (Knipp et al., 2013; Mlynchak et al., 2005). The  $\text{NO}$  cooling peaks around 120–140 km altitude. Models include the  $\text{NO}$  chemistry (Smith-Johnsen et al., 2022) but observations of ion and neutral composition to validate the simulated chemistry are missing in this critical region. As a result, thermospheric cooling via  $\text{NO}$  cannot be directly evaluated.

**Neutral dynamics in the LTI region.** One of the biggest challenges in modeling the processes which couple the lower and upper atmosphere is capturing the neutral dynamic processes across and within the LTI region, such as turbulent mixing and momentum and energy deposition. Lack of quantitative characterization of these processes and their effects has implications on the entire upper atmosphere on large scales, such as the inability to address the composition, neutral density, mean circulation, and the plasma distribution which are all inter-connected. Numerical models still struggle with determining the absolute state of the upper atmosphere beyond relative variations or global averages even on large scales. The LTI region at the base of the upper atmosphere is difficult to include realistically in models since the processes in this region need significantly improved observational descriptions.

In the transition region, gravity waves break and deposit their momentum and energy into the background atmosphere, which then modifies the propagation of other waves, while changing the composition, temperature, and mean circulation. While we have a qualitative understanding of the processes, a quantitative description is missing, preventing a reliable determination of the magnitude and spatial distribution of turbulent mixing and transitioning to the region dominated by molecular diffusion (Becker and Vadas, 2020). High-resolution measurements of neutral wind, temperature and compositional parameters are needed to quantify the heat and momentum flux on various scales, as well as the effective diffusion and changing wave characteristics in the LTI region.



**Summary of LTI modelling needs.** Despite computational improvements and sophisticated coupling in modeling, progress is ultimately hindered by the severely limited knowledge regarding both the properties and physical processes inherent to the LTI including their linkage between the Earth's lower and upper atmospheres as well as the Earth's ionosphere and magnetosphere. Furthermore, there remains an important requirement for the models to include higher-resolution sub-grid scales in both space and time, for example to accommodate the sub-km LTI parameters which quickly vary with altitude, in order to adequately characterize the LTI behavior and coupling to other regions. The broad science areas discussed above where LTI measurements are necessary to improve the accuracy of simulations and models inform the scientific objectives and focus regions described in subsequent chapters of this report.

## 2.5. Summary

In summary, the LTI is marked as a region where cross-field electrical conductivities peak and give rise to a maximum in cross-field currents — a process which is encapsulated by the term **collisional electrodynamics**. Furthermore, neutral-plasma interactions in the LTI regulate the energy transfer — **collisional energetics** — and momentum — **collisional dynamics** — deposited from surrounding geophysical regions. These interactions are elementary for collision-dominated plasmas throughout the universe. Knowledge generated by studying the interactions in the LTI are, therefore, applicable to other collision-dominated plasmas located outside of the near-Earth geospace, such as chromospheric environments and protostellar discs. The presence of currents and the exchange of momentum and energy between the neutral and plasma gases and the electric and magnetic fields have acute consequences for the state of the upper atmosphere and its coupling to the near-Earth geospace environment. Processes in the LTI carry a significant societal impact, an important consideration that is further underscored by the growing societal relevance of the region and its influence on communications, navigation systems and critical space assets.

Building upon the extensive background discussion on the scientific processes in the LTI, in the next chapter we identify a number of key open questions or science objectives, including the scientific justification for the need for measurements to bring closure to these questions. This leads to the specification of a mission concept to gather direct measurements within the LTI needed to address these key science objectives.



### 3. SCIENTIFIC GOALS AND OBJECTIVES

Based on what is known and unknown about the LTI, we adopt an objective-based conceptual framework designed to close on the most important and urgent open aspects of this system. These objectives support the following overarching goal for understanding this critically important region of geospace:

***To understand how the transition from Earth's atmosphere to space is governed by the fundamental neutral-plasma interactions that are intrinsic to the lower thermosphere-ionosphere (LTI).***

This goal is based on several top-level fundamental characteristics. The LTI:

- is to a large extent controlled by the effects of the collisional coupling among neutrals, electrons, and ions.
- is strongly impacted from below by planetary, tidal, and gravity waves that deposit energy and momentum in this region through wave breaking and that thus alter the background wind field.
- is strongly impacted from above at high latitudes by auroral events and magnetic storms caused by solar activity through both the energy deposition of energetic particles and the effects of the intense electric currents that flow within its altitude range.
- is an essential part of the global geospace electrical circuit. At high latitudes, it plays a major role in the closure of electric currents flowing along the Earth's magnetic field lines connecting the Earth ionosphere, magnetosphere and outer space.
- is a principal player in triggering the development of disturbances that ultimately propagate through the upper ionosphere, particularly at mid- and low-latitudes.
- is a critical element for space weather studies and forecasting, and may include important indicators of climate change.

In all these aspects, neutral-plasma interactions, particularly collisions between neutral and charged species, play a central role in the behavior of the LTI. Thus, the overarching goal and its supporting Science Objectives (SOs) are couched in terms of this fundamental physics. Furthermore, the Science Objectives support the goal such that achievement of any or all objectives provides significant progress toward its accomplishment.

To address the overarching goal, we identify three high-level Science Objectives to significantly advance our understanding of the key outstanding LTI science drivers identified in Chapter 2. Each high-level Science Objective is supported by three detailed sub-objectives to investigate how collisional processes affect entities within the LTI and their properties. The Science Objectives are further organized by the relationships to their common physical quantity, the electric current density ( $J$ ), as discussed within the corresponding sections of Chapter 2.

The Science Objectives are presented in Table 3.1 and described in further detail in the subsequent sections. Figure 3.1 places the Science Objectives into a graphical representation of key LTI phenomena.



Table 3.1: High Level Science Objectives (SOs) (top row) including 3 sub-objectives (lower panels)

| SO1<br>Collisional Electrodynamics<br>$J$                                                                                            | SO2<br>Collisional Energetics<br>$J \cdot E$                                                                          | SO3<br>Collisional Dynamics<br>$J \times B$                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Determine how collisions between neutral and charged species affect the electrodynamics of the LTI.                                  | Determine how collisions between neutral and charged species affect the energetics of the LTI.                        | Determine how collisions between neutral and charged species affect the neutral dynamics of the LTI.                              |
| <b>SO1.1</b><br>Determine how electric currents flow and close in the LTI, and thereby couple to the magnetospheric electrodynamics. | <b>SO2.1</b><br>Determine how Joule (frictional) heating depends on scale size, altitude and neutral winds.           | <b>SO3.1</b><br>Determine how winds are accelerated by plasma motions via ion-neutral collisions.                                 |
| <b>SO1.2</b><br>Understand how the various LTI properties and processes act to determine the Hall and Pedersen conductivities.       | <b>SO2.2</b><br>Determine how energy from energetic precipitating particles (EPP) directly heats the LTI.             | <b>SO3.2</b><br>Discover how the exchange of momentum across scales by means of lower atmospheric forcing is manifest in the LTI. |
| <b>SO1.3</b><br>Determine the effect of the neutral winds on the LTI electrodynamics.                                                | <b>SO2.3</b><br>Determine how plasma-neutral collisions cause chemical changes that affect the energetics of the LTI. | <b>SO3.3</b><br>Determine how collisional processes drive vertical transport and cause composition changes.                       |



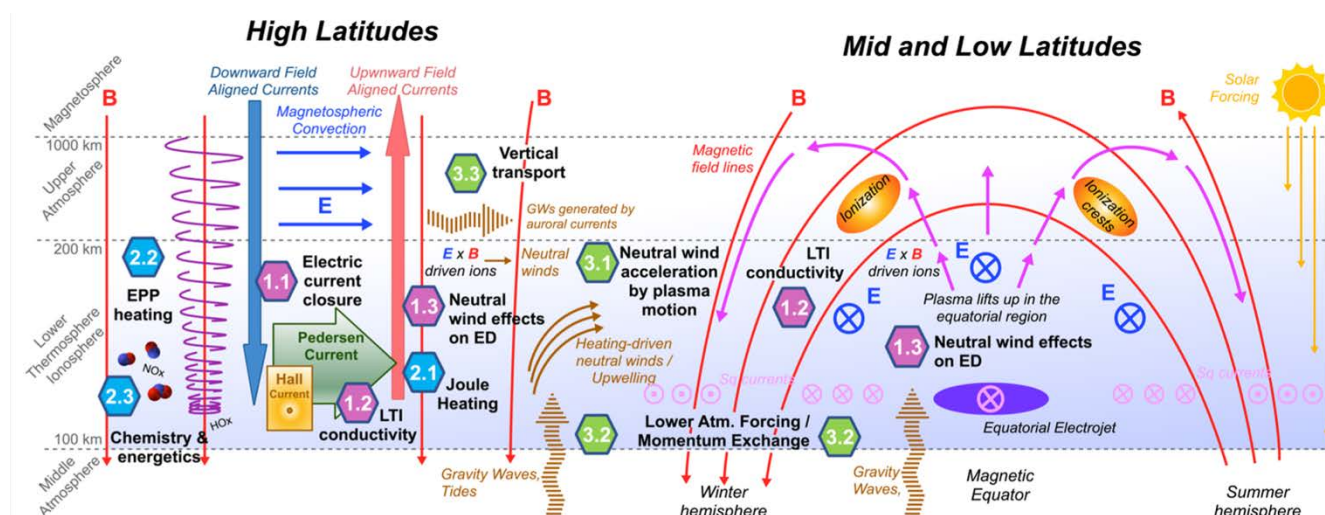
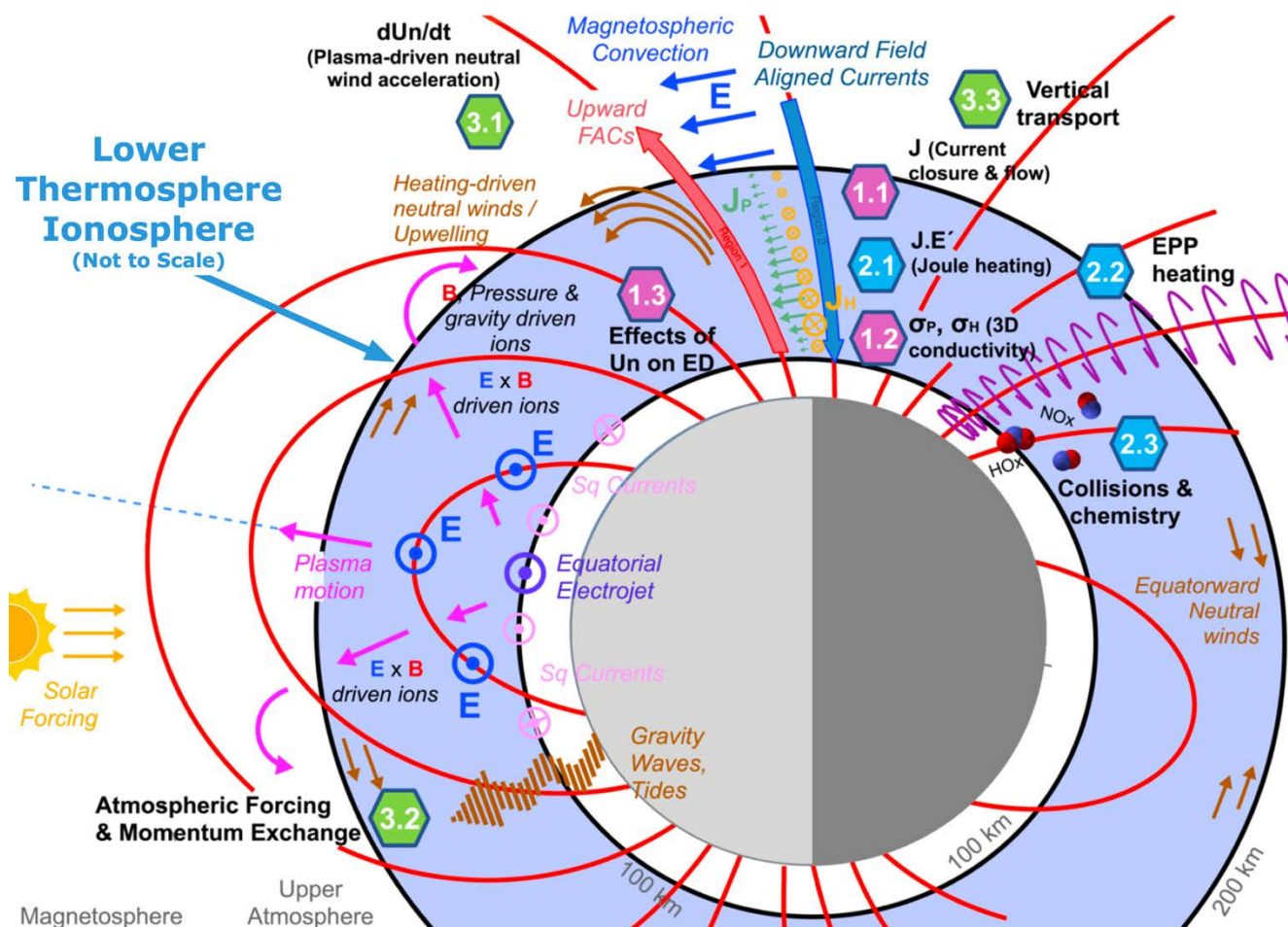


Figure 3.1: Graphical representation of key phenomena in the LTI, with the Science Objectives indicated by the numbers in the colored callouts. Note that the vertical scale of the LTI region is drastically exaggerated in this illustration (credit: DUTH – T. Sarris).

### 3.1. SO1: Collisional Electrodynamics

#### Determine how collisions between neutral and charged species affect the electrodynamics of the LTI.

Electrodynamics refers to the relationships among electric and magnetic fields, charged particles and their motions, and the evolution of those relationships under dynamic conditions. As discussed in Chapter 2, the effects of ion-neutral and electron-neutral collisions and their rapid variations with altitude within the LTI are essential factors when specifying LTI electrodynamics, in particular the relation between electric fields, plasma drifts, and electric currents. How these effects manifest themselves in the electrodynamics of the LTI are poorly understood. Due to their dependence on a large number of local properties and processes, as well as their variability in space and time, the only viable path to greater understanding relies on collocated *in situ* measurements of the germane parameters. These measurements, coupled with dedicated analysis, will lead to new understanding of how collisions in the LTI affect its electrodynamics. The following focused Science Sub-objectives constitute the building blocks used to develop this new understanding.

#### SO1.1 - Determine how electric currents flow and close in the LTI, and thereby couple to the magnetospheric electrodynamics.

The exchange of energy and momentum between the magnetosphere and ionosphere is mediated by electrical currents and depends critically on their closure in the LTI. However, knowledge of the 3D structure of the global currents is severely limited, as illustrated in Figure 3.2 (Heelis and Maute, 2020). This figure emphasizes the main unknown global currents associated with magnetic activity, such as those associated with the disturbance dynamo and penetration fields.

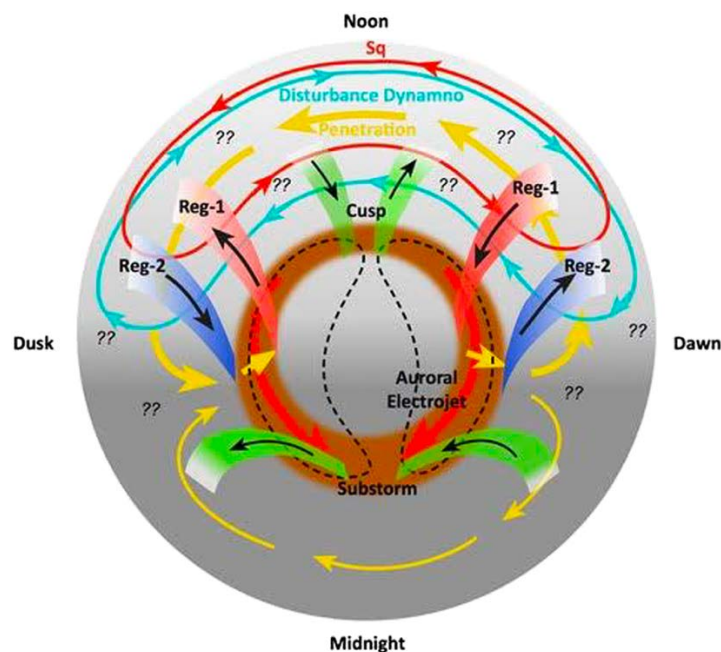


Figure 3.2: Schematic of field-aligned and perpendicular (to the magnetic field) global current systems in the I-T system with the perimeter latitude at the equator (from Heelis and Maute, 2020). Cusp, Region 1, Region 2, and substorm currents originate in the magnetosphere and drive the high-latitude plasma convection shown by the dashed black curves. Question marks indicate regions where the global current path is poorly specified being dependent on the unknown wind and conductivity distribution.

*In situ* measurements in the LTI are required to probe the intricate structure. Above the peak electron density in the ionospheric F region near 300 km altitude, also known as the topside ionosphere, the currents that mediate magnetosphere-ionosphere-thermosphere (M-I-T) coupling are aligned with the magnetic field  $\mathbf{B}$ , and are often sheet-like, allowing them to be estimated from single-satellite measurements of field perturbations,  $\delta\mathbf{B}$ . However, within the LTI (i.e., the ionospheric E region below 200 km altitude), cross-field Pedersen and Hall currents  $\mathbf{J}_P$  and  $\mathbf{J}_H$ , respectively, become important, requiring more advanced analysis to distinguish their contributions in the *in situ* measurements. Obtaining *in situ* measurements of currents statistically under different geomagnetic conditions and as a function of altitude, latitude and local time will allow the determination of the 3D structure of the Hall, Pedersen, and field-aligned currents in the LTI, which is critical to understanding the electric current system as a whole and its coupling to magnetospheric electrodynamics.

### SO1.2 - Understand how the various LTI properties and processes act to determine the Hall and Pedersen conductivities.

Electric currents accompany electric fields in the conducting LTI and its anisotropic conductivity components – Pedersen, Hall and field-aligned. Figure 3.3 contains a representation of the time-averaged structure of the Pedersen and Hall conductivities in the LTI below 150 km altitude based on model simulations constrained by energetic particle measurements from the DMSP satellites.

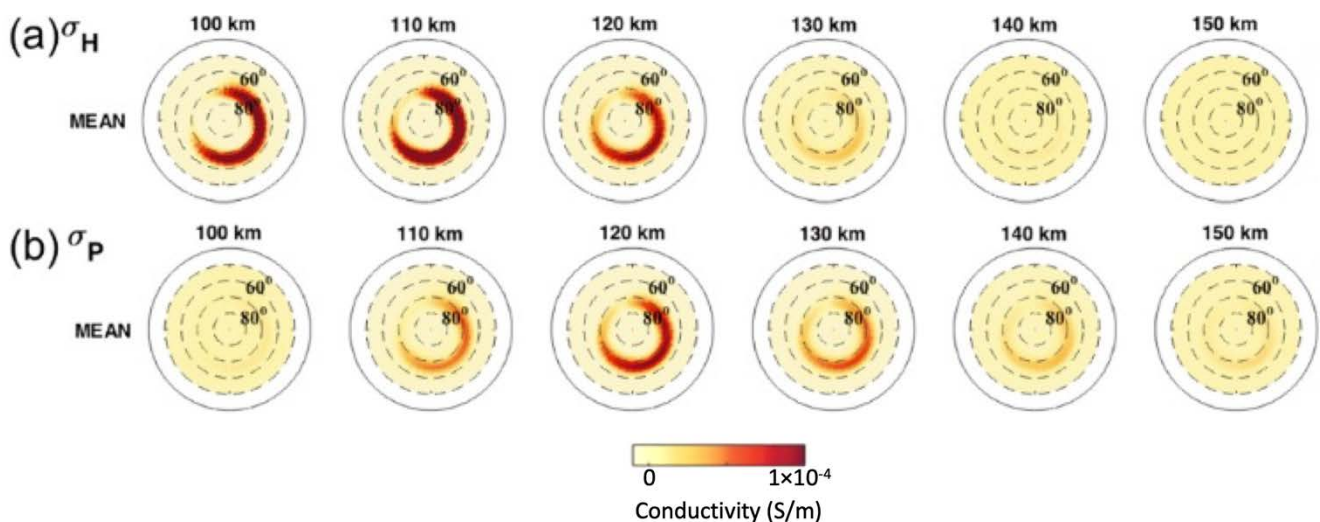


Figure 3.3: Average spatial structure of (a) Hall and (b) Pedersen conductivities based on long-term averages of space-based energetic particle observations and model calculations at different altitude levels in the LTI (from McGranaghan et al., 2016).

The conductivities depend on a number of local properties, including densities of charge carriers (ions and electrons) and their (anisotropic) mobilities, which in turn depend on the density of the neutrals and the collision cross-sections that depend on the temperatures of both neutral and charged species, and the composition of both the neutrals and the ions. Plasma instabilities in the LTI are a source of anomalous resistivity (Papadopoulos, 1977) and contribute to the conductivity. One instability is the Farley-Buneman instability (Buneman, 1963; Farley, 1963), a modified two-stream instability that occurs in E-region where the magnetized electrons flow through the un-magnetized ion gas that is collisional with the neutral background ionosphere. The 3D structure of these constituent properties is not adequately known to specify the structure of the overall conductivity components, and consequently, the pathways of currents in the LTI region. Measuring all of these factors is difficult, so an alternative approach is used: *in situ* measurements of currents and electric fields are utilized to compute conductivities in the region where they are significant. Measuring the bulk conductivities in a statistical



manner by performing simultaneous measurement of the currents and electric fields will result in a conductivity model that is parametrized with respect to LTI processes, properties and other geophysical drivers.

### SO1.3 - Determine the effect of the neutral winds on the LTI electrodynamics.

At mid and low latitudes, winds driven by planetary, tidal and gravity wave forcing drag plasma across magnetic field lines producing currents as the ions and electrons drift in different directions. This results in a global system of dynamo currents that reach their daytime peak in the lower ionosphere between approximately 90–150 km altitude. An example of how the electrodynamic and neutral wind parameters are related in the dynamo is displayed in the vertical profiles measured with sounding rockets (Pfaff et al., 2020) shown in Figure 3.4.

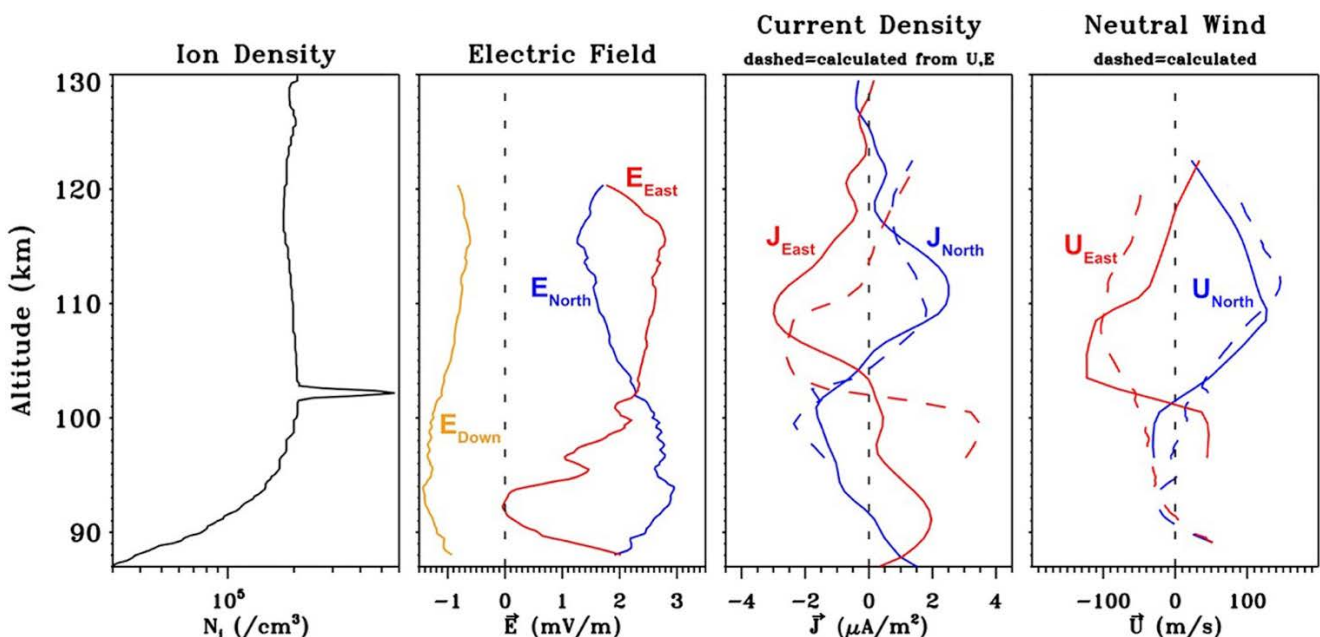


Figure 3.4: Mid-latitude measurements of the daytime ionospheric dynamo gathered by a NASA sounding rocket (from Pfaff et al., 2020) showing how neutral winds ( $\vec{U}$ ) drive the current ( $\vec{J}$ ). Red traces are in the eastward direction, blue traces in the northward direction. Dashed traces are calculated values based on the other measurements and the dynamo equation. The “spike” in ion density near 102 km is a sporadic-E layer, appearing at the same altitude of the shear in the neutral wind.

To keep the currents divergence-free (see section 2.2.1), local and global DC electric fields are created, subsequently influencing ionospheric motions along the magnetic field lines and therefore at higher altitudes. A critical consequence of the global dynamo current system is the daytime equatorial electrojet which flows along the magnetic equator, and which is enhanced by narrowly-confined vertical electric fields set up by the Cowling conductivity, as discussed in Chapter 2. Except for a few sounding rockets, systematic knowledge of how the winds drive the currents and produce the polarization electric fields as a function of altitude is currently lacking. In theory, nighttime equatorial electrojet currents and electric fields reverse direction, but the required measurements needed to probe this feature do not exist. *In situ* measurements of winds, currents, electric fields and conductivities as a function of altitude and latitude are required to illuminate the physics of the wind-driven dynamo and electrojet. These measurements will elucidate the central role played by neutral winds in driving these features of LTI electrodynamics.



## 3.2. SO2: Collisional Energetics

### Determine how collisions between neutral and charged species affect the energetics of the LTI.

Energetics refers to the storage, transport, and transformation of energy into various forms. Beyond effecting simple heat transfer among fluids having differing temperatures, collisions play a central role in several other important LTI energy transfer processes: Joule heating, direct energy deposition from energetic precipitating particles, and chemical processes that are key elements in energetic pathways that both heat and cool the LTI. Figure 3.5 shows how these processes and pathways act to impact the flow of energy within the system. Despite having this basic framework, the distribution of the energy among the different processes is not known, in large part because of limitations in available data sets. An investigation paradigm based on obtaining new measurements through systematic multi-parameter sampling of the LTI is required to improve this situation. The measurements gathered in this manner will reveal how collisions between neutral and charged species affect the energetics of the LTI. This new knowledge will be garnered by accomplishing the focused Science Sub-objectives described here, with the hexagonal callouts in Figure 3.5 indicating where missing knowledge is addressed by this approach.

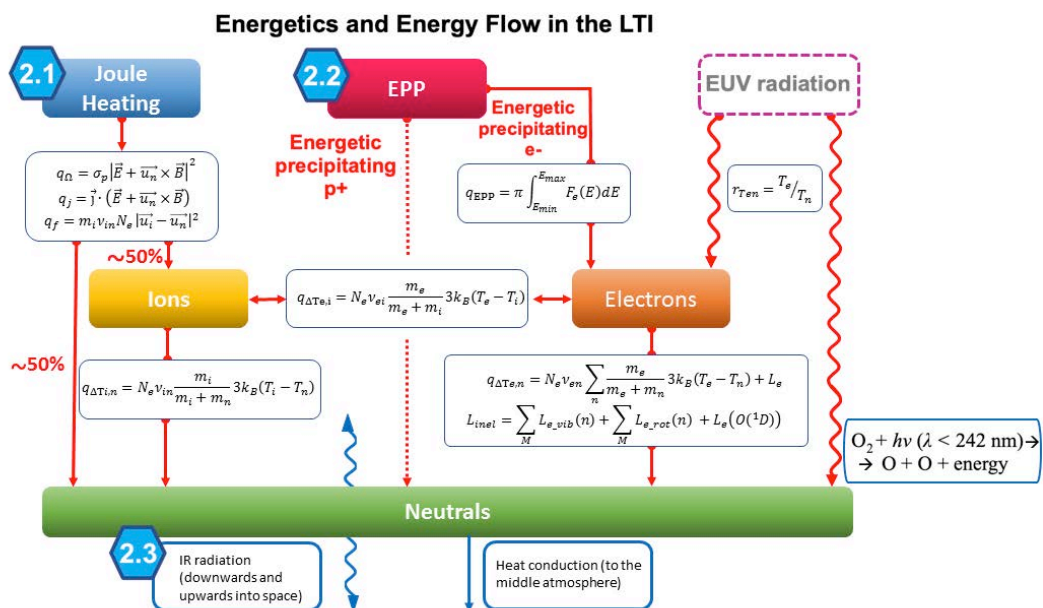


Figure 3.5: Flow chart of LTI energetics showing Joule heating rate ( $q_j$ ,  $q_\Omega$ ), frictional heating rate ( $q_f$ ), heating rate due to energetic particle precipitation ( $q_{EPP}$ ), and heat transfer rates between ions and neutrals ( $q_{\Delta T_{i,n}}$ ), between electrons and neutrals ( $q_{\Delta T_{e,n}}$ ), and between electrons and ions ( $q_{\Delta T_{e,i}}$ ), as well as electron energy dissipation through vibrational, rotational and quantum state excitation of LTI constituents ( $L_{inel}$ ) (from ESA, 2020).

#### SO2.1 - Determine how Joule heating depends on scale size, altitude and neutral winds.

Joule heating is a fundamental process that is proportional to the Pedersen conductivity and the square of the electric field in the neutral wind frame, all of which depend strongly on altitude and horizontal scale size. Joule heating is often greatly enhanced during geomagnetic storms and substorms, as depicted in Figure 2.18 (left). However, large discrepancies between various estimation methodologies

and models indicate that the importance of Joule heating at different scales and its distribution in altitude and latitude are largely unknown. It is also not known how Joule heating influences, and is influenced by, the local transport, thermal structure, and composition within LTI altitudes. The lack of direct measurements of all parameters involved in Joule heating estimations in the regions where it maximizes is the main impediment to our understanding. In particular, how neutral winds, ion drifts and electric fields interact within the conducting ionosphere to generate Joule heating requires simultaneous collocated measurements of all key parameters involved. Via these measurements, Joule heating will be unambiguously determined, and its dependence on neutral winds, altitude, geomagnetic conditions, as well as its importance at different scales, will be established. This determination will enable accurate parametrization of Joule heating in models and improve predictive capabilities.

## SO2.2 - Determine how energy from energetic precipitating particles (EPP) directly heats the LTI.

Understanding collisional energetics in the LTI requires knowledge of a significant source of energy, the direct heating of the LTI fluids by energetic precipitating particles (EPP) through collisions with the LTI constituent particles. Vertical structure in this heating, as shown in the model calculations depicted in Figure 3.6, is controlled by EPP energy and pitch angle distributions as well as variations of the LTI gases with altitude. Similarly, the horizontal structure in this heating is largely controlled by the EPP patterns and boundaries, and the temporal structure of the heating is controlled primarily by variations in the EPP fluxes. Despite plentiful measurements of energetic particles at higher altitudes, their interaction with the lower altitude LTI gases and in particular their heating effect versus altitude are not well known, as demonstrated by the disparate estimates shown in Figure 3.6. Understanding how the EPP and the lower thermosphere gases control and shape direct heating in the LTI therefore requires *in situ* measurements of EPP fluxes and energy distributions, as well as densities and composition of the LTI fluids. Gathering these measurements will reveal the heating effects these EPP interactions have on the LTI through this collisional process.

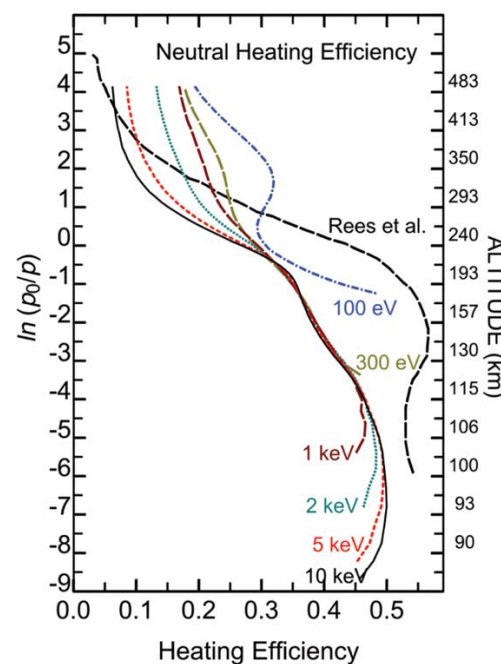


Figure 3.6: Modelled heating efficiencies of energetic precipitating electrons under typical conditions in the thermosphere including thermal electron heating (from Richards, 2013).



### ***SO2.3 - Determine how plasma-neutral collisions cause chemical changes that affect the energetics of the LTI.***

Neutral-plasma collisions significantly influence the energetics of the LTI through various chemical pathways. This chemistry produces a diverse compositional makeup in the LTI and leads directly and indirectly to both heating and cooling of the region. Radiative processes produced by species excited by inelastic collisions act to cool the LTI. Species like CO<sub>2</sub>, NO, and O help offset the heating introduced by neutral-plasma interactions. For example, the chemical pathways for NO production are local to the LTI where states of atomic nitrogen interacting with molecular oxygen undergo exothermic reactions to produce NO. Emissions from NO have been observed remotely for many years by orbiting missions (see Figure 2.19) and have been demonstrated to be vitally important for cooling the LTI. Nevertheless, the formation of NO remains poorly determined. Similarly, atomic oxygen is a known radiative cooler of the LTI, but its concentration is not well determined at these altitudes. In addition, Joule heating and EPP influence these chemical processes, but it is not clear how. *In situ* measurements of composition, as well as the Joule heating and EPP drivers, are required to understand how these chemical processes work throughout the LTI. The knowledge thus obtained will show how neutral-plasma collisions affect the energetics of the LTI through chemical changes.

## **3.3. SO3: Collisional Dynamics**

### **Determine how collisions between neutral and charged species affect the dynamics of the LTI.**

Dynamics refers to how matter moves and is accelerated, with the focus on momentum and transfer of momentum. Collisions allow momentum present in the motions of plasmas to be transported into the LTI to accelerate neutral winds. Atmospheric waves, including gravity waves, incident on the LTI from sources in the lower atmosphere dynamically imprint their signatures on the LTI through collisional interactions. Collisions are the intermediary through which various LTI processes act to transport matter. Significant questions remain concerning how collisions act to produce these behaviors, and multi-parameter *in situ* measurements are required to unravel the operative underlying relationships. Understanding these relationships then reveals how collisions between neutral and charged species affect the dynamics of the LTI.

### ***SO3.1 - Determine how winds are accelerated by plasma motions via ion-neutral collisions.***

Large-scale plasma motions or “convection” in the high-latitude ionosphere drives neutral gas circulation via ion-neutral collisions. This circulation then acts to transport momentum and energy to adjacent regions, as the neutral gas is not tied to the magnetic field. Similarly, intense neutral wind “jets” associated with auroral precipitation and steady, applied electric fields, may also form at low altitudes near 120–150 km, as shown in the model and experimental results depicted in Figure 2.22. How imposed plasma drifts drive neutral motions and structuring at the lower altitudes is not well understood. In addition, the distribution of neutral jet streams across the high-latitude region and their impact on the larger circulation of the neutral gas are very poorly known. *In situ* measurements of  $\mathbf{E} \times \mathbf{B}$ , ion velocity and neutral winds together with conductivity and neutral density on the same platform are the only way to gather the necessary information to reveal how these processes operate. This new understanding will elucidate the role played by ion-neutral collisions in enabling plasma motions to accelerate winds in the LTI.

### ***SO3.2 - Discover how the exchange of momentum across scales by means of lower atmospheric forcing is manifest in the LTI.***

Momentum transport by upward propagating atmospheric tides, planetary waves, and small- to meso-scale gravity waves is a significant contributor to the large-scale momentum balance of the upper atmosphere. Gravity waves, excited in the lower atmosphere, propagate upward, where they dissipate and/or generate secondary gravity waves. This process injects momentum at a range of spatial and temporal scales in the LTI, as exemplified in Figure 3.7. The interaction depends on both the background atmospheric state and the characteristics of the propagating waves. Gravity wave characterization is challenging, and much of what is known about these interactions is derived from models of gravity wave propagation. Few measurements are available to guide development of these models, and the systematic observations that do exist are generally limited to what can be gleaned from ground-based observatories. This situation translates into significant uncertainties in atmospheric general circulation models that seek to describe the behavior of the LTI. *In situ* measurements will improve this situation by resolving perturbations in multiple parameters throughout the three-dimensional domain that is the LTI. The resulting new, measurement-based description of how atmospheric gravity waves interact with the LTI, especially via the momentum flux they transport, is the key to understanding the overall effect of gravity waves on the dynamics of the LTI.

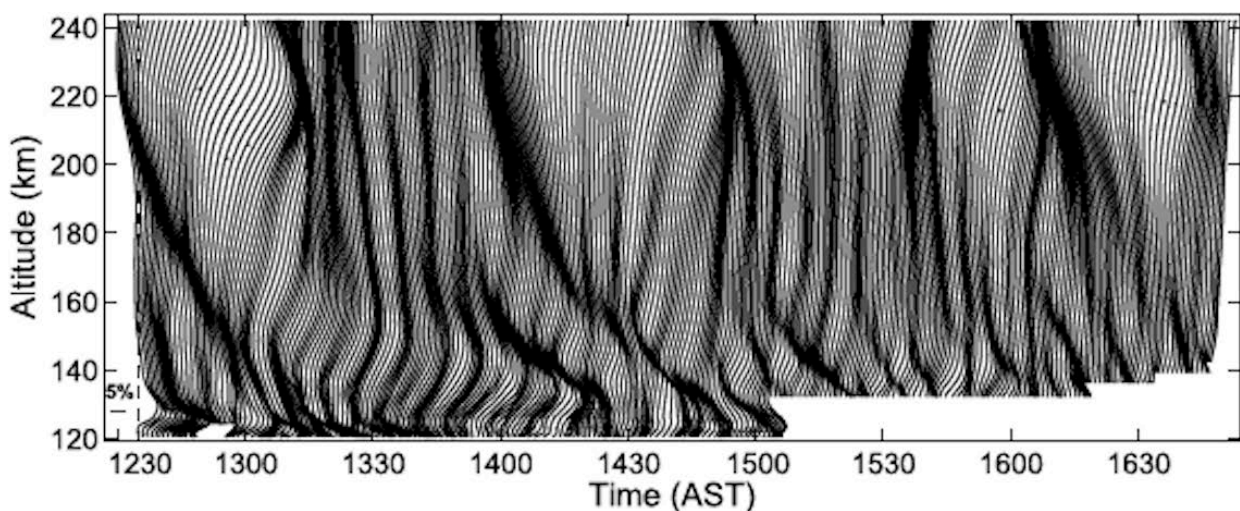


Figure 3.7: Perturbations of ionospheric electron density structured by gravity waves propagating upward into the LTI and beyond, measured using the incoherent scatter radar at the Arecibo Observatory (from Djuth et al., 2010).

### ***SO3.3 - Determine how collisional processes drive vertical transport and cause composition changes.***

A key consequence of LTI dynamics is the variable and structured vertical transport of matter within the LTI and beyond. Important aspects of this transport include upwelling of ionic and neutral species, as well as redistribution of the cooling agents discussed in SO2.3. This upwelling has far-reaching consequences beyond the LTI, including contributing to global circulation and outflow of ionospheric plasma into the magnetosphere where it affects magnetospheric dynamics. While much is known about the processes that lead to vertical transport, significant conceptual challenges remain. For example, the physical mechanisms responsible for some important forms of vertical transport are presently debated. Figure 3.8 depicts two examples in which LTI processes have led to strong restructuring of



the thermosphere due to vertical transport. High-altitude consequences of physical processes that lead to upwelling of the neutral atmosphere at the base of the magnetospheric cusp are depicted in Figure 3.8 (left). The causative processes act primarily within the LTI, but just which processes are at play, and even the altitude at which the processes are most effective, are not settled. Most attempts at resolving these issues have relied on modelling (Brinkman et al., 2016; Lotko and Zhang, 2018; Oigawa et al., 2021, 2022; Sadler et al., 2012), but it is clear that progress in understanding these phenomena require new measurements that will constrain modelling efforts more. Figure 3.8 (right) provides another example of the manifest changes to the neutral atmosphere due to vertical transport. Here nitrogen-rich air from lower altitudes has been transported upward over a large region, changing the chemistry and energetics in the LTI and above. Understanding how vertical transport is set up in the LTI requires spatially resolved measurements of the plasma and neutral flows, as well as densities and composition. Analysis of these measurements will result in understanding how collisional processes drive vertical transport and cause composition changes in the LTI.

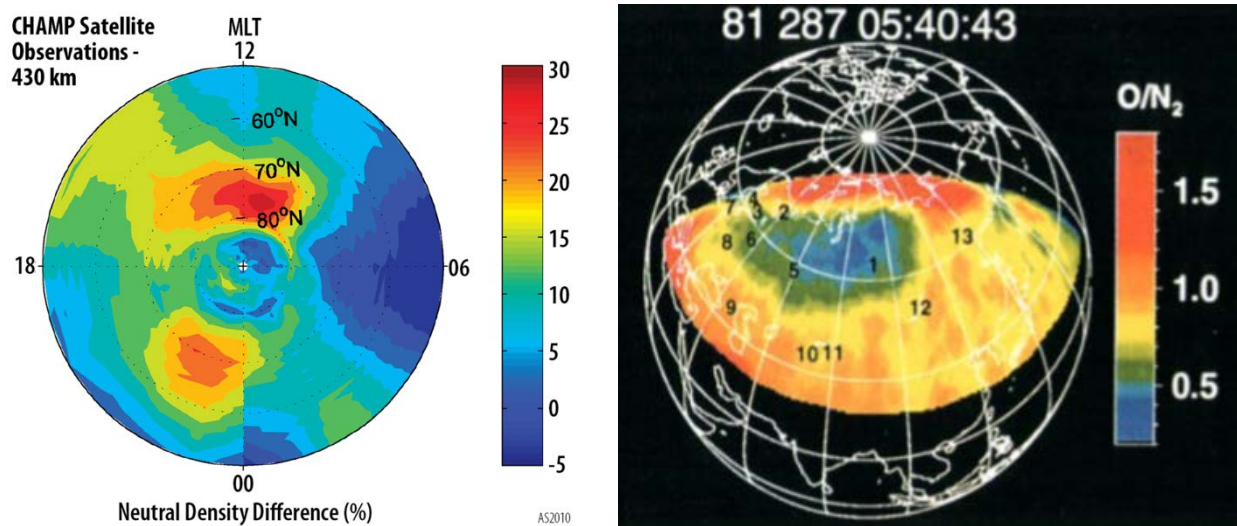


Figure 3.8: Left, thermospheric neutral density measurements from the CHAMP mission revealing regions of strong neutral upwelling at high latitudes (from Liu et al., 2005). The processes thought to be responsible for this upwelling reside in the LTI. Right, thermospheric composition measurements from the Dynamics Explorer 1 mission indicating composition change due to vertical transport (from Strickland et al., 2001). The values represent the columnar O/N<sub>2</sub> above about 150 km.



## 4. SCIENTIFIC STRATEGY AND NEEDS

This chapter develops a strategy by which the science objectives (SOs) described in the previous chapter can be achieved. The **scientific needs** are the approaches and associated data products needed to provide SO closure. Because the current state of knowledge is strongly limited by a lack of LTI measurements, **observational needs** are emphasized. Furthermore, physical models and theoretical concepts are included to support the activity but to also be tested and exercised to achieve the most comprehensive understanding of Earth's LTI. The flow for establishing these scientific and observational needs begins by considering derived parameters and direct observables involved in each SO. A **derived parameter** is a compound property or process that is essential for SO closure and can be obtained from a combination of direct observables and theory. **Direct observables** are fundamental properties of the gas or plasma directly retrieved from a measurement. A **measurement** is provided by an instrument and can produce a direct observable within a required range, precision, and accuracy. Table 4.1 provides a list of key derived parameters for each SO, and the theory and the set of direct observables required to obtain them.

At the highest level, the derived parameters of interest are currents,  $\mathbf{J}(\mathbf{x}, \mathbf{y}, \mathbf{z})$ , and the related electrical conductivity; LTI energy inputs including Joule heating,  $\mathbf{J} \cdot \mathbf{E}'$ , and energetic particle precipitation; and dynamical forcing, including  $\mathbf{J} \times \mathbf{B}$  and gravity wave forcing. These derived parameters cannot be measured directly. Instead, there are well-established theoretical formulations, presented in Table 4.1, that enable their derivation based on other direct observables which require simultaneous, common-volume measurements using *in situ* sensors. Determination of  $\mathbf{J}$ ,  $\mathbf{J} \cdot \mathbf{E}'$ , and  $\mathbf{J} \times \mathbf{B}$  relies on computations or estimations based on combinations of direct observables. Thus, the determination of derived parameters relies on the measurement capabilities for determining the necessary direct observables. Furthermore, observational needs require specification to ensure that the measurements occur with sufficient cadence, coincidence, and coverage under a variety of geophysical conditions. The scientific needs that lead to the derived parameters and direct observables are discussed in Section 4.1. The observational needs with respect to spatial and temporal scales, accuracy, and targeted observing regions are summarized in Section 4.2.

The support from models and complementary ground and space-based observations to achieve the scientific needs are briefly described in Section 4.3. The science strategy matrix (SSM) in Appendix B summarizes the needs levied by this measurement-based strategy. The SSM shows how each SO maps to the derived parameters, direct observables, as well as its observational requirements pertaining to spatial and temporal resolution, regions of interest, geophysical conditions, and modeling. For many of the SOs, the value of the *in situ* measurements can be enhanced significantly by complementary measurements from the ground and from other missions. These are also noted in the SSM.

Table 4.1: Summary of derived parameters, theoretical basis, and primary (direct) parameters (observables) needed. The reader is referred to Chapter 2 for a theoretical development of the derived parameters.

| Themes                                    | Needed parameters (Derived)                                                                  | Equation for determining derived parameters                                                                                                             | Needed observables (Direct)                                   |
|-------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| SO1<br>Collisional Electrodynamics<br>(J) | Perpendicular current <sup>[1]</sup>                                                         | $J_{\perp} = eN_e(\mathbf{V}_{i,\perp} - \mathbf{E} \times \mathbf{B}/B^2)$ $J_{\perp} = (\nabla \times \mathbf{B})/\mu_0$                              | $X = \{\mathbf{V}_i, N_e, \mathbf{E}, \mathbf{B}\}$           |
|                                           | Electric fields in the reference frame of the neutral wind                                   | $\mathbf{E}' = \mathbf{E} + \mathbf{U} \times \mathbf{B}$                                                                                               | $Y = \{\mathbf{U}, \mathbf{E}, \mathbf{B}\}$                  |
|                                           | Field-aligned current <sup>[2]</sup>                                                         | $j_{\parallel} = (d\mathbf{B}_{\perp}/v_{sc}dt)/\mu_0$                                                                                                  | $\delta\mathbf{B}$                                            |
|                                           | Pedersen and Hall currents <sup>[1]</sup>                                                    | $j_p = J_{\perp} \cdot \mathbf{E}'/ \mathbf{E}' ,$ $j_H = J_{\perp} \cdot \hat{\mathbf{b}} \times \mathbf{E}'/ \mathbf{E}' $                            | $X \cup Y$                                                    |
|                                           | Pedersen conductivity <sup>[3]</sup><br>Hall conductivity <sup>[3]</sup>                     | $\sigma_P = j_p/ \mathbf{E}' $ $\sigma_H = j_H/ \mathbf{E}' $                                                                                           | $X \cup Y$                                                    |
|                                           | Parallel conductivity <sup>[3]</sup>                                                         | $\sigma_{\parallel} = \frac{N_e e^2}{m_e(v_{en} + v_{ei})}$                                                                                             | $Z = \{N_e, N_n\}$                                            |
| SO2<br>Collisional Energetics<br>(J · E') | Perturbation Poynting flux <sup>[1]</sup>                                                    | $\mathbf{S}_p = (\mathbf{E}_{\perp} \times \delta\mathbf{B})/\mu_0$                                                                                     | $\mathbf{E}, \delta\mathbf{B}$                                |
|                                           | Joule heating rate <sup>[4]</sup>                                                            | $q_j = J_{\perp} \cdot (\mathbf{E} + \mathbf{U} \times \mathbf{B})$                                                                                     | $X \cup Y$                                                    |
|                                           | Ion-neutral heat exchange rate                                                               | $q_T = \frac{3k_B}{m_n} m_i v_{in} N_e (T_i - T_n)$                                                                                                     | $W = \{T_n, T_i, N_{nx}, N_{ix}\}$<br>$W \cup Z$              |
|                                           | Neutral frictional heating rate <sup>[4]</sup><br>Ion frictional heating rate <sup>[4]</sup> | $q_{fn} = m_i v_{in} N_e (\mathbf{V}_{i,\perp} - \mathbf{U}_{\perp})^2$ $q_{fi} = \frac{m_n}{3k_B} (\mathbf{V}_{i,\perp} - \mathbf{U}_{\perp})^2$       | $W \cup X \cup Y \cup Z$                                      |
|                                           | Impact ionization rate ( $q_{EPP}$ )                                                         | $q_{EPP} = \sum_{n,s} \sigma_n(E) N_n F_{EPP,s}(E)$                                                                                                     | EPP differential energy fluxes<br>( $F_{EPP,s}$ ), $W \cup Z$ |
|                                           | IR cooling rates <sup>[5]</sup>                                                              | $q_{IR} = \frac{\partial F_{Net}}{\partial p}$                                                                                                          | [NO], [O], [CO <sub>2</sub> ], T <sub>n</sub>                 |
| SO3<br>Collisional Dynamics<br>(J × B)    | Ampere force <sup>[1]</sup>                                                                  | $\mathbf{F}_j = J_{\perp} \times \mathbf{B}$                                                                                                            | $X$                                                           |
|                                           | Ion drag force <sup>[1]</sup><br>Neutral drag force <sup>[1]</sup>                           | $\mathbf{F}_i = v_{in} m_i N_e (\mathbf{V}_{i,\perp} - \mathbf{U}_{\perp})$ $\mathbf{F}_n = v_{ni} m_n N_n (\mathbf{U}_{\perp} - \mathbf{V}_{i,\perp})$ | $X \cup Y \cup W \cup Z$                                      |
|                                           | Mechanical work rate <sup>[1]</sup>                                                          | $q_m = \mathbf{U} \cdot (J_{\perp} \times \mathbf{B})$                                                                                                  | $X \cup Y$                                                    |
|                                           | Gravity wave momentum flux                                                                   | $F_M = \frac{1}{2} \langle u' w' \rangle$                                                                                                               | 3D components of $\mathbf{U}$                                 |
|                                           | Gravity wave heat flux                                                                       | $F_H = \frac{1}{2} \langle w' T' \rangle$                                                                                                               | T <sub>n</sub> , 3D components of $\mathbf{U}$                |

[1] (Richmond and Thayer, 2000); [2] (Lühr et al., 2019); [3] (Schunk and Nagy, 2009); [4] (Thayer and Semeter, 2004); [5] (Mlynczak et al., 2005).



## 4.1. Scientific needs

This section expands on Table 4.1 by specifying the scientific needs that lead to the derived parameters and direct observables. The derived parameters are discussed in terms of how they will be determined from accepted theory and direct observables.

### 4.1.1. SO1: Collisional Electrodynamics

#### Determine how collisions between neutral and charged species affect the electrodynamics of the LTI.

Table 4.2: Approaches for closure of SO1.

| Science Objectives                                                                                                         | Scientific needs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SO1.1: Determine how electric currents flow and close in the LTI, and thereby couple to the magnetospheric electrodynamics | Achievement of this objective requires 3D currents to be determined. Field-aligned currents and perpendicular currents are to be determined and current continuity is then applied to determine how the currents connect (close). The horizontal currents are derived parameters which can be determined by different observing methods involving the direct observables: electron density, ion velocity, and electric and magnetic fields. Vertical currents are derived from perturbation magnetic field measurements. 3D neutral winds are needed observables to compute the electric field in the neutral wind reference frame.                                                                                                                             |
| SO1.2: Understand how the various LTI properties and processes act to determine the Hall and Pedersen conductivities       | Achievement of this objective includes horizontal conductivities (Pedersen and Hall) to be determined as a function of SO1.1 derived parameters: currents and electric fields in the neutral wind reference frame. Another approach employs lab expressions that require the direct observables of electron, ion, and neutral densities as well as ion drifts (neutral temperature and composition are secondary observables to constrain the solutions) to determine the horizontal conductivity. Field-aligned conductivity will require measurements of electron and neutral density. Conductivities depend on processes like ionization (via solar UV impingement and particle precipitation), recombination, and transport which will be addressed by SO2. |
| SO1.3: Determine the effect of the neutral winds on the LTI electrodynamics                                                | Achievement of this objective requires knowledge of electrodynamic quantities: electric field, ion drift, current, conductivity, and neutral winds. Current and conductivity are derived parameters as established in SO1.1 and SO1.2 while wind and electric field are direct observables. The effects of winds on the electrodynamics will be determined by analyzing changes in currents across a variety of wind conditions.                                                                                                                                                                                                                                                                                                                                |

Table 4.2 describes the scientific needs to achieve closure for each of the science objectives of SO1. Derived parameters and direct observables are also put forward to meet this need. The following describes the approaches in determining the most pertinent properties and processes associated with SO1.

**3D electric current.** Knowledge of the 3D closure of FACs in the LTI is a first step in understanding the 3D distribution of current-driven energy dissipation, their impact on thermospheric energetics and dynamics, and their role in LTI-mediated feedback on M-I coupling and magnetospheric dynamics. Currently, little is known about the 2D closure of horizontal currents except in the simplest (e.g., sheet-like) geometries. Even less is known about the altitude profiles of these 2D currents, which depend strongly on key but inadequately measured parameters that include ion-neutral collision frequencies





and neutral winds. Electron-neutral and ion-electron collision frequencies can be neglected in the LTI (>100km).

Currents can be measured in several ways. Well above the LTI, currents are B-field-aligned, and can be estimated with multiple-spacecraft measurements of  $\delta\mathbf{B}$  using the curlometer technique (Dunlop et al., 2021), or more typically with a single satellite combined with knowledge or assumptions about the FAC geometry (Luhr et al., 1996).

As the measurement altitude decreases, reconstruction of current systems from  $\delta\mathbf{B}$  (perturbation magnetic field) requires identifying increasing contributions from horizontal (Hall and Pedersen) currents. Fortunately, measurements of Hall and Pedersen currents can be determined using *in situ* magnetometer measurements by examining their different perpendicular components as the satellite traverses the currents and using reasonable assumptions about the current geometry and stationarity, as shown for auroral and equatorial electrojet measurements (Lühr et al., 2004; Vennerstrom and Moretto, 2013).

Furthermore, in the collisional LTI, direct measurements of horizontal currents become possible through the relation:

$$\mathbf{J}_{\perp} = eN_e(\mathbf{V}_{i,\perp} - \mathbf{E} \times \mathbf{B}/B^2) \quad (4.1)$$

or, given adequate knowledge of plasma densities, ion collision frequencies and neutral winds:

$$\mathbf{J}_{\perp} = \sigma_P(\mathbf{E} + \mathbf{U} \times \mathbf{B}) + \sigma_H \hat{\mathbf{b}} \times (\mathbf{E} + \mathbf{U} \times \mathbf{B}) \quad (4.2)$$

The current state of the art does not allow direct measurements of parallel currents:

$$\mathbf{J}_{\parallel} = eN_e(\mathbf{V}_{i,\parallel} - \mathbf{V}_{e,\parallel}) \quad (4.3)$$

due to the difficulty in measuring the parallel flow velocity of electrons at ionospheric (eV) energies. A statistical picture of  $\mathbf{J}(x, y, z)$  can be accumulated by sampling over a range of altitudes, latitudes and local times using magnetometer measurements of  $\delta\mathbf{B}$  and *in situ* observations of electron density, ion motion, and electric fields perpendicular to the magnetic field. Such a database would represent a major advance in knowledge and would provide critically needed information to inform physics-based models of the LTI region.

**Conductivity.** Conductivity relates currents  $\mathbf{J}$  and electric fields  $\mathbf{E}$ , with the important caveat that  $\mathbf{E}$  must be measured in (or translated to) the frame of the neutral gas, where its value is denoted  $\mathbf{E}'$ . Thus, conductivities can be derived with colocated measurements of  $\mathbf{J}$ ,  $\mathbf{E}$ , and  $\mathbf{U}$ , along with measurements or knowledge of  $\delta\mathbf{B}$  to determine the field-aligned, Hall and Pedersen current directions. As described above, direct measurement of  $\mathbf{J}_{\perp}$  can be accomplished by measuring the difference between bulk ion motion  $\mathbf{V}_i$  and perpendicular electron drift, where the electron drift can be assumed to be equal to  $\mathbf{E} \times \mathbf{B}/B^2$  above 100 km in the LTI. The angle between  $\mathbf{V}_i$  and  $\mathbf{E}$  combined with knowledge of  $\mathbf{U}$  has been used to derive Pedersen and Hall conductivities as well as ion-neutral collision frequencies (Burchill et al., 2012; Sangalli et al., 2009).

More commonly, conductivity estimates are derived from ion and neutral density measurements combined with modeled values of ion-neutral collision frequency (see, e.g., equation 5 of Schunk and Nagy, 2009), where the latter is estimated through ion-neutral collision cross-sections. The assumptions that typically underlie this approach, for example the applicability of lab-based collision cross-sections for a range of species, are difficult to test and a more direct conductivity estimate based on *in situ* measurements, as described above and presented in Table 4.1, is badly needed.

**Neutral wind electrodynamics.** In its most basic form, neutral winds affect electrodynamics in a manner that can be handled by evaluating the electric field in the frame of reference of the neutral wind. This frame of reference is the same as that discussed in previous sections, and the primary parameters needed are the same, i.e., electric fields, conductivity, and neutral winds. These must be estimated concurrently within the same volume as the structure and temporal response of these electrodynamic properties are vastly different. The concurrent, common-volume observations of all three properties taken at the cadence of the most rapidly varying property, i.e., electric fields with sub-second resolution, ensures that their combined effects on LTI electrodynamics can be resolved. Furthermore, regions of strong wind gradients make the approach more complex, as shown in Figure 3.4. As discussed in Section 3.1, wind gradients lead to gradients in the electric field and current density which must redistribute plasma to ensure the current system remains divergence free. Thus, gradients in the winds alter the structure of the electrodynamics of the LTI, and adequate spatial sampling of the derived parameters is needed for a complete picture.

#### 4.1.2. SO2: Collisional Energetics

### Determine how collisions between neutral and charged species affect the energetics of the LTI.

Table 4.3: Approaches for closure of SO2.

| Science Objectives                                                                                                 | Scientific needs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SO2.1: Determine how Joule heating depends on scale size, altitude and neutral winds</b>                        | Achievement of this objective requires determination of several electromagnetic heating rate quantities listed in Table 4.1. The approach taken here is to measure these rates and test several assumptions and relationships while improving the estimate of heating to the neutral gas by electromagnetic energy. Poynting flux and Joule heating rates will use SO1 measured and derived parameters. Neutral and ion heat transfer rates will depend on directly observing the ion and neutral temperatures and their masses. Frictional heating rates for neutrals and ions will require, ion and neutral motion, plasma density, and neutral and ion masses. |
| <b>SO2.2: Determine how energy from energetic precipitating particles (EPP) directly heats the LTI</b>             | Achievement of this objective requires determination of precipitating particle energies, pitch angles, and fluxes to reliably estimate their energy deposition into the LTI. Deriving the particle heating rate requires simultaneous determinations of direct observables in densities and temperatures of neutral and charged LTI populations.                                                                                                                                                                                                                                                                                                                  |
| <b>SO2.3: Determine how plasma-neutral collisions cause chemical changes that affect the energetics of the LTI</b> | Achievement of this objective requires determination of the composition of neutral and charged species within the LTI, as well as the temperatures of the neutral gas and ions. The approach to deriving heating and cooling rates from these primary parameters relies on collision rates derived from the determination of conductivities, as well as models used to determine the heating and cooling rates. IR cooling rates will be estimated from radiative models and directly observed neutral concentrations of atomic oxygen, carbon dioxide, and nitric oxide along with neutral temperature.                                                          |

Following the previous section, Table 4.3 describes the scientific needs that lead to the derived parameters and direct observables for SO2. The derived parameters are discussed in terms of how they will be determined from accepted theory and direct observables.

**Electromagnetic heating.** Electromagnetic energy is directed from the magnetosphere into the LTI at high latitudes in the form of Poynting flux and represents a major heat source for the LTI neutral



gas. How this electromagnetic energy is manifested in the LTI depends on several factors including the ionosphere conductivity, Poynting flux scale size, and its partitioning to thermal and kinetic energy of the neutral gas. A common assumption is that all the Poynting flux energy is converted to thermal energy of the neutral gas without considering the real possibility that some of this energy is transferred to the movement of the neutral gas. This needs to be tested to accurately determine this important LTI heat and momentum source. *in situ* measurements of Joule heating, through the relation  $\mathbf{J} \cdot (\mathbf{E} + \mathbf{U} \times \mathbf{B})$ , will advance understanding of its local and global influence on neutral temperature, while its field-line-integrated value can be estimated from the perturbation Poynting vector  $\mathbf{S}_p = (\mathbf{E}_\perp \times \delta \mathbf{B}) / \mu_0$  measured well above the LTI, under assumptions that include sheet-like FAC geometry and quasi-stationarity (Kelley et al., 1991). High-altitude Poynting flux measurements are valuable in both event-based and statistical studies of electromagnetic energy dissipation. This complementary measurement can be performed in the higher-altitude portion of an elliptical dipping orbit, to collect statistics; or if possible, at conjugate points along a field line in case multi-point measurements are available, either as part of a coordinated multi-satellite mission, or in coordination with other missions.

Table 4.1 presents several other descriptions of neutral gas heating rates associated with electromagnetic energy, each having its own set of assumptions and descriptions. The Joule heating rate accounts for internal energy changes in the neutral gas due to currents that dissipate electromagnetic energy as neutral gas thermal energy. However, this estimate assumes all the electromagnetic energy transfers to the neutral gas without consideration of the heat transfer pathway through the plasma. The frictional heating rate describes the energy path to neutral gas heating by considering the differential motion between ions and neutrals that cause both ions and neutrals to increase their temperature. The hotter ions transfer their thermal energy to the neutral gas via random-motion collisions, i.e., conduction – defined in Table 4.1 as the ion-neutral heat exchange rate. Joule and frictional heating can be equated if: 1) the ion energy equation is simplified to a steady state approximation where ion-neutral temperature differences are equal to the square of ion-neutral differential motion, and 2) the ion momentum equation is in steady state and governed only by the Lorentz and neutral drag forces. These assumptions need to be tested in the LTI to determine which neutral gas heating rate is most appropriate. This can be done by measuring the relevant ion and neutral properties concurrently in a common volume and will need to be studied under different geophysical conditions and locations.

Ignoring neutral motion has implications on the local heating rate as the neutral gas is seldom static (Thayer, 1998). The moving neutral gas therefore defines the reference frame (denoted by a prime) for electromagnetic heating calculations involving the electric field  $\mathbf{E}' = \mathbf{E} + \mathbf{U} \times \mathbf{B}$ . Using measurements of  $\mathbf{E}$ ,  $\mathbf{V}_i$  and  $\mathbf{U}$  on the JOULE-II sounding rocket, it has been shown that neglecting  $\mathbf{U}$  would lead to errors of tens of percent when estimating Joule dissipation, and of order 10% when estimating Pedersen and Hall conductivities, even in a relatively quiescent auroral environment (Sangalli et al., 2009). This assumption and impact on neutral gas heating rates needs to be tested by measuring neutral wind motion concurrently with the other heating rates described in Table 4.1.

**Energetic Particle Precipitation.** Energetic particle precipitation (EPP) is an important energy input into the LTI at high latitudes; at scales less than 10 kms it is comparable in magnitude to electromagnetic energy inputs (Vickrey et al., 1982). Furthermore, EPP has a significant impact on conductivities and ionization rates in the 100–200 km LTI transition region and in the mesosphere. Energetic particle precipitation arises through several forms, involving different ion and electron populations that are deposited in the ionosphere in response to different types of solar wind and magnetospheric events. EPP encompasses dynamics ranging from direct access of high-energy solar energetic particles (SEPs) during transient events, such as solar flare-accelerated ions, to more continuous loss of inner magnetospheric electron populations due to wave-particle interactions. To



advance understanding of the heating effects of EPP on the LTI, measurements must capture a diverse range of energies and fluxes of downward moving electrons. Simultaneous and co-located measurements of the composition and temperatures of neutral and charged species are equally important to resolve the thermospheric response as a function of altitude. By sampling at different altitudes, the effect of composition on the heating rate can also be assessed.

In addition to the EPP contribution to collisional heating, the energetic particle precipitation changes the thermospheric composition through ionization. NO is one of the key species that efficiently cools the thermosphere to balance the particle heating. This NO emission process maximizes in the lower LTI at the height of 90–140 km. In particular, the NO emission at 5.3  $\mu\text{m}$  is the dominating thermospheric coolant in the altitude above  $\sim 100$  km, regulating the thermospheric temperature during active times. The spatial and temporal variability of the EPP heating and radiative cooling follows the variability of the precipitation itself with a time decay and spatial spread dictated by the chemical lifetimes of the key species and dynamics. To determine the impact of EPP on LTI populations and neutral gas heating, direct observations of energetic particle fluxes and densities of neutral species are needed for quantifying the EPP-associated heating rate for a given neutral atmosphere constituent (Sinnhuber et al., 2012).

**Chemical energetics.** The compositional makeup of the LTI has many neutral and ionic forms due to chemical processes resulting from plasma-neutral collisions. The rapid reaction rates of the ionosphere E-region result in a multi-step, chemical process producing predominantly  $\text{NO}^+$  and  $\text{O}_2^+$  ions. That production process and the relative abundance of  $\text{NO}^+$  and  $\text{O}_2^+$  has not been well verified and large uncertainties exist in their relevant reaction rates. An ability to directly measure these abundances in the E-region, along with neutral gas constituents, would advance the plasma chemistry processes that are so important in determining the total electron density and subsequently the energetics of the region. Electrical current magnitudes are directly proportional to electron density concentration as are the electromagnetic heating rates.

The high-latitude E-region is also subject to high plasma flows that induce ion heating and electron heating. These thermal affects are known to influence the plasma chemical reactions that seem to reduce the total electron density in the E-region, as inferred from incoherent scatter radar observations. The extent of these plasma depletions in the E-region caused by chemical energetics could be substantial and directly affect LTI conductivities but will require measurements of significant horizontal extent to realize its impact on electron density concentrations. In addition to the plasma and neutral composition measurements, a measure of ion and electron temperatures will be needed to address its effects on E-region plasma chemistry and LTI energetics. Moreover, long-lived metallic ions in the E-region become susceptible to transport processes resulting in vertically narrow, but highly concentrated, E-layer formations that are sporadic around the globe. The makeup of these sporadic E layers has been associated with  $\text{Fe}^+$  and other heavy trace metallics. These layers have electrical and chemical significance in the E-region and are tied to both electric field and neutral wind influences. However, whereas the metallic ions can be measured by remote sensing instruments (Aikin et al., 2004) these influential parameters controlling the metal distributions are not known. Thus, measures of neutral winds and electric fields will help understand this plasma layering process.

Several trace neutral gasses are chemically active and radiatively significant in determining the overall energetics of the LTI. Species like  $\text{CO}_2$ , NO, and O in their respective excited states are radiatively active that serve to offset neutral gas heating in the LTI. NO production enhancement by energetic precipitating particles has been described above but NO production occurs globally. Measuring the concentration of such a species in the LTI would be required to advance understanding of the chemical processes that form NO and its contribution to the LTI global thermal balance. O is chemically active in the LTI being formed through  $\text{O}_2$  photodissociation around 120 km. At these altitudes, O is a minor





species but becomes the major species in the upper thermosphere as it is transported from its chemical origins to higher altitudes through molecular diffusion and neutral circulations. Observing O at LTI altitudes will help in understanding its generation and help constrain the levels of excited O that may be involved in radiative processes. Anthropogenic trace gases, such as CO<sub>2</sub>, will be important to observe to understand how the lower atmosphere source is distributed in the upper atmosphere. Interestingly, CO<sub>2</sub> acts as a cooling mechanism in the LTI while in the troposphere it is a warming agent. The difference lies in the competing timescales of collisions versus radiative de-excitation. The fewer collisions in the LTI than in the troposphere allows for a photon to be released before being quenched by collisions, thus cooling the region. As CO<sub>2</sub> concentrations increase in the troposphere, so follows the LTI, and it has been recorded that the entire thermosphere is experiencing a cooling trend that follows the CO<sub>2</sub> increase. Observing such a trace gas in the LTI would enable a much more thorough investigation into its role in LTI energetics. The direct observables for neutral radiative cooling rate are neutral species (e.g., NO, CO<sub>2</sub>, and O), neutral temperature.

#### 4.1.3. SO3: Collisional Dynamics

### Determine how collisions between neutral and charged species affect the dynamics of the LTI.

Table 4.4: Approaches for closure of SO3.

| Science Objective                                                                                                       | Scientific needs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SO3.1: Determine how winds are accelerated by plasma motions via ion-neutral collision                                  | Achievement of this objective requires knowledge of the force imposed on neutrals by the plasma. Ampere's force can be determined from the parameters derived in SO1.1, i.e. currents and magnetic fields. The force on the neutral gas due solely to ion collisions can be through the combination of direct observables used in SO1, including neutral winds and density, ion drifts, and $\mathbf{E} \times \mathbf{B}$ drifts. These two forces can be evaluated and compared to test assumptions and approaches. Physics-based models will be invoked to address the other terms in the momentum equation. Mechanical work transferred to the neutral gas by the plasma can also be computed with parameters from SO1. This will be compared with Poynting flux estimates to assess how much of that energy contributes to neutral motion. |
| SO3.2: Discover how the exchange of momentum is manifest across scales by means of lower atmospheric forcing in the LTI | Achievement of this objective relies on determination of the rate of momentum transfer in the LTI via gravity waves. A thorough characterization of the gravity wave field is needed, including wavevector, frequency, and amplitude. Several neutral gas direct observables are needed for this characterization, including density, vector winds, temperature, and composition.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| SO3.3: Determine how collisional processes drive vertical transport and cause composition changes                       | Achievement of this objective requires direct observables of vertical winds and neutral composition. A variety of processes cause both the neutral gas and plasma to be driven vertically, as discussed below. Sorting out which processes are active at a given place and time requires knowledge of essentially all the direct observables discussed to this point.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

Following the previous section, Table 4.4 describes the scientific needs that lead to the derived parameters and direct observables for SO3. The derived parameters are discussed in terms of how they will be determined from accepted theory and direct observables.



**Plasma forcing.** The acceleration of neutral winds by plasma motion results from the complex interplay of plasma-neutral collisions. It has been discussed in Chapter 2 how neutral winds can force ions to move in directions that deviate from the Lorentz force causing current generation. Alternatively, currents can impose momentum on the LTI neutral gas via Ampere's force ( $\mathbf{J} \times \mathbf{B}$ ) (Richmond and Thayer, 2000). The Ampere force term imposes change to the neutral gas motion due to the presence of currents. Inherently, currents involve differential ion and electron motion that collide with neutrals and transfer momentum. This physical path is largely through ion-neutral collisions. Ampere's force describes the momentum pathway from currents to neutral gas motion by capturing the physical processes of ion and neutral momentum exchange by collisions. The velocity differential between the neutrals and ions determines the effectiveness of momentum transfer and its direction. Regions of strong electric fields can enhance local ion motion and drive the LTI neutral gas. However, the large inertia of the LTI neutral gas, relative to the ions, makes the neutral response more sluggish and perpetual. This leads to non-local neutral wind effects that can then drive ion motion (Heelis and Maute, 2020).

The form of Ampere's force applied to the neutral momentum equation is dependent on the assumption that the ion momentum equation is in steady state and governed only by the Lorentz and neutral drag forces. This assumption, along with the relative behavior of neutral wind and ion motion at any one time, requires concurrent, common-volume observations of the LTI gas and plasma. Furthermore, this reality effectively makes resolution of this SO a non-local proposition that must be addressed by an appropriate mission design. To elaborate, the inertia of the neutral gas is so large that plasma forcing occurring upstream and over tens of minutes will manifest in an accelerated flow that will be influential downstream, or nonlocally, to the LTI system.

From an energy perspective, mechanical energy is exchanged when work is done on or by the gas. The work rate of currents on, or by, the gas is determined by the vector relation between neutral gas motion and Ampere's force,  $\mathbf{U} \cdot (\mathbf{J} \times \mathbf{B})$ . If work is done on the gas by Ampere's force then it feeds the bulk kinetic energy of the neutral gas and the dot product is positive. This means that winds are accelerated in the direction of the  $\mathbf{J} \times \mathbf{B}$  force. If work is done by the gas via Ampere's force, i.e., negative dot product, then it extracts bulk kinetic energy from the neutral gas and the ions are driven causing the currents to change. The neutral gas in the LTI is also strongly influenced by hydrodynamic forces, thus, the resultant wind is a combination of hydrodynamic and plasma influences. Evaluating the mechanical energy helps determine where in the LTI the neutral wind is accelerating or decelerating in the presence of currents.

The direct observables for plasma forcing and neutral wind accelerations include ion and electron drifts, electron density, neutral winds, and density and composition of ion and neutral species. Neutral acceleration time scales, caused by ion drag, depend on the electron density concentration and the persistence of the plasma motion.

**Atmospheric forcing.** Determination of how atmospheric forcing by gravity waves from below impacts the LTI requires knowledge of both the forcing influence (primarily upwardly propagating gravity waves) and signatures of the forcing in the LTI. The first-order, critical parameters for gravity waves are the wave amplitudes, horizontal and vertical wavelengths, propagation direction, and intrinsic frequency. Ideally, simultaneous and global knowledge of all these quantities is needed, but currently this is not possible at the altitudes of interest (100 – 200 km), and for the scales of interest, notably for meso- $\beta$  (200-20 km) and meso- $\gamma$  (2–20 km) gravity waves. An *in situ* mission that samples the LTI along-track in the 100-200 km altitude range can measure horizontal wavelengths of gravity waves, but not vertical wavelengths or intrinsic frequencies. However, knowledge of several neutral gas properties along a path of an *in situ* mission through a perturbed atmosphere can be used to extract many gravity wave parameters. As an example, amplitudes and relative phases of oscillatory perturbations in density,



winds, temperature, and composition measured by the DE-2 mission can be analyzed to yield amplitudes of vertical gas speed and displacement amplitudes, wave frequency and component phases, as well as upward energy flux (Innis and Conde, 2002). Because only two components of gas velocity were measured by the DE-2 mission, wavelengths could not be unambiguously determined. More recent theoretical work (Vadas and Nicolls, 2012) has gone further to show how including knowledge of the full 3D winds can be used in the compressible, dissipative regime to extract the full wave vector as well.

Knowledge of these gravity wave parameters allows determination of their momentum flux. As the waves propagate and dissipate, their parameters change, allowing the momentum flux to be absorbed in the LTI medium, changing the mean flow. Along-track measurements from the DE-2 mission have been used for the calculation of gravity wave wavelengths (Earle et al., 2008); they further showed an anti-correlation of the neutral and plasma densities consistent with the hypothesis that gravity waves are responsible for the observed plasma perturbations, and found the relative fluctuations between plasma and neutral densities in agreement with theoretical expectations for gravity wave-driven perturbations. DE2 measurements have also demonstrated that the presence of such waves might be quite common at midlatitudes below altitudes of about 300 km. Similarly, measurements from the GOCE have been used to demonstrate propagation characteristics of gravity waves as well as an inverse relationship between gravity wave activity and increasing solar flux (Garcia et al., 2014, 2016).

**Vertical transport.** Collisional dynamics is central to understanding the distribution of composition in the thermosphere. Thus far, modeling has been the only means to test the various processes responsible for composition transport and change, with little observational validation. Turbulent mixing, molecular diffusion, and wind advection are the fundamental, but very different, dynamical processes involved in affecting thermosphere composition change (Burns et al., 1989). These processes compete within the domain of the LTI and have very different temporal, spatial, and altitudinal influences. Altitude is particularly consequential when assessing which process is influential in changing thermosphere composition. This is because the time response of the various processes to cause composition change is strongly altitude dependent.

The turbopause altitude (90-110 km) is variable and poorly known across the globe because it is influenced by the local and global terrestrial wave activity that has not been adequately specified. The long-held presumption that composition changes with altitude transition directly from eddy to molecular diffusion is not in line with the dynamic activity known to take place in the LTI. However, it has been difficult to observe concurrently dynamical and thermal processes in the lower thermosphere where composition is strongly influenced. How dynamical processes can influence thermosphere composition is through large-scale circulation cells (1,000 – 10,000 km) that induce mixing by advection of air masses with different fractions of compositional makeup. This dynamically induced mass-exchange process changes how individual species distribute themselves horizontally and vertically in ways that deviate from diffusive behavior. Clear indications of this occurring can be seen in inert species such as helium whose maximum concentration in the upper thermosphere occurs on the nightside completely out of sync with thermally influenced diffusive equilibrium (Sutton et al., 2015). Lower thermosphere circulation is driven by internal forces within the region but also by tidal dissipation and geomagnetic forcing. There is evidence that the dynamical mixing process is sufficiently efficient below 200 km to compete with molecular diffusion when eddy diffusion is no longer effective, i.e., between 100-200 km altitude, which is often described as the thermospheric spoon concept (Fuller-Rowell, 1998; Jones Jr. et al., 2018).

Most of our observations of neutral composition below 200 km come from the early days of exploratory missions, such as Atmospheric Explorer, with limited capability and coverage. Rocket measurements and other ground-based resources have provided restricted views of composition behavior in the LTI



over the years. Mass density measurements in the upper thermosphere by accelerometers over the past two decades have opened new inquiries into how the cumulative effect of composition change caused by geomagnetic and terrestrial activity affects the observed behavior in total mass density (Liu et al., 2014b; Prölss, 2011). However, as mentioned in Chapter 2, the composition changes that constitute mass density change are not well known. Far ultraviolet (FUV) imaging of the Earth limb and disk has enabled exploration of composition behavior for O and N<sub>2</sub>. Although informative, these missions lack the necessary complementary 3D temperature and wind observations to explore the mechanisms that produce composition change. Furthermore, understanding the behavior of a wide variety of species in the thermosphere, ranging in molar mass from 2 AMU to 80 AMU across minor to major concentrations, requires knowledge of the dynamical processes that can cause composition change.

## 4.2. Observational needs

In determining the scientific needs of a mission designed to resolve the science objectives outlined in Chapter 3, we must ultimately identify the **observational needs** to capture the direct observables. Since the LTI is highly variable in space and time due to changing geophysical conditions, this environment needs to be characterized statistically, requiring sampling across many different regions and under different geophysical conditions: magnetic/solar local times, latitudes, seasons, and solar/geomagnetic/terrestrial activity. Thus, over its lifetime, an *in situ* mission will need to provide global coverage of LTI observations in latitude and local time, within the altitude ranges of interest, and under various solar/geomagnetic/terrestrial activity conditions. These needs must be considered when developing aspects of any future mission concept, such as the total mission lifetime and the orbital precession rates, as will be discussed further in Chapter 5.

As a preliminary step, this section provides additional information on scales, regions of interest, and geophysical variability relevant to some of the derived parameters in Table 4.1 and associated LTI processes and phenomena, from which direct observational requirements can then be derived. This section is organized by the science objectives with a focus on the observational needs to close those objectives:

Section 4.2.1 documents the spatial and temporal scales of key LTI phenomena and processes, and some of the associated derived parameters.

Section 4.2.2 describes the latitudinal, local time, and altitudinal regions of interest for these phenomena.

### 4.2.1. Spatial and temporal scales of LTI phenomena

The relevant scales of key LTI processes and phenomena (expressed by structure in an observed property or process) are shown in Figure 4.1. In this figure, ovals approximate the range of relevant scales. Many of the associated derived parameters require the combination of several direct observables at small spatial (< 0.1 km) and temporal scales (< 0.1 s). This fact establishes the need for concurrent and common-volume measurements, driving the approach towards *in situ* sensors.

Owing to strong gradients and rapid responses in almost every property of the LTI, spatial and temporal scales for the same process or phenomenon are known to vary greatly from meters to tens of kilometers and sub-seconds to hundreds of seconds. Whereas the smallest scales drive the sampling needs, the largest scales contribute to the coverage needs of a mission. Targeted scales (“resolution”) for the processes and phenomena under investigation are drawn on the figure to indicate drivers for sampling and coverage needs. Also indicated are the present-day approximate resolution limits of General



Circulation Models (GCMs) in the LTI region. The model limits indicate that many LTI processes and phenomena are not adequately captured by existing modeling capabilities and tend to occur on sub-grid scales. Observations of higher resolution are therefore needed to advance their general understanding and their representation in predictive models of the LTI.

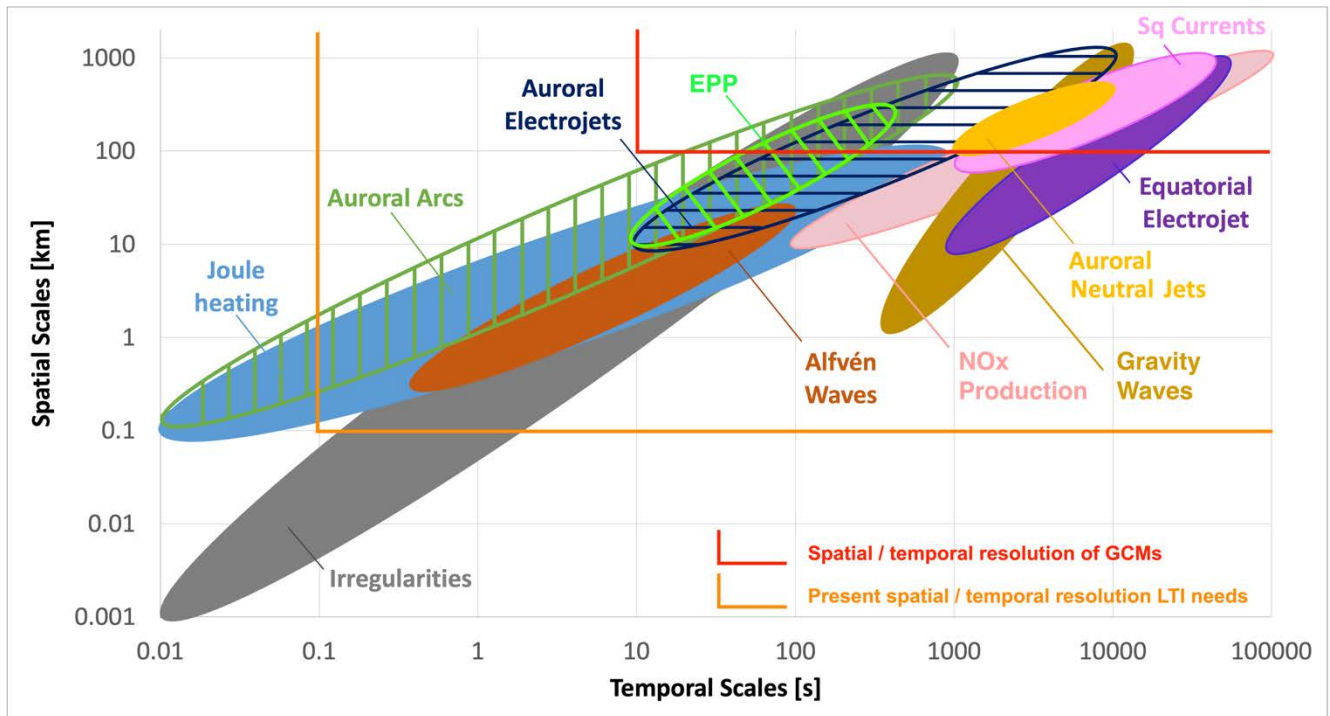


Figure 4.1: Spatial and temporal scales of key phenomena and processes in the LTI (credit: DUTH – T. Sarris).

In the following, scales for key derived parameters are addressed. Also, LTI phenomena are provided to give context to the derived parameters. The scales for the derived parameters represent the required scales in the Science Strategy Matrix presented in Appendix B.

**Field Aligned Currents (FACs) and auroral arcs.** The scales of FACs studied by (Johansson et al., 2007) were reported to be on the order of 5–10 km, whereas (Gjerloev et al., 2011) reported field-aligned currents with scale-sizes greater than 200 km (mapped to ionospheric altitudes), with lifetimes of the order of a minute, consistent with the result of a smaller study by (Le et al., 2009). Using the Swarm satellites and ground-based cameras, FAC sheets associated with single and multiple quasi-static arcs range from tens to hundreds of kilometers in width, as documented by (Wu et al., 2017). Downward current regions that were dominated by current filaments of 10–20 km scale size in the ionosphere have also been reported (Wright et al., 2008), whereas the upward current region has been found to be 1–4 times narrower than the downward current-region, or as small as 2.5 km. FAC structures are known to exist down to scales as small as 100 m (Stasiewicz and Potemra, 1998). As Karlsson et al. (2020) note, the disparate results of these studies, as well as the unclear exact relation to discrete auroral arcs means that there is a need for more detailed statistical studies of spatial properties of FAC structures clearly identified with discrete arcs.

**Auroral electrojets.** In the high-latitude ionosphere below ~125 km, where the ion drifts are significantly impeded by collisions with the neutral gas, the  $\mathbf{E} \times \mathbf{B}$  motion of the electrons drives a system of **enhanced** Hall currents referred to as the auroral electrojets. The width of these enhanced Hall currents is governed by the characteristics of the driving DC electric fields. The width of auroral electrojets span



from tens of km to over 1000 km with time scales of tens of seconds to hours (e.g., Vennerstrom and Moretto, 2013). The length, aligned along the auroral oval, can span local times from 14 – 9 local time, and a high-inclination mission should ensure that the orbit intersects the auroral electrojet.

**Equatorial electrojet.** At the equator, the equatorial electrojet is another **enhanced** Hall current that flows during the daytime, with the most intense current typically extending in longitude over scales of 1000's of km corresponding to several hours of local time. By contrast the latitudinal width of the equatorial electrojet is on the order of 1-2 degrees or 100-200 km. The altitude thickness of the equatorial electrojet corresponds to 10-15 km, confined to the vertical extent of enhanced Cowling conductivity (Pfaff et al., 1997; Richmond, 1973).

**$S_q$  currents.** Solar quiet or  $S_q$  currents associated with the atmospheric dynamo can be quite large as they typically encompass the entire dayside, middle-latitude ionosphere. Smaller scale currents could exist associated with localized neutral winds. Hence, the horizontal scales of  $S_q$  currents might vary from 100 km to 5000 km with time scales from tens of minutes to hours (Yamazaki and Maute, 2017).  $S_q$  currents obtained from supporting modeling and complementary ground-based observations are helpful to interpret observations of 3D electric currents.

The direct observables for 3D electric current are perturbation magnetic fields, electric fields, neutral winds, electron and ion density, and electron and ion velocity. These findings indicate that 2D electric currents must be acquired at spatial scales from 1 to 1000s kilometers and a temporal scale of tens of seconds to one hour. The third dimension concerns the altitude range where each current system peaks. The horizontal Pedersen component of the current in all current systems peaks near ~125 km altitude, while the other horizontal component of the current, i.e., the Hall current, peaks near 110 km. Note that the Hall current equals the Pedersen current at the altitude where the Pedersen current peaks. This can be used to extrapolate Hall current behavior downward when observing at Pedersen peak altitudes.

**Ionospheric conductivity.** The 3D variation of ionospheric conductivity is another key component to understanding ionospheric electrodynamics. Direct measures of conductivity are difficult to establish due to inadequate knowledge of the mobility tensor and the associated ion-neutral and electron-neutral collision frequencies, as discussed in Chapter 2. The required observational scales are similar to those for the current systems with 2D variations to be resolved down to spatial scales of 1 km and a temporal scale of 10s of seconds. The horizontal Pedersen and Hall conductivities are derived parameters based on direct observables required for current and electric fields in the neutral wind reference frame. Parallel conductivity is a derived parameter reliant on direct observations of electron and neutral densities. Vertical variations resolved down to ~1 km are of interest.

**Neutral wind dynamo.** As discussed in Chapter 2, atmospheric currents are set up by neutral gas motions or winds which, via ion-neutral collisions, drag the ionosphere plasma across the ambient magnetic field, producing a current density,  $\mathbf{J}$ , as the ambient ions are collision dominated within the lower ionosphere (90–140 km) and the electrons are not. Rocket measurements (Pfaff et al., 2020) as well as theoretical work and modeling (Jin et al., 2008; Yamazaki and Maute, 2017) have revealed how the winds and currents both change markedly with 1 km vertical gradients, in some cases showing spiral patterns indicative of tidal forcing. Recent measurements of horizontal winds in the LTI gathered by the ICON satellite (England et al., 2022) also show sharp gradients with altitude as well as variations with latitude corresponding to 1000 kms or more. The sharp vertical gradients can also create distinct layering in the plasma density that are known (from sounding rocket and radar measurements) to vary within 1 km vertical spacing.



The direct observables for 3D neutral wind-driven electric current, which is determined with the dynamo equation,  $\mathbf{J} = \underline{\sigma} \cdot (\mathbf{E} + \mathbf{U} \times \mathbf{B})$ , include all direct observables required for currents and electric fields in the reference frame of the neutral winds. The 3D neutral winds must be resolved down to horizontal spatial scales of  $\sim 10$  km and vertical scales of  $\sim 1$  km, both with a temporal scale of tens of seconds.

**Plasma motion.** Plasma convection affects the flow and closure of  $\mathbf{J}$ , energy dissipation by Joule heating  $\mathbf{J} \cdot \mathbf{E}'$ ; and driving thermospheric winds via the ion drag force. At the largest scales, i.e., 10,000 kms or more, plasma motion is structured at high latitudes according to the well-known two-cell pattern for a southward interplanetary magnetic field (IMF). This cell pattern breaks into smaller cells for northward IMF (Heelis, 1984) with horizontal scales of 1000 km. Intense plasma convection jets occur on scales of tens to hundreds of km in association with sub-auroral ion drifts (Anderson et al., 2001), “STEVE” events (MacDonald et al., 2018), FAC closure (Archer et al., 2017) and auroral arcs. These meso-scale convection structures map effectively along B-field field lines and can affect the ionosphere at all altitudes. Furthermore, they can persist on time scales of tens of minutes, which is the time scale required to “spin up” the neutral wind through plasma-neutral coupling (Walterscheid and Lyons, 1992).

**Plasma irregularities.** Plasma irregularities affect processes relevant to current closure and energy dissipation. The LTI is replete with plasma irregularities resulting from wave processes associated with plasma gradients and local plasma instabilities. One source of wave-related plasma irregularities unique to the lower ionosphere is intense, meter-scale two-stream waves associated with the electrojets created when the relative ion-electron drift velocity exceeds roughly the acoustic velocity. For the auroral zone case, layers of such intense waves may extend to altitudes of 125 km or higher depending on the strength of the DC electric field whereas two-stream waves in the equatorial electrojet exist generally below 110 km (Pfaff, 2012). These meter-scale electrostatic waves can be an important source of plasma heating (St.-Maurice et al., 1981) with amplitudes sometimes as large as the driving DC electric fields (Pfaff et al., 1992). Longer wavelength irregularities include gradient drift waves, with broad spectra that include long wavelengths of a few to tens of km that cascade to meter scales (Kudeki et al., 1987) as well as Kelvin-Helmholtz waves associated with neutral wind shears, with scales of tens of km (Larsen, 2000; Pfaff et al., 2005). Larger scale structures of hundreds of km are believed to also exist in the LTI, such as those associated with MSTIDs with E-region origins (Kelley et al., 2011).

Typical time scales for LTI plasma instabilities may range from msec (for two-stream waves) to minutes or tens of minutes for the larger K-H and MSTID driven instabilities. However, particular features such as their temporal variability on minute scales, are still unsolved. They are most likely unidentified consequences of the interplay between plasma physics, photochemistry, and lower atmospheric dynamics. In support of this, low-latitude 150-km altitude irregularity models need to account for gravity wave modulation and forcing (Reyes et al., 2020), which they currently do not.

**Joule and frictional heating rates.** The spatial and temporal distribution of electromagnetic heating is known to occur over large areas of the auroral zone/polar cap spanning all local times during geomagnetically active periods. Of known importance but poorly observed are the characteristic scale sizes of the electric field in the neutral reference frame that contribute to the heating. The contribution of small-scale variability or structuring of the electric field, e.g., on scales less than 10 km, is known to contribute significantly to electromagnetic heating (Billett et al., 2022; Codrescu et al., 1995). Electric field structures with scale sizes of roughly 1 km to 300 km can make a comparable contribution to Joule heating (Matsuo et al., 2003; Matsuo and Richmond, 2008; Richmond, 2021), whereas the variability of the large-scale field about the climatological average also affects the calculation of Joule heating. Previous studies based on DE-2 data (Matsuo et al., 2003; Matsuo and Richmond, 2008) have shown that this variability in the electric field can be comparable in magnitude or can even exceed the average. The effect of small-scale electric field structures does not average to zero since Joule heating is proportional to the square of the electric field.



It must be emphasized that electromagnetic heating requires calculation in the frame of the neutral wind  $\mathbf{U}$ , which can vary on scales as small as tens of kilometers horizontally. Moreover,  $\mathbf{U}$  can change on time scales of tens of minutes to hours. Therefore, measurements of electrodynamical parameters in the LTI must include simultaneous and collocated measurements of the neutral wind. The electromagnetic heating scales for the derived parameters of Poynting flux and Joule and frictional heating are all similar.

**Energetic Particle Precipitation.** Variations in EPP are known to occur on spatial scales from  $\sim 100$  km to as small as  $\sim 1$  km (Akasofu et al., 2010; Angelopoulos et al., 2008; Miles et al., 2018); a subset of these EPP scales from 50 to 100 km have been associated with timescales on the order of a few minutes (Boudouridis and Spence, 2007). Some specific EPP types typically vary on the order of tens of seconds (Nishimura et al., 2020). Auroral arcs are known to be important conduits for energy input and are visible manifestations of EPP, FACs, and large plasma flows. Auroral precipitation, a specific form of EPP, is structured on scales from  $\sim 100$  m to 500 kms (Borovsky and Suszcynsky, 1993; Maggs and Davis, 1968; Newell et al., 1996). Quiet discrete auroral arcs, the east-west elongated forms that dominate the pre-midnight and midnight sectors, are known to extend over hundreds to thousands of kilometers of longitude (Davis, 1978), with typical widths of a few tens of kilometers (Aikio et al., 2002; Jiang et al., 2015; Knudsen et al., 2001; Nishimura et al., 2011; Wu et al., 2017). Meso-scale structures of the order of 1 km are also observed (Partamies et al., 2010). Smaller scales are seen during active periods such as substorm breakup (Kataoka et al., 2021), where timescales of 0.1–10 s are common. The separation distance between adjacent arcs has been found to be between 30–50 km, and there are some arcs separated by more than 100 km. The lifetimes of quiet arcs can be from 2 to 5 minutes (Moen et al., 1994) to tens of minutes (Knudsen et al., 2001) or even span several hours (Galperin, 2002; Lessard et al., 2007).

**E-region plasma structure.** The E-region plasma is most structured at high latitudes due to production by particle precipitation and losses by fast chemical recombination processes. The formation of the ionospheric constituents within the LTI involves several multi-step, chemical processes producing predominantly  $\text{NO}^+$  and  $\text{O}_2^+$  ions. Chemical time constants to produce these ion species are much faster than transport, thus the effects of EPP on the LTI electrodynamics are localized but strongly altitude dependent. The LTI vertical structure of plasma density span from 100–180 km altitude and determine Pedersen and Hall conductivities. EPP-produced E-region plasma has similar spatial and temporal scales to auroral EPP. The chemical pathways to produce E-region plasma is fundamental to estimating the efficiency of electromagnetic heating of the neutral gas.

**Alfvén waves.** Any changes to the otherwise quasi-static FAC and convection structures propagate along magnetic field lines as Alfvén waves. For timescales longer than 10 s, Alfvén wavelengths are much longer than the thickness of the ionosphere and the associated electric fields can be treated as quasi-static, meaning they map along  $\mathbf{B}$  (Knudsen et al., 1992). At higher frequencies, wave behaviors become important; these include standing wave patterns and ducting over large distances (Greifinger and Greifinger, 1973). When the Alfvén wavelength perpendicular to  $\mathbf{B}$  reaches kilometer scales or less, wave-particle interactions lead to electron acceleration, and breaking and damping become important (Goertz and Boswell, 1979; Lysak and Song, 2003; Seyler et al., 1995). For waves and quasi-static fields with perpendicular spatial scales of order 10 km and less, sampling from LEO leads to Doppler shifting which becomes mixed with inherently time-varying fields. While it is known that highly structured and dynamically varying magnetic and electric fields (Alfvén waves) can direct significant amounts of Poynting flux into the ionosphere. Furthermore, the few observations of Alfvén waves below 250 km from sounding rocket flights have confirmed that resonant Alfvén waves exist in the high latitude LTI with significant Poynting Flux (Akbari et al., 2022; Cohen et al., 2013). Understanding the altitude distribution of the dissipation of these waves is central to LTI energetics at high latitudes.





**Auroral neutral jets.** Associated with steady magnetosphere-driven convection and relatively long-lasting conductivity enhancements in the LTI associated with auroral precipitation, localized neutral wind systems may be set up at altitudes near 110-200 km, referred to as auroral neutral jets (Brinkman et al., 1995). Such localized winds can exist for time scales on the order of an hour or longer, with typical widths of hundreds of km and lengths corresponding to long-lived auroral forms on horizontal scales of several hundred to several thousand kilometers as observed by vapor trails released by sounding rockets (Larsen et al., 2022).

Direct observables include ion and electron motion, neutral winds, electron density, density and composition of ion and neutral species. Neutral acceleration time scales strongly depend on the electron density concentration and the persistence of the plasma motion. Time scales of 10 minutes are expected to initiate a response in the neutral gas. Plasma scales needed are similar to those for plasma convection, while neutral winds and composition can be on the order of 10 kilometers or more.

**Mechanical energy transfer.** Sustained plasma motion has a direct influence on the bulk kinetic energy of the neutral gas. The persistence of the plasma motion and density structure is important because the neutral gas has significant inertia in the LTI and the plasma forcing will need to persist to influence the gas bulk kinetic energy. Thus, ten minutes or more of sustained plasma forcing is necessary to influence the neutral flow. This “ion drag” force can be separated into Pedersen and Hall drag forces. The Pedersen drag force is much more important than the Hall drag force in the LTI. Above 130 km, the Pedersen drag force varies with height in a manner similar to the electron density. The effects are global with the strongest influence occurring at high latitudes. Thus, concurrent and collocated measurements of the neutral wind, plasma motion and density, along with conductivity estimates are needed to resolve the mechanical energy transfer to the neutral gas.

**Gravity waves.** Spatial scales of gravity waves are classified as meso- $\alpha$  (2000-200 km), meso- $\beta$  (200-20 km) and meso- $\gamma$  range (2–20 km). The relevant wavelengths of gravity waves influencing the LTI range from tens to thousands of km (Fritts and Lund, 2011), spanning all three classification ranges of gravity waves with corresponding time scales on the order of tens to hundreds of minutes. Earle et al. (2008) demonstrated gravity wave observations with *in situ* measurements using DE-2 data and showed a persistent wavelength of about 200 km along the track of the satellite. Through their analysis they demonstrated the need to measure winds with a cadence of 3 seconds (or shorter), providing measurements every 25 km intervals along the track.

As discussed in Chapter 2, the upwards propagation of gravity waves is influenced by both the background atmosphere and by nonlinear interaction between waves. Their amplitudes grow exponentially with altitude due to the decrease in atmospheric density which consequently effects both the spatial and temporal scales of the waves. For example, these large wave amplitudes lead to wave breaking and hence momentum deposition, often occurring in the LTI, exhibiting non-linear turbulent-like cascades. Direct evidence of neutral structuring of the LTI measured on a low altitude U.S. DoD satellite shows high time resolution neutral density structuring of 10's of km in data from the Streak instrument (Clemmons et al., 2008) gathered near 150 km. Neutral structures with scale sizes less than a kilometer have also been reported in the LTI in the Streak data.

**Composition changes.** Important compositional changes associated with vertical processes in the LTI of unknown origin have been observed in images of the O/N<sub>2</sub> ratio from auroral imagers on the DE-1 and Global Geospace Science Polar satellites as well as in neutral density variations reported by the CHAMP satellite (see previous discussion associated with Figure 3.8). In general, the few satellite observations suggest large-scale phenomena corresponding to 100's of km to 1000's of km in horizontal scales. Here, the vertical scales are unknown. However, sounding rocket data gathered in conjunction with ground-based photometers (Hecht et al., 2018) suggest vertical winds in the LTI region possibly



associated with turbulent diffusion which may account for the rapid compositional changes observed in that experiment, which also included a high-time resolution imager on the rocket. Such data suggest temporal changes on the order of 10's of seconds with spatial variations on the order of 10's of km. Processes responsible for composition change involve large-scale wind circulations, temperature distributions, and eddy mixing rates on the horizontal order of 1000 - 10,000 kms.

**Sporadic E layers.** The long-lived metallic ions in the E-region become susceptible to transport processes resulting in vertically narrow, but highly concentrated, E-layer formations that are a few kilometers thick and sporadic around the globe (see Figure 2.28). They occur most frequently at mid-latitudes but have also been observed at low (Arras et al., 2022) and high latitudes (Kirkwood and Nilsson, 2000). The compositional makeup of sporadic E layers has been associated with Fe<sup>+</sup> and other trace metallics. The mechanisms responsible for their formation can be resolved with local measurements of vertical shears in the horizontal wind, electric fields, and ion composition. The thin layers descend in altitude from 200 – 90 km altitude and have horizontal extents of 100 km or more and persist for minutes to hours.

**Traveling Ionospheric Disturbances.** Often, atmospheric gravity waves are associated with ionospheric signatures known as a travelling ionospheric disturbance (TIDs) consisting of a horizontal modulation of the ionospheric electron density propagating over large distances. Such ionospheric structures are easily observed in ground-based GPS data. The coupling of the neutral and ionized gases in the LTI in which plasma structuring serves as a proxy for the neutral gas variations due to collisions between the neutral and ionized gases and are at the core of the collisional physics described in this report. Accordingly, the significant advances in research with respect to TIDs can provide important information regarding the scales and time constants of the upper atmosphere. TIDs vary in wavelength (hundreds to thousands of km), period (tens of minutes to hours), and wave speed (~100 m/s to over 800 m/s).

#### **4.2.2. Regions of Focus of LTI phenomena**

An overview of altitude and latitude ranges where key LTI processes and phenomena tend to arise is provided in Figure 4.2. The ovals (left) provide rough spatial extents of their occurrence, with the most pronounced variations commonly observed in altitude (right). The exact spatial distribution of these phenomena is not fully known due to limited observational data. Below, we propose latitudinal focus regions to which the targeted LTI phenomena can be assigned.

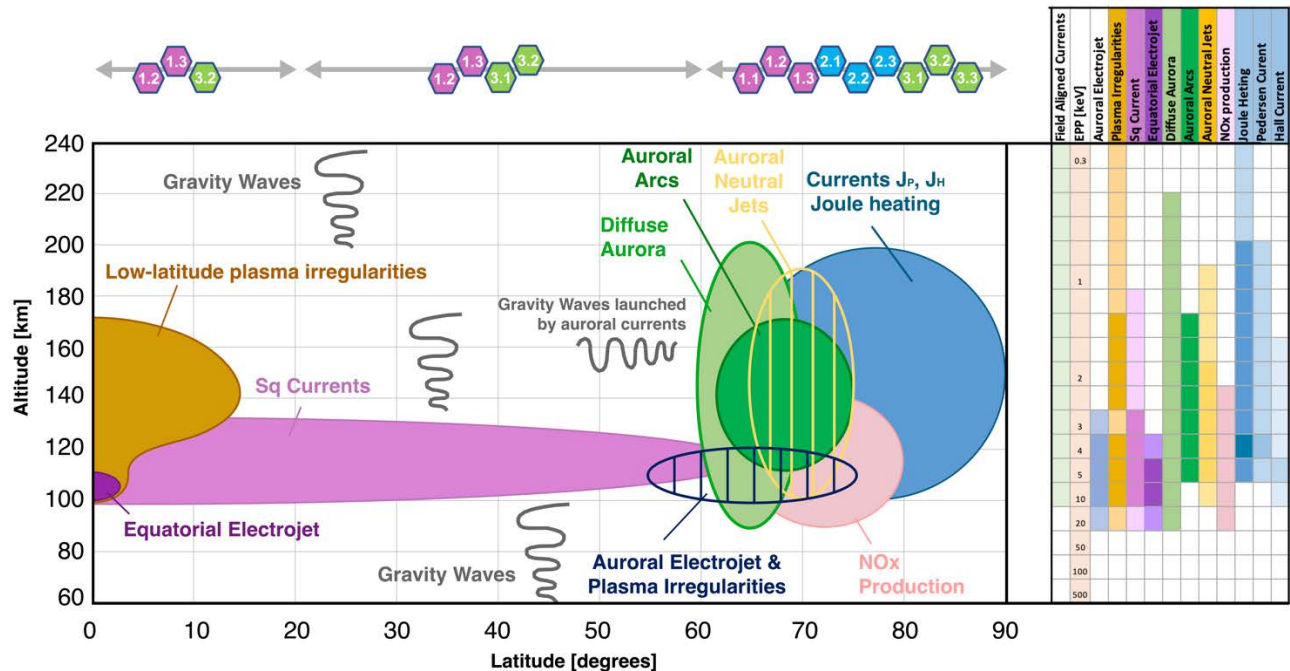


Figure 4.2: Approximate altitude and magnetic latitude extents of key phenomena and processes in the LTI. Latitude bands (regions of focus) associated with the SOs are on top (credit: DUTH – T. Sarris).

#### 4.2.2.1. High-latitude Regions of Focus

**Field Aligned Currents.** Large-scale regions of field-aligned currents, including region 0 (R0), region 1 (R1), and region 2 (R2) current systems, can cover a range of 60° to 80° magnetic latitude depending on geomagnetic activity. The latitudinal widths of the R1 & R2 currents increase by 20-30% during active times and shift equatorward by several (2-5°) degrees (Iijima and Potemra, 1976). Small-scale FACs (horizontally spanning hundreds of meters to kilometers) are embedded in the large-scale FACs (Rother et al., 2007). The FAC flow does not change much above 400 km (apart from magnetic field geometry), but below ~200 km, horizontal currents change the altitude variation of the FAC.

**Auroral electrojet.** The auroral electrojet is part of the auroral E-region current system, driven by coupling to the magnetosphere (Milan et al., 2017), and it is thus an important link in atmosphere-magnetosphere transfer of energy. The auroral electrojet is an **enhanced** Hall current that flows predominantly in the east-west direction at altitudes where the Hall conductivity is largest, at about 105-120 km. The auroral electrojet occurs between 60° and 75° magnetic latitude, extending to lower latitudes (<55°) during geomagnetic activity, and is prominent from dusk through midnight and into the dawn local time sector.

**Cusp.** The cusps are funnel-shaped magnetic field regions in the northern and southern hemispheres on the dayside of the earth directly connected with the solar wind. As such, they permit direct entry of solar wind particles and energy into the high-latitude ionosphere/upper atmosphere. The cusp aurora thus represents a distinct region of the near-earth plasma at high latitudes. Indeed, the ion diffuse aurora peaks in the cusp (Newell et al., 2009). Observations at ionospheric altitudes in the cusp (Pfaff, 2012) show highly correlated electric and magnetic field signatures, field-aligned currents and downward directed Poynting Flux. Dynamics Explorer-2 observations of electric and magnetic fields suggest that the strongest low-frequency structures are present in the cusp (Heppner et al., 1993). The FACs in the cusp are believed to close in a much more complex manner than in the auroral oval and

indeed show complicated small-scale structures in satellite data, Measurements in the LTI are needed to understand how the FACs and soft energy particles associated with the solar wind deposit their energy and close in the cusp regions.

**Pedersen and Hall currents and conductivities.** A practical consequence of collisions between ions and neutrals in the bottommost LTI, i.e. the ionospheric E-region, is separation of ion and electron drifts and peaks in Pedersen and Hall conductivities. Whereas the transition in the behavior of ion motion and the resulting electric currents is theoretically well-described, it has not been systematically explored to date, with the exception of a few sounding rocket missions (Sangalli et al., 2009). Based on this small set of *in situ* measurements and ion-neutral collision cross sections derived from laboratory measurements, it is inferred that Pedersen and Hall currents occur primarily at altitudes between 100 and 200 km with the daytime Hall current peak in general between 105 - 110 km, and the Pedersen current peak around 125 km. While these currents and conductivities are always present at high-, mid- and low latitudes, they are enhanced and highly variable in the auroral oval.

**Electromagnetic heating.** Even though electromagnetic heating and its thermospheric effects are enhanced during strong M-I coupling, it is always present at high latitudes and impacting the LTI structure. Most of the electromagnetic heating occurs below approximately 150 km (Figure 4.3b) where the Pedersen conductivity peaks, with only 18-34% of electromagnetic heating deposited above 150 km (Deng et al., 2011; Huang et al., 2012). However, the thermosphere and ionosphere response to the electromagnetic heating above approximately 150 km can be significant due to the smaller mass density (Brinkman et al., 2016). Electromagnetic heating associated with M-I coupling predominantly occurs from early evening to late morning in the 60° to 75° magnetic latitude region associated with the auroral oval (Figure 4.3b), but also in the dayside cusp at higher magnetic latitudes.

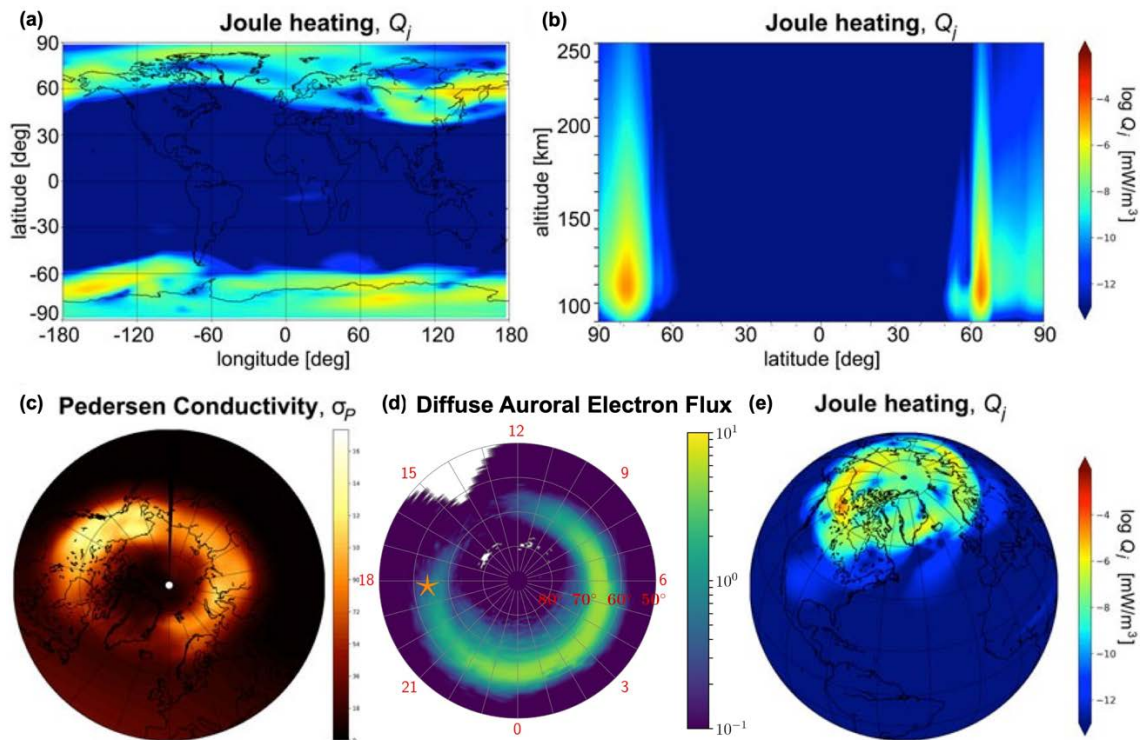


Figure 4.3: (Top), distribution of Joule Heating (model results) as a function of latitude and longitude (left), as well as altitude and latitude (right). Bottom, polar distributions of Pedersen conductivity at 120 km (left), diffuse auroral electron energy flux [ $\text{mW/m}^2$ ] from OVATION-Prime model for 20 Jan, 2016 at 18:08:11 UT (Palmroth et al., 2021) (middle), and Joule heating at 120 km (right) according to model calculations (adapted from Sarris et al., 2023b).





**Auroral arcs.** Auroral arcs represent the lower-energy part of EPP, and are most frequently observed in the afternoon to pre-midnight sector of the main auroral oval between 62-75° magnetic latitude (Gillies et al., 2014), but also are prevalent in the pre-noon sector (Newell et al., 2009). Most arc observations show the highest electron fluxes at electron energies of a few keV, and therefore a peak emission height of about 110 – 170 km (Karlsson et al., 2020). Auroral arcs are typically found at or close to the boundary of oppositely directed FACs. Auroral arcs lasting for several minutes are associated with a region of enhanced electric field with approximately four times the width of the arc (mesoscale ~1-10s km) (Aikio et al., 2002).

**Energetic Electron Precipitation.** Energetic particle precipitation (EPP) is an important energy input into the LTI at high latitudes and its deposition altitude depends on the energy of the precipitating particles (Mironova et al., 2015). A typical peak energy of auroral electron precipitation is a few keV, which deposits energy at the height of about 110-130 km. Although the higher-energy (several tens of keV and above) radiation belt electrons reach LTI heights, their fluxes are low, resulting in low total energy input. The latitude distribution of EPP follows the auroral oval latitudes at all local times but the highest energy inputs are often found from the midnight to late morning hours.

**Auroral neutral jets.** Associated with steady magnetospheric-driven convection and relatively long-lasting conductivity enhancements in the LTI associated with auroral precipitation, localized neutral wind systems may be set up at altitudes between 110-190 km, referred to as auroral neutral jets (Brinkman et al., 1995). The most definitive examples of these jets are from vapor trails released from sounding rockets (Larsen et al., 2022). The vapor trails have revealed intense accelerated winds near 130 km altitude with speeds of over 300 m/s in the auroral oval. Their local time and latitude extent are not known.

**Vertical transport.** As described in the previous section, neutral composition change is influenced the most by vertical transport. However, turbulent mixing, molecular diffusion, and neutral circulation are all capable of causing vertical transport but produce different outcomes. The high-latitude region produces dramatic changes in the temperature, winds, and turbulent mixing that modify composition in ways unresolved. Strong regions of heating will enhance the molecular diffusion of neutral species, while large-scale accelerations of the winds produce divergent and convergent behavior that result in strong vertical motion and mixing of composition. The regions influenced are the cusp, polar cap, auroral region and sub-auroral region. Each domain can produce different composition behavior as the competing processes differ in these regions. Latitudinal and local time extent is strongly influenced by the onset time of a geomagnetic storm.

**Nitric oxide production.** The chemical pathways for NO production are local to the LTI where states of atomic nitrogen interacting with molecular oxygen undergo exothermic reactions to produce NO. Interestingly, LTI NO can be transported downward where it impacts ozone concentrations in the middle atmosphere – particularly at high latitudes. Vibrationally excited nitric oxide is produced by direct impact of NO with aurorally precipitating electrons leading to enhanced cooling of the neutral gas via infrared emissions. The emission occurs globally and is influenced by the temperature, atomic oxygen concentration, and EPP. During geomagnetic activity, NO emissions spike in the auroral region. Direct observables needed to understand NO production are NO and atomic oxygen concentrations, neutral temperature, and EPP energies in the region of the auroral oval. NO emissions can then be modelled to ascertain the amount of cooling produced through this emission process. The latitudinal and local time extent of NO production and emission are similar to energetic electron precipitation regions which are strongly dependent on levels of geomagnetic activity.

**High-latitude plasma irregularities.** The high-latitude ionosphere is the seat of numerous plasma instabilities which create electrostatic irregularities throughout the LTI. The most intense irregularities



in the high-latitude LTI are the meter-scale two-stream waves which are generated whenever the relative electron-ion speed exceeds approximately the acoustic velocity. At high latitudes, this is a very common mechanism since typical sound speeds are on the order of 400-600 m/s and  $\mathbf{E} \times \mathbf{B}$  driven electrons associated with magnetospheric electric fields that “map” into the high-latitude LTI “stream” through the collisionally-bound ions at speeds that are often in excess of 1000 km. In the auroral electrojet, such supersonic, relative ion-electron drifts occur at altitudes between 95 and 125 km where the enhanced Hall current is dominant. For typical magnetic field strengths at these altitudes within the high-latitude regions, the instability threshold is met for DC electric fields of approximately 20–25 mV/m. Other processes at high latitudes can produce irregularities such as those associated with plasma density gradients at the edge of the auroral oval, structured polar cap patches, and plasma in the cusp region.

#### 4.2.2.2. Middle-latitude Regions of Focus

**Sq currents.** Solar quiet or *Sq* currents are driven through the ionospheric wind dynamo and dominate at low and middle magnetic latitudes up to  $\sim 65^\circ$  (Yamazaki and Maute, 2017) where the coupling to the magnetosphere is less significant. *Sq* currents are thus an important link in the atmosphere-ionosphere coupling. Regions of focus include the dayside mid- and low-latitude E region. *Sq* currents typically form an anticlockwise current circuit in the Northern and a clockwise current circuit in the Southern hemisphere. They have a strong Hall component and flow at about 100-130 km altitude. At low latitude, below approximately 200km, the Hall current forms a complex 3D current system with the Pedersen and field-aligned currents.

**Neutral winds.** Neutral winds play a central role in the physics of the LTI. Vast amounts of momentum and inertia are contained within the motions of the lower thermosphere with particular influence at middle and low latitudes. Collisions between the neutral gas constituents and the plasma constituents have large effects on the behavior of the LTI when the neutral gas and plasma are not moving together. Winds in the LTI are quite variable, and are the consequences of many factors, notably including ion drag, propagation of tides and waves, and deposition of energy in the thermosphere. Typical horizontal thermospheric winds are in the range of 50-150 m/s, while the strongest horizontal LTI winds tend to peak in the altitude range of about 110–150 km at speeds of about 400 m/s (Larsen et al., 2022). Recent measurements of horizontal winds in the LTI gathered by the ICON satellite (England et al., 2022) reveal sharp gradients in the 100 to 135 km altitude range as well as variations with latitude.

**Traveling Ionospheric Disturbances.** Traveling Ionospheric Disturbances (TIDs) are associated with disturbance waves in the neutral atmosphere. There are various sources of the disturbance such as enhanced energy deposition due to M-I coupling (Lin et al., 2022), solar flares, solar terminator, and upward propagating atmospheric gravity waves due to tornadoes, weather fronts, and earthquakes (Tsuchiya et al., 2020). TIDs always travel away from the source region over long distance. They are frequently observed in GNSS data at  $10^\circ$ - $60^\circ$  latitude but their vertical structure and evolution in the altitudes around 160 to 200 km are not well determined.

**Gravity waves.** With respect to the altitudinal distribution of gravity waves, it is well known that gravity waves generated in the troposphere can propagate vertically and reach LTI altitudes before they undergo dissipation. However, the altitudes of wave dissipation (and secondary gravity wave generation) are still topics of active research in the LTI. Based on theory, gravity waves in the LTI could be the result of a multistep and altitude-dependent vertical coupling mechanism: according to this mechanism, primary GWs generated in the troposphere (e.g., orographically generated GWs) can lead to the generation of secondary GWs; when secondary GWs break and deposit their momentum (and energy), tertiary GWs are generated. Both secondary and tertiary GWs are believed to be very significant in the thermosphere (Liu et al., 2019; Vadas et al., 2018, 2023), although their effects are

not well quantified or understood. In that respect, all LTI altitudes are of interest in the study of gravity waves. With respect to their latitudinal extent, upward propagating gravity waves can be observed at low, middle and high latitudes. Furthermore, horizontally propagating gravity waves can propagate from the auroral regions towards mid-latitudes (Fritts and Alexander, 2003). Thus, sampling at all latitudes is of interest for the characterization of gravity waves.

#### 4.2.2.3. Low-latitude Regions of Focus

**Equatorial Electrojet.** The equatorial electrojet (EEJ) flows during daytime along the magnetic equator between the two mid-latitude *Sq* circuits. The EEJ is an **enhanced** Hall current band that peaks at altitudes near 105-110km (Figure 4.4) with latitudinal width of 1-2° during daytime. Although it is primarily directed eastward, the equatorial electrojet can sometimes be directed westward ("counter electrojet") when the wind system at low latitudes is perturbed, typically by atmospheric planetary waves and tides in the morning or afternoon sector, as recently shown with Swarm and ICON observations (Yamazaki et al., 2021). Reversed zonal currents at the magnetic equator have also been observed in the pre-dawn sector, although their altitudes are not known. How such westward or counter-electrojets are set up remains an unresolved question.

**Plasma irregularities.** Low-latitude plasma irregularities in the LTI are linked to equatorial electrojet two-stream instabilities which occur on spatial and time scales of meters and milliseconds. Larger wavelength (100 m to 10's of km) gradient drift waves are also commonly observed in the equatorial electrojet region, appearing where the ambient DC electric field is aligned with the ambient ionospheric gradient. Common low-latitude LTI regions where a variety of intense plasma irregularities are located are shown in Figure 4.4.

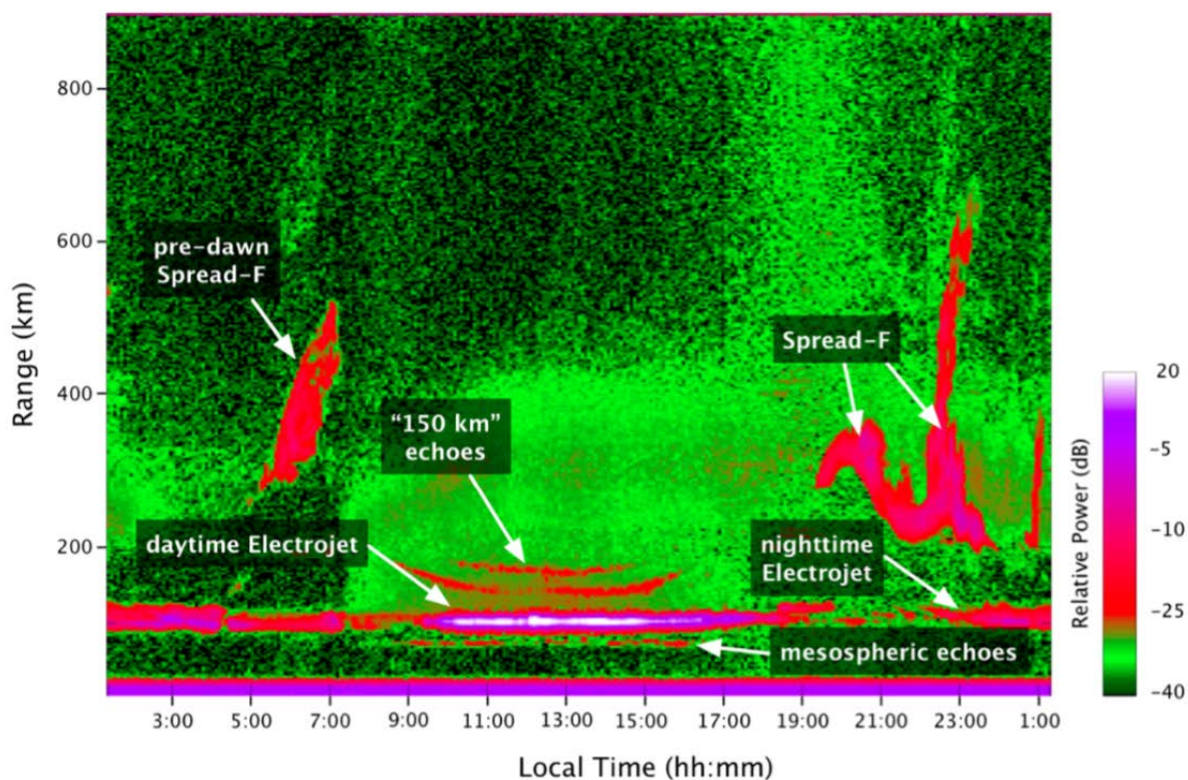


Figure 4.4: Common low-latitude LTI regions where a variety of intense plasma irregularities are located, measured on 14<sup>th</sup> September 2010 at the Jicamarca Radio Observatory, Peru (see, e.g., Hysell, 2022).





The so-called 150 km radar echoes (Chau et al., 2023) have time scales between 12 hours and few minutes and occur in the valley between the E- and F-region at around 120-180 km altitude and within  $\pm 10^\circ$  magnetic latitude. Low-latitude E-region plasma instabilities are dayside phenomena due to the present ionization. Variations in the neutral atmosphere, such as gravity waves, occur both day and night. They, too, strongly disturb LTI structures and may seed irregularities in both the LTI and the F-region. Some nighttime F-region irregularities, such as those associated with the bottomside spread-F are preconditioned in the LTI. Hence, much of our understanding of F-region irregularities depends on an improved knowledge of structures in the LTI.

### 4.3. Supporting models and observations

The advancement resulting from a highly capable *in situ* mission to the LTI will take place within an extensive existing scientific infrastructure that provides complementary and enabling capabilities. This infrastructure consists of synergistic orbiting assets, focused observations provided by sounding rockets, ground-based instrumentation, and modeling capabilities. Section 2.4 discusses current shortcomings in model capabilities that have been difficult to address in the absence of *in situ* measurements within the LTI. Improving this situation is thus a significant motivation for performing an *in situ* LTI mission. However, the capabilities of these models will be used to place the point measurements from the mission into a larger geophysical context, as is typically done with such missions. In particular, models will be used to expand the volume of relevance of a given set of measurements. An important example is presented in Section 2.1 with Figure 2.6 and the surrounding text. Here a sequence of single-point measurements of auroral electron fluxes was used to estimate ionization profiles over hundreds of kilometers of altitude by invoking a well-vetted electron transport model. As discussed above, knowing the plasma density in the LTI is key to understanding the conductivities and how they are structured in three-dimensional space. A mission to the LTI will take the next step in refining such estimates because it will provide samples of other parameters (neutral density and composition, plasma density and composition) that will further constrain such models. Similarly, models of the neutral atmosphere and ionospheric plasma will be used to “propagate” point measurements into the local environment. Further examples abound – modeling must be employed to close on the SOs through estimating heating and cooling rates, chemical processes, composition transport, gas accelerations, current directions and magnitude, ion-neutral collision frequencies, gravity wave dissipation, and other related processes. Thus, such a mission will not only further the development of models as discussed in Section 2.4, but the models will provide important enhancements to the returned *in situ* measurements.

In parallel to the modeling needs, the advancements provided by an *in situ* LTI mission will be compounded by measurements from other assets. These observational sources provide complementary spatial and temporal context by providing information on 2D variations and/or altitudinal profiles leading up to and following an LTI-mission satellite pass. Presented here is list of the most important observational tools capable of providing significant support for an LTI mission as it seeks to complete its scientific objectives.

- **All-sky cameras (ASCs)** provide maps of precipitation-driven features on scales of several to hundreds of km, and on 1-s time scales (and faster if required). ASCs are particularly valuable for establishing the morphology of auroral forms (2D EPP distribution), and therefore the geometry of current sheets, which is needed information to interpret *in situ*  $\delta B$  measurements. Furthermore, multi-spectral measurements can be used to generate 2D conductance maps.
- **Fabry-Perot interferometers** provide neutral wind patterns at meso-scales (tens to hundreds of km), at altitudes of the main emission layers and with time cadences of minutes.





- **Orbiting constellations and ground-based arrays of GNSS receivers** provide global- and regional-scale observations of the ionosphere.
- **Incoherent-scatter radars** provide altitude profiles of plasma density and ion/electron temperatures, and regional maps of ion drift velocity, on spatial scales of several tens to hundreds of km, and time scales on the order of 1 minute.
- **Coherent-scatter radar arrays** provide regional and global plasma convection maps on scales of tens to hundreds of km and at cadences of minutes.
- **Active radio experiments** (e.g., HAARP) provide an estimate of LTI plasma parameters, e.g., conductivities.
- **Sounding rockets** provide vertical profiles of plasma and neutral parameters, as well as electric and magnetic fields and energetic particle precipitation, among other parameters, which can be used to validate orbital measurements, assimilated schemes, and physics-based models. Importantly, sounding rocket missions can provide important correlative measurements simultaneous with horizontal overflights of a low perigee satellite.
- **Vapor trails** released by sounding rockets provide vertical neutral wind profiles through the LTI region including evidence of shear with altitude and the presence of large-scale instabilities.
- **Conjunctive observations** from upcoming missions like GDC and DYNAMIC, for example, could provide gas and plasma measurements at different altitudes or complementary remote sensing measurements that enable broader vantage points in connecting the LTI to the lower atmosphere and other regions of the ionosphere.
- **Current mission datasets** operating concurrently with an LTI satellite mission would provide key parameters for assessing inputs into the LTI from forcing above and below the region. This includes, among others, continuous, high-latitude AMPERE  $\delta B$  observations, and global Swarm measurements of magnetic field and plasma characteristics. These observations can provide large-scale context as well as complementary local LTI observations.

#### 4.4. Science Strategy Matrix

The needs derived above through this measurement-based strategy are summarized within a generalized science strategy matrix (SSM) shown in Appendix B, Table B.1. As done with the sections above, the SSM flows from Science Goals and Objectives in the left columns to science needs, observational needs, modeling needs, and complementary measurements in the right columns. The SSM is put forward here as a strategy that relates science definitions with science and observational needs in anticipation of a subsequent science traceability matrix that would quantitatively define requirements and identify any urgently needed new methods for a particular LTI mission.



## 5. NOTIONAL MISSION ELEMENTS AND EXPECTED CHALLENGES

The previous chapter outlined the need to gather a suite of key scientific measurements within the Lower Thermosphere-Ionosphere in order to address the science objectives essential to closing on the major unanswered questions that pertain to this region of geospace. To this end, the development of a scientific observatory is considered to obtain simultaneous measurements of all the necessary physical parameters concurrently in the same volume, and spatially resolved to investigate the relevant scales of the processes at play. This observatory will carry out three main functions:

1. Journey *into* the LTI on repeated orbits *to directly sample* the region where the key physical processes occur.
2. Gather *simultaneous measurements* of the key physical parameters outlined in Chapter 4.
3. Accumulate *statistically significant data* in key regions of interest, *under different* solar and geomagnetic *conditions*.

In this chapter, we outline a number of preliminary mission elements, including potential instruments and orbits, which together describe a notional observatory that would enable the necessary data to be gathered, thus opening the door to a new understanding of the LTI. Below, we first remark on how the measurements can be met with existing instruments followed by comments on the spacecraft design and notional orbit parameters that together provide a concept or “way forward” regarding a viable observatory that could achieve these goals. These are followed by expected challenges related to the successful gathering of accurate scientific measurements in the LTI.

### 5.1. Mission elements

***In situ measurements of all relevant parameters.*** As discussed previously in Chapters 3 and 4, the quantification of the physical processes related to collisional ion-neutral interactions and their driving sources of free energy in the LTI requires the simultaneous knowledge of key state parameters gathered in the same volume. These include ion drifts ( $\mathbf{V}_i$ ), ion temperature ( $T_i$ ), ion number density ( $N_i$ ), electron temperature ( $T_e$ ), electron density ( $N_e$ ), ion composition ( $N_{ix}$ ), neutral winds ( $\mathbf{U}$ ), neutral number density ( $N_n$ ), neutral temperature ( $T_n$ ), neutral composition ( $N_{nx}$ ), electric fields ( $\mathbf{E}$ ), magnetic fields ( $\mathbf{B}$ ), and energetic precipitating particles (EPP). These simultaneous measurements must be gathered on the same platform at the needed temporal and spatial scales, as discussed in Chapter 4.

Although some important aspects of the LTI can be revealed using remote observations, including those from both space-based platforms situated at higher altitudes as well as from ground-based instruments, these data do not provide the direct, multi-parameter information at the scales needed to resolve the science questions outlined in this report. On the other hand, continuous, remote observations provide important information regarding the context and 2D “picture” of several important processes, including, in some cases, invaluable profiles of key parameters as a function of altitude. When such remote sensing observations are merged with direct measurements in the LTI as discussed herein, the resulting *combined* data set is particularly powerful, and much greater than the sum of its parts.

For the remainder of this discussion, we proceed under the assumption that the most efficacious measurement scheme to achieve the science objectives is that of one or more spacecraft equipped with an instrument suite capable of providing comprehensive *in situ* measurements simultaneously in the same volume. These measurements are summarized in Table 5.1, which includes examples of



commonly flown scientific instruments that have successfully carried out such measurements on past ionosphere/thermosphere orbiting missions.

Table 5.1: Measurement parameters, commonly used instruments, and heritage examples from ionosphere/thermosphere orbiting missions.

|                           | Symbol   | Geophysical Observable           | Commonly used Instruments                                         | Heritage                                                              |
|---------------------------|----------|----------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|
| Ionospheric plasma        | $N_i$    | Ion number density               | Thermal ion imager, ion drift meter, retarding potential analyzer | Swarm, DE-2, AE, ROCSAT, ICON, C/NOFS, DMSP                           |
|                           | $V_i$    | Ion drift                        |                                                                   |                                                                       |
|                           | $T_i$    | Ion temperature                  |                                                                   |                                                                       |
|                           | $N_{ix}$ | Ion composition                  | Ion mass spectrometer, retarding potential analyzer               | Swarm, DE-2, AE, MAVEN, Pioneer Venus, CHAMP, C/NOFS, San Marco, DMSP |
|                           | $N_e$    | Electron number density          | Langmuir probe, impedance probe                                   | Swarm, DE-2, AE, MAVEN, Pioneer Venus, CHAMP, C/NOFS, San Marco, DMSP |
|                           | $T_e$    | Electron temperature             |                                                                   |                                                                       |
| Thermospheric neutral gas | $N_n$    | Neutral gas number density       | Ionization gauge, neutral mass spectrometer                       | AE, DE-2, Streak, MAVEN, Pioneer Venus Orbiter                        |
|                           | $U$      | Neutral winds                    | Wind & gas temperature sensor                                     | DE-2, AE, C/NOFS                                                      |
|                           | $T_n$    | Neutral gas temperature          |                                                                   |                                                                       |
|                           | $N_{nx}$ | Neutral gas composition          | Neutral mass spectrometer                                         | AE, DE-2, MAVEN, Pioneer Venus Orbiter                                |
|                           | $\rho_n$ | Neutral gas mass density         | Accelerometer                                                     | CHAMP, Swarm                                                          |
| Electro-magnetic fields   | $E$      | Electric field                   | Electric field detector                                           | DE-2, C/NOFS, DEMETER, San Marco                                      |
|                           | $B$      | Magnetic field                   | Magnetometer                                                      | CHAMP, Swarm, DE-2, C/NOFS, MAGSAT                                    |
| Precipitating particles   | $EPP$    | Energetic particle precipitation | Charged particle spectrometer                                     | DE-2, DMSP                                                            |

Importantly, all the measurement capabilities exist with strong heritage and high technical readiness levels (TRLs) at least for higher regions of the thermosphere and ionosphere. The notional instruments are envisioned to be placed on a non-spinning platform, ideally with body-mounted solar arrays, such as was carried out on AE-C, AE-D, AE-E, and DE-2 (see later discussion). Here, the instruments that measure the ambient gas properties (neutral and plasma) are placed in the forward or ram direction and the fields instruments notionally placed in the aft. The energetic particle detector is notionally placed on the side. The spacecraft essentially spins once per orbit such that the forward instruments

are continuously in the ram direction along the orbit. Additional remarks on a notional spacecraft design are provided further on below.

Despite the fact that instruments which measure all of the required physical parameters exist and have been shown to provide accurate measurements higher up in the ionosphere and thermosphere, the performance of many of these instruments has not been demonstrated on orbiting platforms at the lowest altitudes of the LTI. On the other hand, sounding rocket measurements have shown that such scientific instruments perform well and gather excellent data within all altitudes of the LTI (including in some cases below 100 km), but at lower platform velocities. As a result of the combination of the much larger, supersonic velocity of an orbiting vehicle ( $\sim 8$  km/s compared to 1 km/sec, typically encountered on a sounding rocket) with the exponentially increasing atmospheric density with decreasing altitude, the likelihood of successful, accurate measurements below 120 km altitude has not been established for many of the physical parameters. Atmospheric density as a function of altitude is provided in Figure 5.1 (left), showing the significant increase below 120 km. Besides posing a challenge for the performance of some instruments travelling at high velocities, the increased density contributes significant drag effects on the spacecraft orbit, as discussed below.

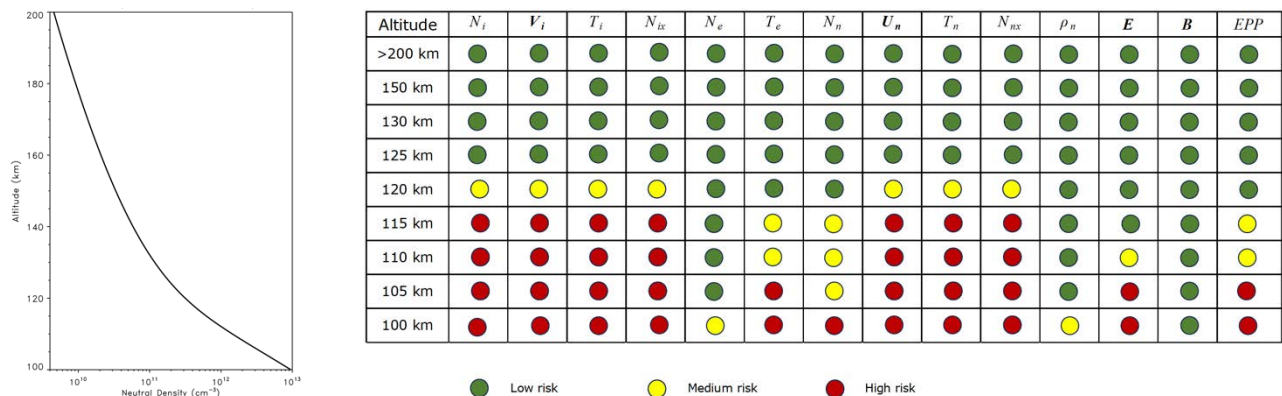


Figure 5.1: Left, atmospheric neutral density from the MSIS model for mid-latitude, daytime conditions. Right, expected risks to successful measurements of the targeted geophysical observables from on an orbiting satellite (at  $\sim 8$  km/s orbital velocity) within the LTI (green: low, yellow: medium, and red: high risk). Note that the lower half of the altitude bins represent the lowest 20 km of the LTI (credit: NASA/GSFC – R. Pfaff).

At the end of this chapter, we discuss some of the challenges that the large velocity and increased atmospheric density pose for the operation of scientific instruments orbiting within the lowest altitudes ( $< 125$  km) of the LTI, particularly for measurements of the ionized and neutral gas properties. Figure 5.1 (right) provides an indication of the general likelihood of measurement success for each of the observables shown in Table 5.1. This figure is highly qualitative and is intended to portray notional expectations of measurement success for commonly flown instruments in the Earth's ionosphere and thermosphere.

**Notional concept: Measurements at low altitudes on orbiting missions.** Addressing key science objectives in the LTI requires performing the measurements within the theatre where the LTI processes take place, i.e., below the 200 km altitude range. Traversing this high-drag region at high speeds dissipates a significant amount of energy. Most low Earth orbiting satellites without propulsion begin to “re-enter” when they encounter this altitude range, where their orbits decay precipitously due to atmospheric drag. Based on current technology of propulsion subsystems, circular orbits below 200 km are expected to be very short-lived, again due to atmospheric drag, thus precluding the accumulation of statistically significant measurements under a range of solar and geomagnetic conditions. Compiling the required *in situ* measurements in the LTI then requires an approach that carefully budgets energy



usage, whether it be stored as chemical or gravitational energy, or possibly electrical energy, against mission data-gathering requirements.

To achieve effective sampling at the lower altitudes while ensuring adequately long lifetimes, the standard approach is to base such missions on the employment of elliptical orbits with high apogees such that the satellite can gather the necessary data along the perigee portion of the orbits that are in the LTI region. In this manner, the gravitational potential energy at apogee compensates for the loss of kinetic energy due to atmospheric drag. The enhanced overall satellite lifetime compared to those with circular orbits comes at the expense of limited sampling time per orbit within the lower altitude range of interest. The excursions in the LTI, however, still cover sufficiently large horizontal expanses to enable significant observations to be obtained.

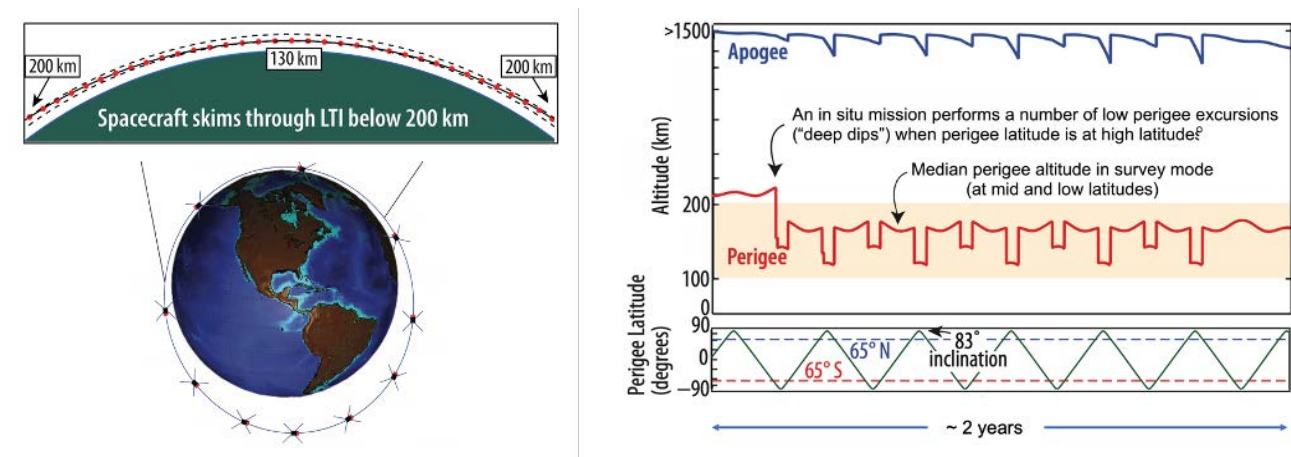


Figure 5.2: Left, elliptical orbit showing how the spacecraft skims through the LTI region at perigee. Right, orbit history of a notional mission concept slightly modified from the ASTRE orbital study, showing dips to low altitudes and subsequent changes in apogee. Here, the perigee alternates between “deep dip” excursions and “survey mode” (adapted from Pfaff et al., 2022).

To illustrate how a satellite gathers data in the LTI at and near its perigee altitude, Figure 5.2 (left) shows a sample orbit with a perigee of 130 km and an apogee of 1500 km. An enlargement of the lower altitudes sampled on this orbit is shown in the top part of the figure. Note that during this time, the satellites “skims” through the perigee region along a largely horizontal orbit. After repeated low perigee orbits, the spacecraft apogee decreases such that after about two weeks or 250 orbits (depending on the apogee), perigee must be increased to avoid re-entry. At this time, using propulsion, both perigee and apogee are re-established to their nominal “survey orbit” values. Importantly, only one propulsion burn is needed to initiate the orbit maneuvers to provide the lower perigee as well as to raise it again, and thus continuous firings are not needed to establish the orbits which passively sample the LTI for the extended “dipping” periods. As discussed later, this is a critical design parameter which ensures that the ambient LTI environment that is being sampled is not perturbed or contaminated by continuous propulsion firings. A notional mission modified slightly from the ASTRE orbital study (Pfaff et al., 2022) showing apogee and perigee history is shown in Figure 5.2 (right). Note that during any several weeks-long period after the perigee is lowered, the apogee decreases to compensate for drag at perigee. Eventually, the perigee and apogee are re-boosted as described above.

An important aspect of such elliptical orbits is that, due to Earth’s oblateness, precession of the semi-major axis occurs via the laws of orbital dynamics, and hence the perigee latitude precesses. The precession of perigee allows for the sampling of all (low, mid, and high) latitudes for a high inclination, polar orbiting satellite or, for a low inclination satellite, the sampling of all local times every orbit. The notional mission profile in Figure 5.2 (right) is for a high inclination mission that focused on high latitudes

only. Due to the periods defined by its elliptical orbit properties, the spacecraft performs uninterrupted low perigee excursions for two weeks (about 250 orbits) every two months when the perigee alternately precesses over the northern and southern polar regions.

An important design consideration with respect to the orbit parameters include the choice of inclination. For high inclination missions that address the polar and auroral processes selecting an inclination somewhat less than  $90^\circ$  (e.g.,  $83\text{--}85^\circ$ ) instigates a precession in local time which enables sampling at all local times and seasons, while still providing sampling of the magnetic poles which are offset from the geographic poles.

**Previous low perigee *in situ* measurements.** The only systematic *in situ* observations within the Earth's LTI region to date were gathered by instruments on the NASA Atmosphere Explorer AE-C, AE-D, and AE-E satellites (1973–1981) with inclinations of  $68.1^\circ$ ,  $90.1^\circ$ , and  $19.7^\circ$ , respectively. These satellites carried instruments that successfully measured plasma density, ion drifts, neutral and ion composition and temperatures, and one component of the neutral wind on their platforms (Burgess and Torr, 1987; Spencer et al., 1973).

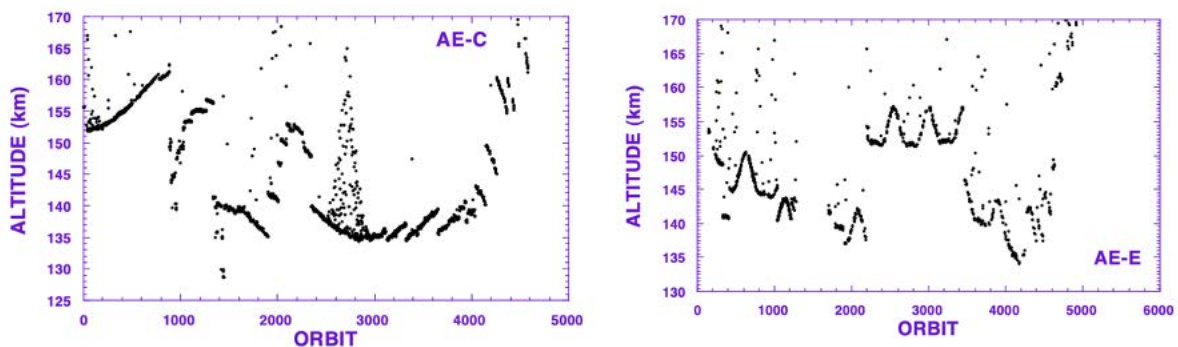


Figure 5.3: Perigee altitudes for the eccentric orbits of the AE-C (left) and AE-E (right) satellites (credit: NASA/GSFC).

Using onboard propulsion, AE-C and AE-E each included about one year of operations to perigees between 130 and 150 km. Figure 5.3 shows the lowest altitudes for each orbit where AE-C and AE-E gathered data below 170 km. The lowermost altitude was 128 km achieved by AE-C.

The AE measurements constitute an important dataset for many parameters included in current empirical models of the upper atmosphere, and great expansions of knowledge were achieved by these limited missions. However, the total sampling time from these missions combined amounts to only  $\sim 60$  hours at altitudes below 200 km, with a small fraction of this time associated with measurements at high latitudes. Furthermore, the AE spacecraft did not carry instrumentation to measure vector neutral winds, electric and magnetic fields, and energetic precipitating particles. Nevertheless, the AE spacecraft served as important scientific and technological pathfinders by revealing the large variability within the LTI region and by demonstrating the feasibility of low-perigee operations of both the satellites and instruments.

Comprehensive scientific measurements were included on the NASA Dynamics Explorer (DE-2) satellite (1981–1983) (Hoffman, 1980). However, this satellite did not include propulsion. DE-2 was launched into a 300 km by 1000 km orbit and lasted for 18 months prior to re-entry. It did provide some sampling below 250 km towards the end of the mission, but did not provide extensive measurements in the critical LTI region.

Planetary missions to Venus and Mars also included neutral and ion gas measurements at low perigee and are also important for demonstrating the feasibility of satellite measurements within the Earth's LTI.



The Pioneer Venus Orbiter (PVO) mission (Kasprzak et al., 1993) included a Langmuir probe and mass spectrometer and executed numerous periapsis journeys to ~150 km altitude. The density of the upper atmosphere of Venus is somewhat greater than that of the Earth, and includes diurnal variations. The lowest altitude measurements of PVO were 129 km at the end of the mission. The pressure associated with this altitude corresponds to that of an altitude of 120 km around Earth

More recently, NASA's MAVEN mission has been successfully performing measurements in the Martian thermosphere–ionosphere since 2014 (Jakosky et al., 2015). The MAVEN mission included several successful “low perigee dips” as part of high-apogee elliptical orbits. Data from these orbits demonstrated the richness of processes at low altitudes in the Martian upper atmosphere (Bougher et al., 2015; Collinson et al., 2020). MAVEN utilized similar instruments as those envisioned for a future Earth LTI mission, shown in Table 5.1. MAVEN successfully gathered data to altitudes as low as 125 km on Mars. Although the density of the Martian upper atmosphere is only about 1% of that of the Earth, the successful dipping performance (with associated high-apogee elliptical orbits) demonstrated the viability of this approach for carrying out repeated sampling of the Martian LTI.

**Previous *in situ* mission studies.** A number of low-perigee mission concepts have been studied and proposed over the past several decades, including many with detailed, mature designs that resulted from extended Phase A analyses. These include the following:

- In the early 1990s, NASA's Thermosphere-Ionosphere-Mesosphere Energetics and Dynamics (TIMED) mission was conceived. It consisted of two satellites with dipping capability including both a high-inclination and a low-inclination orbit journeying to altitudes as low as 120 km. A Science Definition Team report was published (NASA, 1991) and an Announcement of Opportunity soliciting instrument proposals was released. Ultimately, the *in situ* portion of the mission was descoped, and the TIMED mission was carried out using only remote sensing measurements of the ionosphere-thermosphere-mesosphere on a single satellite orbiting at 625 km altitude.
- In 2000, NASA initiated the Geospace Electrodynamic Connections (GEC) mission which was to include only *in situ* measurements, and which was designed to study the high-latitude, low-altitude ionosphere/thermosphere. A Science Definition Team report and numerous technical studies were published (NASA, 2001). GEC consisted of four identical satellites that would perform *in situ* measurements with perigees as low as 130 km. The spacecraft were to be placed on a “string-of-pearls” formation with variable spacings, enabling the investigation of, and differentiation between, temporal and spatial effects. A “petal” formation was also envisioned that enabled simultaneous sampling of different altitudes within the LTI of the four GEC satellites.
- The Daedalus mission concept (ESA, 2020; Sarris et al., 2020) was proposed in 2018 to ESA's Earth Explorer program and successfully completed an ESA Phase 0 mission definition study in 2020. Daedalus, as initially proposed, included a main satellite with a 150 km by 2000-3000 km elliptical orbit, with deep dipping campaigns down to 120 km.
- A number of P.I.-led proposals to explore the LTI with *in situ* measurements were submitted to NASA's Explorer program, beginning with the Low Perigee Explorer (LOPEX) in 1993, followed by Dipper in 1998, and the Magnetic Atmosphere Explorer (MAX) in 2000, the latter two contributing to the formation of the GEC initiative mentioned above. These proposed missions were reformulated as the Atmosphere-Space Transition Region Explorer (ASTRE) mission, proposed to NASA in 2008 as a Small Explorer and again in 2011 as a full Explorer mission (with greater technical margins), which included a detailed, Phase A Concept Study Report



submitted in 2012 (Pfaff et al., 2022). These proposed missions consisted of a single spacecraft in elliptical orbits with dipping “campaigns”, with perigee altitudes that varied from 120 km to 150 km, depending on the proposal.

## 5.2. Mission design features and challenges

Several key aspects of the LTI affect the performance of the spacecraft, orbital elements, and scientific instruments that deserve mention here. Given the extensive studies of efforts to explore this region with *in situ* probes, we include remarks on various mission “challenges” inherent to the LTI, particularly at the lowest altitudes.

Three main areas of technical concern for LTI missions include:

- Large atmospheric density which increases exponentially with decreasing altitude, creating drag forces on the orbital platform and shock-like conditions (at satellite velocities of  $\sim 8$  km/s) for measurements at the lowest altitudes ( $< 125$  km);
- Aerodynamic heating of both the sensors and the spacecraft at the lowest altitudes;
- Increased atomic oxygen whose reactive properties can erode sensor surfaces and electrostatic coatings, such as on solar arrays.

### 5.2.1. Unique mission elements: spacecraft design

We describe here a notional “top level” spacecraft design to illustrate the approach that was developed to meet the unique requirements and challenges of gathering data with *in situ* instruments during low perigee excursions to the Earth’s LTI.

**General guidelines.** Figure 5.4 provides an illustration of actual, studied or proposed satellites designed to explore the LTI. The designs for future (proposed) missions can take into account notional instrument accommodation requirements (Pfaff et al., 2022; Sarris et al., 2020).

Importantly, the spacecraft bus of an LTI *in situ* mission must be tightly integrated with the instruments to enable high-accuracy, scientific measurements. It is not simply a platform upon which instruments are attached. Rather, it must conform to important guidelines to optimize the observatory’s performance in the LTI region. In particular, it must minimize its interactions with the neutral and ionized medium and enable accurate measurements of electric and magnetic fields and energetic particles, to ensure that highest quality data are obtained.

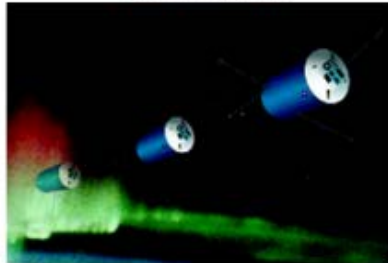
For notional designs, the instruments are essentially divided between ram pointing instruments that sample the neutral and ionized gases, the electric and magnetic field instruments deployed on booms, and the energetic particle detector which notionally is situated on the side of the spacecraft to enable a 360-degrees field-of-view. For continuous ram pointing, a three-axis, inertially stabilized spacecraft is required with momentum wheels to ensure that the forward instruments that gather measurements of the ambient plasma and neutral gases are continuously oriented along the ram direction in the prime data-taking portion of the orbit (i.e., below 500 km altitude.)



**Atmosphere Explorer  
C, D, E**



**Geospace Electrodynamic  
Constellation**



**Dipper**



**Pioneer Venus Orbiter**



**ASTRE**



**Daedalus**



Figure 5.4: Renderings of actual (blue), studied, and proposed (red) satellites designed to explore the LTI (credits: NASA and Sarris et al., 2020).

Some notional spacecraft design considerations include the following general but important guidelines to optimize performance in the LTI:

1. A cylindrical shape with body-mounted solar arrays with no appendages (except for instruments) would minimize drag, aerotorquing, aeroheating, and atomic oxygen fluence:
  - a. Most importantly, the absence of appendages would greatly minimize drag and thus conserve Delta-V and propellant.
  - b. The absence of appendages would greatly minimize aerodynamic torques imparted on the spacecraft when atmospheric perturbations are encountered along the orbit, thus minimizing asymmetric forcing such as to maintain the spacecraft axial direction along the velocity vector. The severity and abruptness of neutral density variations during magnetic storms is not known, and ram-pointing instruments may be subject to tight attitude maintenance requirements (e.g.,  $< 2^\circ$ ).
  - c. A cylindrical “bullet shape” would minimize perturbations of the spacecraft to the space environment which could impact accurate scientific measurements, particularly those of the neutral and ionized gas that may be influenced by shocks.
  - d. Appendages create both solar and magnetic shadowing of instrument apertures and/or sensors, which can be variable at different sun/magnetic field angles, and should be minimized.
  - e. A sleek cylindrical spacecraft design with body-mounted solar arrays without sharp edges would help to assure uniform electrostatic potentials around the spacecraft.



- f. A round or cylindrical shape would reduce discontinuities in power production when incident solar illumination transfers between adjacent (angled) solar cell panels as the spacecraft slowly rotates once per orbit, particularly at sunrise and sunset. Varying shadows on solar arrays generate current perturbations creating unwanted magnetic signatures.
2. Ram-mounted instruments should be situated on the forward-most face of the observatory to allow measurement of unperturbed incoming gas and plasma populations.
3. The spacecraft volume should be optimized within the launch vehicle fairing to accommodate an adequate propulsion system (e.g., a large propellant tank) and maximize the solar array area, while otherwise keeping the cross-section small.
4. The center of mass should be forward of the center of pressure for stability, for instance by placing the propulsion tank near the spacecraft center and the electric field booms in the aft section of the spacecraft.

**Aerodynamic heating and thermal control.** Although aerodynamic heating is not typically a concern for low earth orbit (LEO) satellites that do not venture below 200 km, it becomes an important design consideration for satellites with perigees at lower altitudes. The situation is compounded by the fact that the spacecraft thermal design must maintain temperatures within acceptable limits while many proposed designs accommodated body-mounted solar arrays covering most of the external surface, where radiators might otherwise be located. Importantly, at LTI altitudes, the aerothermal heat flux on a spacecraft body can be significant and much greater than direct solar fluxes (Caruso and Naegeli, 1976) and, therefore, can result in significant thermal gradients on the spacecraft.

Temperature monitors on NASA's AE-C satellite, which journeyed to perigees as low as 128 km, demonstrated that aerodynamic heating was considerably less than predicted by conventional modeling (Caruso and Naegeli, 1976). Nevertheless, advanced modeling efforts to specify the thermal stresses imposed on a low-perigee spacecraft mission provide a critical element of the mission. Such modeling work has been incorporated into design elements of recent low-perigee mission proposals such as the Atmosphere-Space Transition Region Explorer (ASTRE) (Pfaff et al., 2022) and Daedalus (ESA, 2020).

As an example, the main approach for the ASTRE designs is to use a mostly passive system consisting of coatings, insulating blankets, heat pipes, and heaters to provide the necessary thermal control. The design approach is simplified by putting most of the heat dissipating electronics in the forward section behind the ram plate and surrounded by radiators (not solar cells), as shown in particular for the notional ASTRE spacecraft rendering in Figure 5.4. The ram plate is also where aerodynamic heating is greatest and thus thermal control of the ram deck is very important. One approach is to use heat pipes connecting to radiators which have sufficiently large area to reject ram heating.

Instruments deployed from the aft end of the observatory are isolated from the bus and provide their own thermal control. To maintain their straightness the electric field booms are designed to maintain stable temperatures resulting from both solar illumination and aerodynamic heating (Maynard et al., 1981; Pfaff et al., 2021). The solar arrays, which are decoupled from the ram section, have a system of linear and circumferential heat pipes that maintain their thermal environment within acceptable limits.

The ASTRE team developed a thermal model which demonstrated that this thermal approach provides a stable thermal platform for the ram instruments, maintaining the ram deck between 5° C and 25° C for all beta angles and perigee encounters including worst-case atmospheric density. Aerothermal heating was also a primary constraint on the low-altitude coverage of the proposed Daedalus mission. Simulations revealed heat flux values of the order of 20 kW/m<sup>2</sup> on the ram face of the Daedalus

spacecraft at 115 km altitude. As a result, traditional design and manufacturing techniques were modified to account for the expected heating, including needs for additional shielding on the ram side of the spacecraft and surrounding some of the instruments.

### 5.2.2. Propulsion and low perigee excursions

Repeated excursions into the LTI region (i.e., below 200 km) require a propulsion system to counter the effects of the high drag environment. When coupled with a high apogee (elliptical orbit) and other key spacecraft features as noted above (e.g., bullet shape, limited deployables, high mass, good thermal design), such propulsion enables data collection down to altitudes of at least 125 km and probably 120 km. Figure 5.5 shows simulations of numerous excursions to the LTI for a number of proposed satellites, including for the Dipper and ASTRE (Pfaff et al., 2022) Explorers missions proposed to NASA, and for the original Daedalus Earth Explorer mission idea proposed to ESA (Sarris et al., 2020).

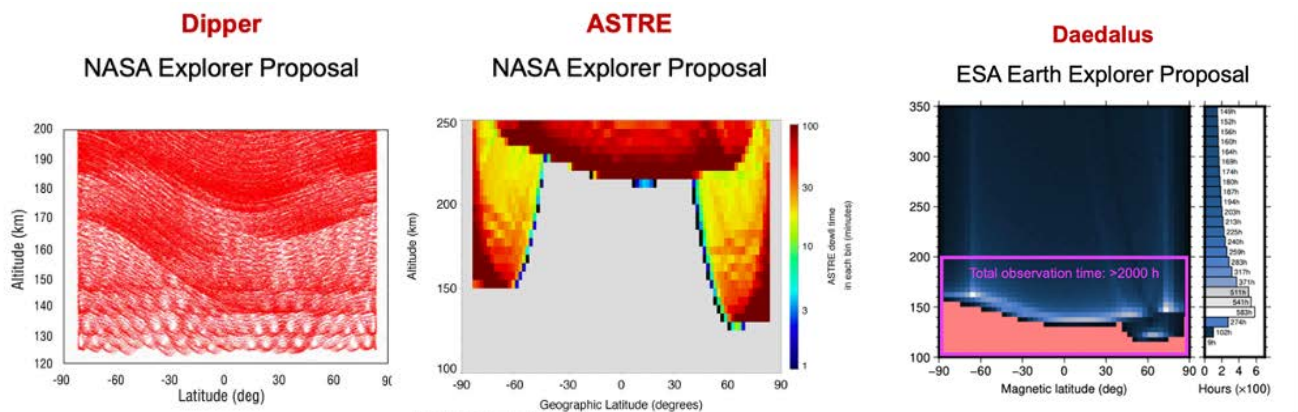


Figure 5.5: Representative simulations for various low-perigee mission concepts and ideas to explore the LTI region: (left) Dipper proposal to NASA; (middle) ASTRE proposal to NASA (adapted from Pfaff et al., 2022); (right) Daedalus proposal to ESA (from Sarris et al., 2023a, 2020). The simulations shown here are for satellite concepts with different apogees, masses, drag coefficients, and propulsion systems.

### 5.2.3. Scientific instrument challenges

#### Measurement perturbations resulting from neutral density build-up in the ram direction.

Performing particle measurements in a rarefied atmosphere is a technical and scientific challenge, particularly when there is a build-up of neutral density ahead of the sensor due to the motion of the sensor through the ambient environment. In some cases, the combination of increased ambient neutral density and the high orbital velocity creates a diffuse shockwave around the satellite. To best prepare for such conditions, a better understanding of the anticipated instrument performance can be obtained from both past measurements and from simulations.

Representative data from large neutral number densities at altitudes extending down to 135 km are available from the AE satellites (Burgess and Torr, 1987; Spencer et al., 1973) to 150 km from PVO (Brace and Kliore, 1991; Niemann et al., 1980), and to 130 km from the Streak satellite which included an Ionization Gauge (Clemmons et al., 2008).

Plasma sensors providing measurements on AE and PVO, in particular the Langmuir probe, the Retarding Potential Analyzer and the ion mass spectrometers recorded effects of secondary electron emission and ion sputtering on the lowest perigee passes, but except in rare instances, the measurements were apparently not compromised. The situation was more severe during the periapsis



conditions on the PVO mission. Whereas the peak atmospheric pressure encountered at the lowest altitudes by PVO was comparable to that for AE-C at lowest perigee, the higher orbital speed of PVO and the dominance of heavy CO<sub>2</sub> in the Venusian atmosphere led to the build-up of higher pressure inside the PVO gas instruments and provided more impact energy for secondary electron emission than was the case for AE-C. These effects compromised some of the PVO measurements at the very lowest altitudes, suggesting that they would appear to degrade measurements for similar sensors on terrestrial spacecraft that attempt to dip near or below 120 km.

Rarefied flow simulations with the Direct Simulation Monte Carlo (DSMC) technique were carried out during the Daedalus Phase 0 studies (ESA, 2020). Local effects such as diffuse shocks were shown to be more pronounced at altitudes of 120 km and below, although the simulations suggested that they still persist, in part, at 140 km or above.

The creation of a ram cloud and the density build-up in the ram direction of the spacecraft impacts the ability to measure neutral temperatures in the LTI. More specifically, factors affecting the accuracy of neutral temperature measurements include the applicability of the kinetic theory used in extracting neutral temperatures, in particular at lower altitudes where a shorter mean-free-path of the measured particles might affect the measurements, and gas-surface interactions, which are also dependent on altitude and neutral density (Spencer et al., 1973). All chemical species will experience a density build-up on the ram face, but the relative composition (molar fractions) also changes. This is partly due to thermalisation (and slowing down) of the particles at the ram face as they interact with the surface, which depends on the particle mass, but also to chemical changes across the diffuse shock.

In addition to density and temperature measurements, the perturbation of velocity distribution functions of particles can also be evaluated to characterize these interactions across the forward shock, in particular the diffusion of the incoming particles off the ram face. Fortunately, measurement techniques that retain the bulk energy or velocity of the flow through energy filtering should enable the unperturbed geophysical observables to be retrieved after post-processing. Preliminary assessments performed during the Daedalus Phase 0 studies (ESA, 2020) suggest much smaller perturbations to the distribution functions than those associated with the density enhancements. The simulations have shown that perturbations to the distribution functions are expected to be near negligible above 140 km, and increasingly more important with decreasing altitudes below that altitude.

**Measurement effects of temperatures and sheaths in the vicinity of the spacecraft.** Due to the high spacecraft velocity and the interaction of the spacecraft with the surrounding plasma and neutral environment, there are several potential sources of uncertainty that can lead to systematic and random errors in *in situ* measurements in space. For example, factors that affect the accuracy of *in situ* measurements include the acceleration of plasma by a charged surface, the generation of a complex plasma cloud that surrounds the spacecraft and interacts with the environment, and impact ionization and reflection of particles off the spacecraft and the subsequent inclusion of reflected ions in the measurements (Ergun et al., 2021; Hanley et al., 2021). The removal or correction of such errors has been the topic of multiple studies over the past decades (DeForest, 1972; Ergun et al., 2021; Hanley et al., 2021; Hastings, 1995; Whipple, 1981).

Sheath and wake effects around the spacecraft affect the performance of the electric field probes, particularly when the ambient plasma density is low. Simulations have shown the plasma wake extending to several meters in low-density environments (10<sup>2</sup>/cm<sup>3</sup>) that could easily be encountered in the winter polar cap and below the bottomside F-region ledge in the nighttime LTI. Electric field double probes included individual element lengths of 11 m on DE-2, providing baselines of 21.4 m sensor-to-sensor (Maynard et al., 1981) and 10 m on C/NOFS providing baselines of 20 m tip-to-tip (Pfaff et al., 2022). To decrease drag effects, double probes of 8 m (17 m tip-to-tip) were used in many of the



aforementioned proposals to NASA's Explorer program. Long electric field boom lengths are also needed to minimize shadowing of the main spacecraft as well as magnetic shadowing effects.

**Bending of stiff electric field booms.** Electric field double probe booms undergo bending at the lowest perigees due to the significant atmospheric densities, particularly below 150 km. Figure 5.6 shows anticipated bending of 8-m booms that utilize the same design as those flown on the DE-2 and C/NOFS spacecraft. The anticipated bending is approximately  $0.5^\circ$  at 130 km for a worst-case atmosphere. Other stiff booms, such as helically wound booms, are expected to show similar properties, based on their respective stiffness. Boom bending of several degrees can certainly be accommodated in the electric field analysis, as shown by the analysis of thermal bending on the order of  $1^\circ$  in the C/NOFS data set which maintained the  $\pm 1$  mV/m DC electric field accuracy (Pfaff et al., 2021).

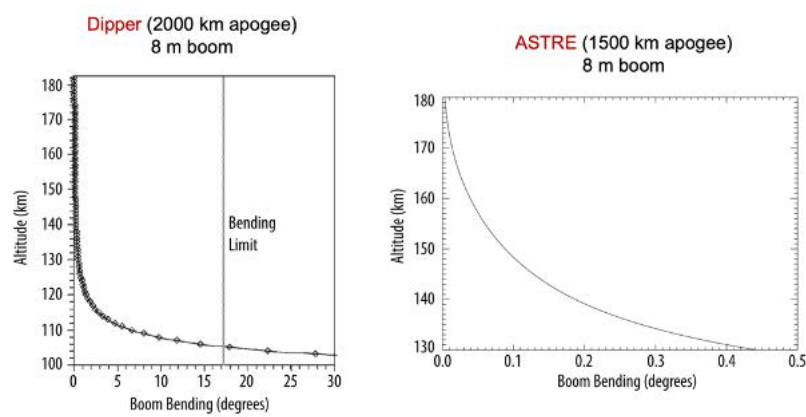


Figure 5.6: Predicted boom bending of 8-m, stiff electric field booms of the same type as used in DE-2 and C/NOFS for two LTI mission proposals (left, Dipper; right, ASTRE, from Pfaff et al., (2022)), and for a worst-case atmosphere.

**Increased erosion by atomic oxygen.** At altitudes below 200 km, the orbiting satellite environment is dominated by  $O_2$ ,  $N_2$ , and atomic O. Some of these gases are reactive and well-known to attach to spacecraft surfaces, and many studies have been directed to revising surface accommodation coefficients and derived neutral densities to account for gas-surface interactions (Moe et al., 2011; Pilinski et al., 2013, and references therein). Oxidation due to atomic/ionized oxygen is a challenge in particular for electrically conductive materials, such as are used, for example, on the probe surfaces of electric field sensors. Although the thickness of the traditional carbon-based coating, DAG-213 (a resin base graphite dispersion), can be increased to accommodate this erosion, other conducting surfaces with uniform work functions that do not react with atomic oxygen are sought. To this end, titanium nitride (TiN) coatings on electric field probes have a long flight history on sounding rockets for achieving a uniform surface conductivity and could be considered, although it is believed that for MAVEN's Langmuir probe measurements, the TiN coating became oxidized after the spacecraft dipped into the Mars upper atmosphere, degrading the measurements due to the reduction in surface conductivity (Ergun et al., 2015).

### 5.3. Towards a notional observatory

To summarize, this chapter provided a general overview of some past and proposed satellite missions and concepts designed to study the LTI. The intent was to provide evidence that such missions are feasible using well-established scientific instruments with excellent heritage in the ionosphere/thermosphere, and viable for LTI research, although challenges remain for repeated excursions into the LTI at and below 120 km. These challenges are primarily within the realm of formidable atmospheric drag, limiting repeated excursions below 120 km, as well as concerns regarding



instrument performance of detectors targeting accurate measurements of gas properties (both neutral and ionized) below 120 km, on platforms with orbital velocities which are expected to experience shock fronts and thus to perturb these measurements.

A pathfinder mission to explore the LTI to 120 km would be an important first step to demonstrate the feasibility of such measurements, including their performance during occasional forays to even lower altitude (e.g., below 120 km). This would serve as an opportunity to resolve outstanding technical challenges and lower possible risk for a potential future strategic or flagship mission, while providing key information on LTI properties and processes that has long been sought by the international geospace community.



## 6. SYNERGIES AND INTERNATIONAL CONTEXT

The science objectives of the LTI region defined by this report lie at the interface between atmospheric and space physics. They combine two distinct scientific disciplines that do not regularly interact, yet, as illustrated in this report, a cross-disciplinary approach is necessary to advance our understanding of this important transition region between Earth's atmosphere and the geospace plasma. Likewise, the potential future applications enabled by a mission addressing these science objectives span a wide range of distinct user communities and domains, from climate research, to Earth system and whole atmosphere modeling, to global space weather modeling and services. Bringing together these diverse interests and needs offers unique opportunities for cross-disciplinary research, development, and solutions. The broad range of science and application areas also brings into play a very large number of relevant resources, both current and planned, with immense potential to yield synergistic advances. This chapter is aimed at describing some of these programs, initiatives, projects, facilities, and missions that an LTI investigation would both benefit from and add great value to.

### 6.1. International programs and collaborations

The need for unified modeling of the whole atmosphere, from ground to space, including the mesosphere, thermosphere, and ionosphere, is emerging within atmospheric and space science and an international community is forming to support and promote this development (Jackson et al., 2019). Relatedly, system-level understanding and modeling of the Earth system are pursued as high-priority research and development goals by many national and international organizations and partnerships (e.g., Climateurope, ESA's Living Planet Programme, the US's National Earth System Prediction Capability). We argue here that extending the boundary of the Earth system to encompass the LTI transition region is a natural evolution of this endeavor. Finally, investigations related to the solar influence on climate, including effects of energetic particle precipitation, are conducted as part of the SOLARIS-HEPPA activity within the *Stratosphere-troposphere Processes And their Role in Climate* (SPARC) framework, which is one of four core projects of the World Climate Research Programme. A holistic atmosphere-space activity that connects Earth science with heliophysics is inherently cross-disciplinary. A mission addressing the science objectives indicated in this current report will enable and foster an upper atmosphere community with the capabilities to take on these formidable and important challenges.

Long-standing international fora for coordination and collaboration in upper atmosphere research include, notably, the *Committee on Space Research* (COSPAR) and the *International Association of Geomagnetism and Aeronomy* (IAGA). Recently, the *Sub-Committee on Solar-Terrestrial Physics* (SCOSTEP) embarked on a new program for the period 2020-2024: *Predictability of the Solar-Terrestrial Coupling* (PRESTO). One of the pillars of this program concerns "Space Weather and the Earth's Atmosphere" and targets the connectivity of the thermosphere and ionosphere (Daglis et al., 2021).

In 2018, COSPAR constituted a Panel on Space Weather with the purpose to oversee and accelerate the implementation of the recommendations presented in the roadmap it had commissioned earlier: *Understanding space weather to shield society: A global roadmap for 2015–2025* (Schrijver et al., 2015). This roadmap provides an assessment of the field of space weather from the scientific and end-user viewpoints, and identifies missing links in our scientific understanding, modeling capabilities and observational coverage, several of which will be addressed by a mission to the LTI region. An important component of the implementation is the formation of the *International Space Weather Action Teams* (ISWAT) initiative, which will serve as a global hub for community-coordinated collaborations



addressing challenges across the field of space weather. Several action teams have already formed around topics pertaining to LTI science and applications.

## 6.2. Current and planned synergistic observations

### 6.2.1. Space missions

In the realm of ESA's Earth Observation Programmes Directorate, an LTI mission would add a unique upper atmosphere science perspective to ESA's EO mission portfolio. With the *Swarm* mission, ESA/EOP is already making a highly successful venture into research of the near-Earth space environment. The *Swarm* constellation has three satellites in polar circular orbits at 450-530 km altitudes and each carries high-precision vector and scalar magnetometers, as well as Langmuir probes and Thermal Ion Imagers, along with accelerometers and GNC related instruments. While the primary goal of *Swarm* concerns the full description of sources to the geomagnetic field from inside the Earth, it also provides unprecedented information on the electric currents flowing in the ionosphere. Increasingly, *Swarm* data have been used for upper atmosphere studies, yielding new insights into parameters, processes, and their scale lengths at *Swarm* altitudes. The satellites were launched on 13 November 2013; a dedicated LTI mission would build on, continue, and expand *Swarm* achievements, addressing critical questions concerning the LTI from its lower altitude orbit. Together, a dedicated LTI mission and *Swarm* (or *Swarm*-like measurements, including those by the forthcoming Scout NanoMagSat constellation) would constitute a powerful systems approach to investigating the upper atmosphere from several vantage points. In addition, high-precision magnetic field measurements from a mission with orbit segments below 250 km altitude have the potential to tremendously enhance our knowledge of Earth's lithospheric magnetic field beyond what is possible with *Swarm*.

In the realm of NASA's Heliophysics Division, an LTI mission would greatly complement and expand the science of the planned NASA *Geospace Dynamics Constellation* (GDC) and *DYnamical Neutral AtMosphere-Ionosphere Coupling* (DYNAMIC) missions. Both strategic missions were recommended by the 2013 Decadal Survey for Solar and Space Physics. GDC consists of 6 identical spacecraft, carrying *in situ* plasma and fields instrumentation, in an orbit of ~400 km. The science goals of GDC are: a) to understand how the high latitude ionosphere-thermosphere system responds to variable solar wind/magnetosphere forcing, and b) to understand how internal processes in the global ionosphere-thermosphere system redistribute mass, momentum and energy. DYNAMIC is designed to answer the question how the lower atmosphere variability affects geospace. In effect, GDC is designed to study forcing from above, while DYNAMIC would concurrently study forcing from below. GDC and DYNAMIC are designed to operate concurrently, thereby maximizing science output while reducing costs. An LTI mission, addressing the science questions outlined in this report, would both greatly enhance the science return from a combined GDC+DYNAMIC mission while also leveraging the global and mesoscale observations afforded by the combined GDC+DYNAMIC observations.

The *Solar wind Magnetosphere Ionosphere Link Explorer* (SMILE) is a joint mission between ESA's Directorate of Science and the Chinese Academy of Sciences (CAS) with expected launch in 2025. As part of its mission, SMILE will image the aurora (soft X-ray and UV) from very far away (third of the distance to the moon), providing prolonged views of Earth's northern auroral region. The remote sensing data from SMILE is expected to provide global context for the science objectives in this report connected to the auroral region.

In 2018, the China National Space Administration (CNSA) in collaboration with the Italian Space Agency (ASI), launched the *China Seismo-Electromagnetic Satellite* (CSES) for the purpose of investigating possible correlations between perturbations in geospace and the occurrence of seismic events. The satellite orbits at 500 km altitude at fixed 02/14 local time. It carries an extensive suite of particle, fields,





and plasma instruments and from its different vantage point could complement the ionospheric studies of a dedicated LTI mission. The program foresees several satellites to be sequentially launched, with a second follow-on satellite already in advanced stage of development.

The AMPERE project is based on the commercial Iridium-NEXT constellation of 66 spacecraft launched in 2017-2019 into orbits at 800 km altitude and distributed in 6 orbital planes to provide uniform global coverage. AMPERE combines magnetic field measurements from all satellites to produce global maps that refresh every 9 minutes of geomagnetic variations and the corresponding field-aligned currents flowing into and out of the ionosphere. These measurements add a tremendous global context and high-altitude complement to *in situ* LTI measurements.

NASA's Electrojet Zeeman Imaging Explorer (EZIE) mission is slated to launch in late 2024 or early 2025. Using a fleet of 3 spacecraft at ~500 km, EZIE will study the currents flowing within the LTI using innovative remote sensing instruments that measure the Zeeman splitting of oxygen in the ionosphere near ~95 km, and infer the intensity of currents flowing near 110 km. By taking measurements so near to the location of the currents, EZIE will be able to study smaller scale dynamics than can be measured from ground-based observatories. In addition to the compelling science of EZIE, this innovative technique is a pathfinder for future observations of this type, providing contextual information for an *in situ* LTI mission.

Following recommendations from the US National Research Council's Committee for a Decadal Strategy in Solar and Space Physics 2013-2022, several current and planned NASA missions are aimed at the science of the upper atmosphere. Two NASA missions target the thermosphere and ionosphere interactions: GOLD (*Global-scale Observations of the Limb and Disk*) was launched in 2018 into geostationary orbit and ICON (*Ionospheric Connection Explorer*) in 2019 into a low inclination, 560-km altitude orbit. From their different vantage points, the duo performs remote sensing of airglow emissions to estimate thermosphere density, temperature, and composition during daytime and density variations in the low-latitude ionosphere at night. Observations and insights from GOLD and ICON provide an excellent remote sensing complement to *in situ* measurements of the LTI. ICON recently failed on orbit after completing its prime mission, however analysis of ICON data continues to provide new insight into the LTI region. Recently, NASA has launched the *Atmospheric Waves Experiment* (AWE), which uses remote sensing performed from the International Space Station of airglow emissions to investigate how atmospheric gravity waves impact the transport of energy and momentum from the lower atmosphere. Likewise, the Swedish *Mesospheric Airglow/Aerosol Tomography and Spectroscopy* (MATS) mission investigates mesospheric waves around 80-100 km using tomographic analysis.

Potential strong synergies with an LTI mission exist with JAXA FACTORS (*Frontiers of Formation, Acceleration, Coupling, and Transport Mechanisms Observed by the Outer Space Research System*) mission concept, which is in the early planning stages. The mission is proposed as a next-generation formation flight mission with two satellites in the altitude range from ~300 km to 4000 km measuring small-scale plasma parameters and simultaneous auroral imaging to resolve dynamical spatial and time variations in Magnetosphere-Ionosphere coupling. A third satellite to the constellation is being sought through a collaboration with the Swedish InnoSat programme.

While these current or planned missions promise extensive and valuable complementarity and synergy with an LTI mission, none of them will study *in situ* the key transition region of the lower thermosphere-ionosphere. Potentially, *in situ* complementary measurements could come from the development and utilization of very small satellite systems. The last decade has seen an almost explosive growth in this area, including cubesats that weigh on the order of only 10 kg and with physical dimensions of the order of a few tens of cm. Though only a fraction of the small satellite missions launched to date has been for scientific exploration, several missions have already delivered or demonstrated significant Earth and



space science discoveries. A notable example of this promise is the QB50 mission launched in 2017, which aims to provide the first multi-point measurements of the density, composition, and conductivity of the lower thermosphere using a distributed constellation of sensors on board cubesats (Wicks and Miles, 2019). Many more missions are planned or envisioned for the future (Lal et al., 2017; Millan et al., 2019). Such missions, many of which are launched from the International Space Station, offer significant opportunities for contributing to the investigation of the LTI, yet do not obviate the need of a dedicated LTI mission with *in situ* measurements.

One category of observations that would add a highly valuable complement to an LTI mission, but for which limited capability exists involves measuring the abundances of HO<sub>x</sub> and NO<sub>x</sub> species through the middle atmosphere. Such observations would directly record and verify the expected impact of the new and much improved energetic particle precipitation data from a dedicated LTI vantage point. However, these observations are currently made by an aging group of limb sounding satellites, which are unlikely to continue in operation through to the late 2020s. Some current capability in measuring HO<sub>x</sub> and NO<sub>x</sub> species includes NASA's TIMED/SABER limb-scanning infrared radiometer, the Canadian SciSat-1/ACE solar occultation Fourier Transform Spectrometer, and the Swedish-led ODIN/SMR sub-millimetre wave radiometer. Future limb sounders might plug this observational gap, which is one amongst the science objectives targeted by the CAIRT EE11 mission candidate (NO<sub>x</sub>) to ESA's Earth Explorer program (ESA, 2023) and the Keystone EE12 mission candidate (HO<sub>x</sub>, NO).

In conclusion, an LTI mission as outlined here would provide unique and unprecedented measurements in the lower thermosphere and ionosphere to directly address and quantify local processes, complement and calibrate remote sensing measurements, and help develop much-improved models of the upper atmosphere. By combining assets and expertise from ESA Earth Observation with NASA Heliophysics, an LTI mission also enables an exciting cross-disciplinary program at the interface of Earth and space science.

### 6.2.2. Ground-based observations

An extensive network of ground-based instruments provides supplementary and context measurements for the LTI, which can be used together with *in situ* measurements for great additional benefits, including cross-calibration in some cases. Observational resources include existing networks of ionosondes, Incoherent Scatter Radars (ISR), Coherent Scatter Radars, auroral imagers, photometers, lidars, and Fabry-Perot Interferometers (FPI). The coordination and full exploitation of these resources globally is greatly aided by the *Coupling, Energetics and Dynamics of Atmospheric Regions* (CEDAR) program, which is a focused component of the US Global Change program with the goal to enhance the capability of ground-based instruments to measure the upper atmosphere and to coordinate instrument and model data for the benefit of the scientific community.

Particular synergistic promise is offered by the state-of-the-art volumetric *EISCAT\_3D* ISR operated by the international scientific EISCAT Association. Initial components of this new facility will go online in 2024. The radar will provide electron density, electron and ion temperature and vector ion drift velocity between 70 and 1000 km within a 3D cone with a diameter of 500 km at an altitude of 150 km (McCrea et al., 2015). Together, an *in situ* LTI mission and EISCAT\_3D would constitute the “microscope-telescope” approach of combined *in situ* and remote sensing measurements, providing a powerful tool for upper atmosphere studies.

### 6.2.3. Sounding rockets

Sounding rockets are a main source of *in situ* measurements and altitude profiles of the full set of parameters characterizing the LTI. In particular, they provide essential means of gathering data at



lower altitudes (< 110 km) that are difficult to sample directly with satellite probes. Indeed, seminal results for the formulation and refinement of the science objectives in this report were provided by numerous sounding rockets. A multitude of sounding rocket flights have been performed at various latitudes, including the geomagnetic equator, mid-latitude locations, the auroral zone, and cusp. The sounding rocket community continues to actively investigate the LTI regions with innovative payloads and instruments and with launches at all latitudes.

In the operational phase of a future LTI satellite mission, sounding rocket campaigns can provide unique complementary measurements to those gathered along the largely horizontal satellite orbital transects. Such combined datasets would yield corresponding vertical and horizontal profiles of LTI characteristics and parameters, for example, of the Joule heating distribution. In addition, sounding rockets can also provide valuable context information about temporal and spatial variability.

### 6.3. Uniqueness and complementarity

Summarizing from the above, a dedicated mission to the LTI, providing truly unique *in situ* low-altitude measurements, has excellent **complementarity** with a number of current and planned missions and ground-based observation facilities, in a wide international context, including notably:

- ESA's Swarm mission, launched in 2013, in relation to which an ESA-involved LTI mission would continue and expand the highly successful venture into research of the near-Earth space environment for the ESA EO program. Such an LTI mission, from its lower altitude vantage point, would greatly aid upper atmosphere investigations and would furthermore add a capability to resolve lithospheric sources to the Earth magnetic field;
- ESA-CAS SMILE scheduled for launch in 2025, which will provide global auroral imaging context for LTI investigations;
- NASA's GOLD mission, launched in 2018 to geostationary orbit, performing remote sensing of Earth airglow emissions to investigate ionosphere-thermosphere-mesosphere coupling;
- NASA's GDC mission which will provide magnetosphere-ionosphere-thermosphere coupling and dynamics context from a high-altitude perspective as compared to *in situ* LTI sampling;
- NASA's DYNAMIC mission, which will complement GDC by gathering remote sensing measurements to reveal how forcing from below affects the variability of the upper atmosphere;
- The AMPERE project utilizing the Iridium-NEXT global constellation of telecommunication satellites to provide global field-aligned current context for the LTI;
- EISCAT\_3D and extended network of incoherent scatter radar and ionosonde facilities, which would provide a "microscope-telescope" capability in conjunction with *in situ* LTI measurements;
- Extensive existing and planned networks of optical, radio, lidar, and magnetic ground-based observatories, which will provide remote sensing and, in some cases, global context for local LTI observations;
- Sounding rockets, including dedicated campaigns, to provide complementary *in situ* measurements and adding invaluable altitude profiles as well as information on the spatial and temporal variability of the LTI observations;



- Cubesats, many of which are launched from the International Space Station and offer significant opportunities for contributing to the investigation of the LTI;
- Limb-sounders to provide observations on the abundances of  $\text{HO}_x$  and  $\text{NO}_x$  and further chemical species through the middle atmosphere.





## 7. SUMMARY AND DIRECTION FORWARD

This report presents the science case for defining and developing a spaceflight mission focused on understanding the Earth's lower thermosphere-ionosphere (LTI). The report demonstrates the importance of the LTI in its role as a key element of the coupled Earth-Sun system, and elucidates how improved understanding of the behavior of its commingled neutral (un-ionized) and ionized gases (plasma), electromagnetic fields, and charged particles is required to advance upper atmospheric and space science goals and space-related societal needs. An overarching goal with specific science objectives together describes a vision for this much-needed new understanding, and provides the basis for specifying the observations required to make progress. A systematic, comprehensive, *in situ* observation methodology based on numerous visits of one or more instrumented space platforms to the LTI is adopted as the central component of the science strategy. A notional mission paradigm based on the concepts employed by previous missions and developed further by several pre-formulation studies is documented. This report provides the framework for more detailed and thorough pre-formulation activities that comprise the next step toward a mission dedicated to advancing understanding of the LTI and its role within the Earth-Sun system.

**Challenges for LTI missions.** The special challenges in improving LTI understanding stem in large part from two circumstances, as discussed in Chapters 2 to 4 of this report. First, this region exhibits numerous phenomena (see Figure 7.1) varying over a large range of spatial and temporal scales. Second, the relatively high atmospheric densities within the LTI region (e.g., below 200 km altitude) give rise to significant environmental effects (aerodynamic heating, plasma shock effects, increased exposure to atomic oxygen, etc.) that make systematic *in situ* measurements from an orbital platform challenging. In the past, this second challenge has been a serious impediment for any mission that must fly at low altitudes. However, considerable study of this issue has been enjoined over the past decades, as discussed in Chapter 5, including successful instruments and operations on the Atmosphere Explorer satellites as well as the Pioneer Venus and MAVEN planetary missions.

Approaches for mitigating these challenges benefit from some of the previous work noted in Chapter 5 of this report. Further work necessary to lead to a new LTI mission should include the following topics:

- Effects of high ram pressure on measurements
- Atmospheric drag at low altitudes
- Aerothermal heat flux at low altitudes
- Effects of atomic oxygen influence on spacecraft materials
- Effects of plasma wakes on measurements

**The need for *in situ* measurements.** Chapter 4 of this report makes the case that *in situ* measurements are necessary to provide the comprehensive, systematic observations that are needed for LTI science to make the next big advance in understanding. While a great deal of new knowledge has resulted from remote observations and short-duration *in situ* missions, these are either not fully comprehensive or long-lived enough to examine important parameters that vary on secular timescales. A revolution in understanding will be realized by a dedicated *in situ* LTI mission concept, sampling all required parameters concurrently at the relevant scales.

**Historical context of a new LTI mission.** The ENLoTIS working group was born of a long-standing desire within the international science community to take the next big leap forward in LTI science,

starting with the AE missions, progressing to the TIMED and GEC technical studies, and extending to the recent ASTRE and Daedalus mission concepts. This report provides the impetus for a new space-based investigation into the LTI realm through a collaborative effort between ESA and NASA. The scientific need has grown over time while the barriers to performing such a mission have been reduced to such an extent that now is the time to proceed with the further development needed to make it possible.

**Science context of a new LTI mission.** Chapter 6 of this report discusses how a new *in situ* LTI mission would relate to a large contingent of Earth and space science investigations, including past, present and planned future missions, both large and small, as well as with ground-based investigations. The report discusses how a new LTI mission could complement the science return of investigations in the operational or planning stages right now.

**Science return as a function of altitude.** A key parameter of any mission that would examine the LTI *in situ* is the altitudes which must be sampled. This mission parameter carries special significance because of the challenges associated with sampling within a high-drag environment. In short, the higher the drag, the more difficult the mission. Thus, it is important to weigh the level of difficulty of the mission against the value of observations at a given altitude. The idea of “science return as a function of mission altitude” will need to be carefully evaluated and balanced against scientific, technical and programmatic constraints in future trade studies.

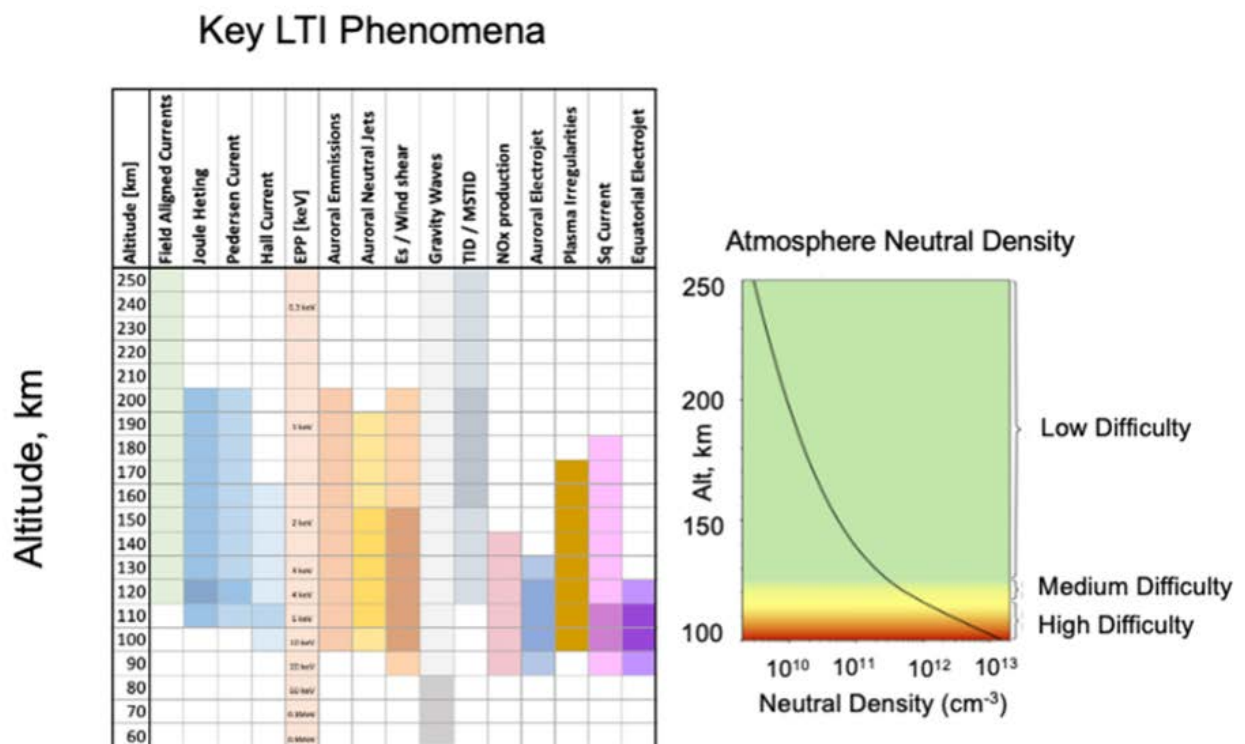


Figure 7.1: Left, key LTI phenomena vs. altitude (credit: DUTH – T. Sarris). Right, atmospheric density with color shading corresponding to difficulty levels anticipated for a notional *in situ* LTI mission (credit: NASA/GSFC – R. Pfaff).

Figure 7.1 provides information on both sides of this trade. The table on the left shows how the occurrence of the various phenomena of interest are distributed as a function of altitude. The columns are color-coded to indicate the relevant altitude ranges, with the darker shades highlighting the region where LTI observations yield the most valuable insights and thus lead to the highest scientific



knowledge gain. It shows that most LTI phenomena exhibit steep vertical gradients. In some cases, phenomena are shown that exist throughout the LTI, such as Joule heating, but are strongest in regions where the Pedersen conductivity is enhanced, such as between 110 and 140 km. Ideally, a comprehensive LTI mission would cover all of these altitude ranges. However, traveling to altitudes below about 120 km, for example, has a high difficulty factor, with the plot on the right containing a proxy for the relative difficulty based on atmospheric density. This forms the basis for the trade study, as coverage of the entire altitude range of interest is traded against the difficulty factor.

**Next steps.** This report represents the first step toward potential ESA/NASA cooperation on development of a new *in situ* LTI mission, resulting in scientific advancements in the understanding of plasma-neutral interactions that connect our upper atmosphere with solar and geomagnetic drivers. The scientific goals and objectives established here lead to a set of next steps to be undertaken as soon as possible within the existing ENLoTIS working group framework:

1. Flow-down of the scientific needs expressed in this report into firm mission requirements supported by analyses, tools and capabilities to justify and verify these requirements;
2. Assessment of instrumental capabilities needed to meet measurement requirements;
3. Study of orbits, propulsion, as well as mission and science operational concepts capable of meeting sampling requirements;
4. Further progress on the mitigation approaches to the special challenges that affect the measurement techniques, data processing, spacecraft design choices and concepts of operations.

In addition to these next steps, longer-term planning by both agencies could consider the benefits and risks of developing a low-cost pathfinder mission based on existing concepts and technology versus providing a testbed for new instrumentation and spacecraft operations (e.g., single platform vs. series of small satellites). This planning could also consider the importance of both ground-based LTI observation infrastructure (e.g., rockets, radar, etc.) and advanced geospace modeling capabilities to maximize the science return from a dedicated LTI mission concept.

In summary, a dedicated LTI mission to perform direct measurements of the interactions between the ionized and neutral gas will provide the first detailed and systematic investigation of this important unexplored region, vastly improve and constrain models of the upper atmosphere, and fill a critical gap in our understanding of the fundamental processes that define Earth's interface with space.



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## ACRONYMS

|         |                                                                                      |
|---------|--------------------------------------------------------------------------------------|
| AE      | Atmosphere Explorer (mission)                                                        |
| AMPERE  | Active Magnetosphere and Planetary Electrodynamics Response Experiment (mission)     |
| AMU     | Atomic Mass Unit                                                                     |
| ASTRE   | Atmosphere-Space Transition Region Explorer (mission)                                |
| AWE     | Atmospheric Waves Experiment (mission)                                               |
| CHAMP   | Challenging Minisatellite Payload (mission)                                          |
| C/NOFS  | Communications/Navigation Outage Forecasting System (mission)                        |
| DC      | Direct Current                                                                       |
| DE-2    | Dynamics Explorer 2 (mission)                                                        |
| DEMETER | Detection of Electromagnetic Emissions Transmitted from Earthquake Regions (mission) |
| DMSP    | Defense Meteorological Satellite Program (mission)                                   |
| DoD     | Department of Defense                                                                |
| DSMC    | Direct Simulation Monte Carlo                                                        |
| DYNAMIC | Dynamical Neutral Atmosphere-Ionosphere Coupling (mission)                           |
| EEJ     | Equatorial Electrojet                                                                |
| EISCAT  | European Incoherent Scatter Scientific Association                                   |
| ENLoTIS | ESA-NASA Lower Thermosphere-Ionosphere Science                                       |
| EO      | Earth Observation                                                                    |
| EPP     | Energetic Particle Precipitation / Energetic Precipitating Particles                 |
| ESA     | European Space Agency                                                                |
| ESA/EOP | ESA Earth Observation Programmes                                                     |
| EUV     | Extreme Ultra-Violet                                                                 |
| EZIE    | Electrojet Zeeman Imaging Explorer (mission)                                         |
| FAC     | Field Aligned Current                                                                |
| FPI     | Fabry-Perot Interferometer                                                           |
| FUV     | Far Ultra-Violet                                                                     |
| GCM     | General Circulation Model                                                            |
| GDC     | Geospace Dynamics Constellation (mission)                                            |
| GEC     | Geospace Electrodynamics Connections (mission)                                       |
| GIC     | Ground-Induced Current                                                               |
| GLOW    | Global Airglow (model)                                                               |
| GNSS    | Global Navigation Satellite System                                                   |
| GOCE    | Gravity field and steady-state Ocean Circulation Explorer (mission)                  |
| GOLD    | Global-scale Observations of the Limb and Disk (mission)                             |
| GW      | Gravity Wave                                                                         |



|           |                                                                            |
|-----------|----------------------------------------------------------------------------|
| HAARP     | High-Frequency Active Auroral Research Program                             |
| ICON      | Ionospheric Connection Explorer (mission)                                  |
| IMF       | Interplanetary Magnetic Field                                              |
| IRI       | International Reference Ionosphere (model)                                 |
| ISR       | Incoherent Scatter Radar                                                   |
| LEO       | Low Earth Orbit                                                            |
| LTI       | Lower Thermosphere-Ionosphere                                              |
| MAVEN     | Mars Atmosphere and Volatile Evolution (mission)                           |
| MIGHTI    | Michelson Interferometer for Global High-resolution Thermospheric Imaging  |
| M-I-T-M   | Magnetosphere-Ionosphere-Thermosphere-Mesosphere                           |
| MLT       | Mesosphere and Lower Thermosphere                                          |
| MSTID     | Medium-Scale Traveling Ionospheric Disturbance                             |
| NASA      | National Aeronautics and Space Administration                              |
| NASA/SMD  | NASA Science Mission Directorate                                           |
| NRLMSIS   | Naval Research Laboratory Mass Spectrometer and Incoherent Scatter (model) |
| PRE       | Pre-Reversal Enhancement                                                   |
| PVO       | Pioneer Venus Orbiter                                                      |
| ROCSAT    | Republic of China Satellite (mission) / FORMOSAT                           |
| SABER     | Sounding of the Atmosphere using Broadband Emission Radiometry             |
| SEP       | Solar Energetic Particle                                                   |
| SO        | Science Objective                                                          |
| Sq        | Solar quiet                                                                |
| SuperDARN | Super Dual Auroral Radar Network                                           |
| TAD       | Travelling Atmospheric Disturbances                                        |
| TID       | Travelling Ionospheric Disturbances                                        |
| TIE-GCM   | Thermosphere-Ionosphere-Electrodynamics General Circulation Model (model)  |
| TIMED     | Thermosphere-Ionosphere-Mesosphere Energetics and Dynamics (mission)       |
| TRL       | Technology Readiness Level                                                 |
| UV        | Ultra-Violet                                                               |
| VLEO      | Very Low Earth Orbit                                                       |
| WACCM(-X) | Whole Atmosphere Community Climate Model (-eXtended) (model)               |
| 3D        | 3-Dimensional                                                              |



# APPENDICES

## Appendix A Definition of Terms

Table A.1: Symbols, terms and definitions of LTI physical properties and processes used throughout this document. Terms in bold font denote vectors.

|                              | Term and definition                                                                                                 | Symbol [unit]                           | Equation                                                                                                                      |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Plasma properties</b>     |                                                                                                                     |                                         |                                                                                                                               |
|                              | Drift velocity: <i>plasma collective motion forced solely by the Lorentz force</i>                                  | $\mathbf{V}_e$ and $\mathbf{V}_i$ [m/s] | $= \mathbf{E} \times \mathbf{B} / B^2$                                                                                        |
|                              | Ion velocity: <i>ion direction and speed</i>                                                                        | $\mathbf{V}_i$ [m/s]                    |                                                                                                                               |
|                              | Electron and ion temperature                                                                                        | $T_e$ and $T_i$ [K]                     |                                                                                                                               |
|                              | Ion number density                                                                                                  | $N_i$ [m <sup>-3</sup> ]                |                                                                                                                               |
|                              | Plasma density: <i>under charge neutrality, volumetric concentration of all ions equates to electron population</i> | $N_e$ [m <sup>-3</sup> ]                | $= \sum N_i$                                                                                                                  |
|                              | Ion mass                                                                                                            | $m_i$ [kg]                              |                                                                                                                               |
|                              | Electron mass                                                                                                       | $m_e$ [kg]                              |                                                                                                                               |
| <b>Neutral properties</b>    |                                                                                                                     |                                         |                                                                                                                               |
|                              | Neutral wind                                                                                                        | $\mathbf{U}$ [m/s]                      |                                                                                                                               |
|                              | Neutral temperature                                                                                                 | $T_n$ [K]                               |                                                                                                                               |
|                              | Neutral number density                                                                                              | $N_n$ [m <sup>-3</sup> ]                |                                                                                                                               |
|                              | Neutral mass                                                                                                        | $m_n$ [kg]                              |                                                                                                                               |
|                              | Neutral mass density                                                                                                | $\rho_n$ [kg/m <sup>3</sup> ]           |                                                                                                                               |
| <b>Electrical properties</b> |                                                                                                                     |                                         |                                                                                                                               |
|                              | Pedersen conductivity                                                                                               | $\sigma_P$ [mhos/m]                     | $= \frac{eN_e}{B} \left( \frac{v_{in}\Omega_i}{v_{in}^2 + \Omega_i^2} + \frac{v_{en}\Omega_e}{v_{en}^2 + \Omega_e^2} \right)$ |
|                              | Hall conductivity                                                                                                   | $\sigma_H$ [mhos/m]                     | $= \frac{eN_e}{B} \left( \frac{\Omega_e^2}{v_{en}^2 + \Omega_e^2} - \frac{\Omega_i^2}{v_{in}^2 + \Omega_i^2} \right)$         |
|                              | Parallel conductivity                                                                                               | $\sigma_{  }$ [mhos/m]                  | $= \frac{N_e e^2}{m_e(v_{en} + v_{ei})}$                                                                                      |
|                              | Cowling conductivity                                                                                                | $\sigma_c$ [mhos/m]                     | $= \sigma_P + \frac{\sigma_H^2}{\sigma_P} = \frac{eN_e v_{in}}{B\Omega_i}$                                                    |
|                              | Electric field                                                                                                      | $\mathbf{E}$ [V/m]                      | $= -\mathbf{V}_e \times \mathbf{B}$                                                                                           |
|                              | Electric field in neutral frame                                                                                     | $\mathbf{E}'$ [V/m]                     | $= \mathbf{E} + \mathbf{U} \times \mathbf{B}$                                                                                 |
|                              | Pedersen current density                                                                                            | $j_P$ [A/m <sup>2</sup> ]               | $= \sigma_P \mathbf{E}'_{\perp}$                                                                                              |
|                              | Hall current density                                                                                                | $j_H$ [A/m <sup>2</sup> ]               | $= \sigma_H  \hat{\mathbf{b}} \times \mathbf{E}'_{\perp} $                                                                    |
|                              | Field-aligned current density                                                                                       | $j_{  }$ [A/m <sup>2</sup> ]            | $= \sigma_{  } \mathbf{E}'_{  } \hat{\mathbf{b}}$                                                                             |
|                              | Magnetic field (potential field and perturbations)                                                                  | $\mathbf{B}$ [T]                        | $= -\nabla\Phi_0 + \delta\mathbf{B}$                                                                                          |



| <b>Coupling properties</b> |                                                                                                                                                       |                             |                                                                                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | Ion-neutral collision frequency                                                                                                                       | $\nu_{in} [\text{s}^{-1}]$  |                                                                                                                                                                                                                             |
|                            | Neutral-ion collision frequency                                                                                                                       | $\nu_{ni} [\text{s}^{-1}]$  |                                                                                                                                                                                                                             |
|                            | Electron-neutral collision frequency                                                                                                                  | $\nu_{en} [\text{s}^{-1}]$  |                                                                                                                                                                                                                             |
|                            | Electron gyrofrequency                                                                                                                                | $\Omega_e [\text{s}^{-1}]$  | $= eB/m_e$                                                                                                                                                                                                                  |
|                            | Ion gyrofrequency                                                                                                                                     | $\Omega_i [\text{s}^{-1}]$  | $= eB/m_i$                                                                                                                                                                                                                  |
| <b>Energy input</b>        |                                                                                                                                                       |                             |                                                                                                                                                                                                                             |
|                            | Poynting flux vector                                                                                                                                  | $\mathbf{S} [\text{W/m}^2]$ | $= (\mathbf{E} \times \delta \mathbf{B})/\mu_0$                                                                                                                                                                             |
|                            | Particle kinetic energy flux vector                                                                                                                   | $\mathbf{K} [\text{W/m}^2]$ | $= (\mathbf{V}_{\text{epp}} \rho_{\text{epp}}  V_{\text{epp}} ^2)/2$                                                                                                                                                        |
| <b>Energy transfer</b>     |                                                                                                                                                       |                             |                                                                                                                                                                                                                             |
|                            | electromagnetic energy transfer rate<br><i>under steady-state assumption:</i>                                                                         | $q_E [\text{W/m}^3]$        | $= \mathbf{J} \cdot \mathbf{E} = \sum_i e N_i (\mathbf{V}_i - \mathbf{V}_e) \cdot \mathbf{E}$ $= \mathbf{J} \cdot \mathbf{E}' + \mathbf{U} \cdot \mathbf{J} \times \mathbf{B}$ $= -\nabla \cdot \mathbf{S}$                 |
|                            | Joule heating rate                                                                                                                                    | $q_{j/V} [\text{W/m}^3]$    | $= \mathbf{J} \cdot \mathbf{E}' = \sum_i e N_i (\mathbf{V}_i - \mathbf{V}_e) \cdot \mathbf{E}'$ $= \sigma_p (\mathbf{E} + \mathbf{U} \times \mathbf{B})^2$ $= \frac{j^2}{\sigma_c} = \frac{j^2 B \Omega_i}{e N_e \nu_{in}}$ |
|                            | Specific Joule heating rate                                                                                                                           | $q_{j/M} [\text{W/kg}]$     | $= q_{j/V} / \rho_n$                                                                                                                                                                                                        |
|                            | frictional heating rate                                                                                                                               | $q_f [\text{W/m}^3]$        | $q_f = m_i \nu_{in} N_e (\mathbf{V}_i - \mathbf{U})^2$                                                                                                                                                                      |
|                            | mechanical energy transfer rate                                                                                                                       | $q_m [\text{W/m}^3]$        | $= \mathbf{U} \cdot (\mathbf{J} \times \mathbf{B})$                                                                                                                                                                         |
| <b>Momentum transfer</b>   |                                                                                                                                                       |                             |                                                                                                                                                                                                                             |
|                            | Plasma Lorentz acceleration: <i>the force per unit mass exerted on a charged particle moving with velocity through an electric and magnetic field</i> | $a_L [\text{N/kg}]$         | $= \frac{e}{m_i} \mathbf{E} + \frac{e}{m_i} \mathbf{V}_{i,e} \times \mathbf{B}$                                                                                                                                             |
|                            | Neutral Ampere acceleration: <i>the force per unit mass exerted on the neutral gas by momentum transfer through collisions with the plasma.</i>       | $a_A [\text{N/kg}]$         | $= \frac{\mathbf{J} \times \mathbf{B}}{\rho_n}$ $= \frac{N_e m_i \nu_{in}}{\rho_n} (\mathbf{V}_i - \mathbf{U}) + \frac{N_e m_e \nu_{en}}{\rho_n} (\mathbf{V}_e - \mathbf{U})$                                               |





## Appendix B Science Strategy Matrix

| Goal                                                                                                                                                                                        | Science Themes                                                                                                                                     | Science Objectives                                                                                                                          | Scientific Needs                                                                                                                       |                                          | Observational Needs                                                                                  |                                                                                                                       | Supporting Modeling                                                                                                                           | Supporting Observations                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                             |                                                                                                                                                    |                                                                                                                                             | Derived Parameters                                                                                                                     | Direct Observables                       | Spatial/Temporal Scales                                                                              | Regions                                                                                                               |                                                                                                                                               |                                                                                                                                                                                                                                                                                                                   |
| To understand how the transition from Earth's atmosphere to space is governed by the fundamental neutral-plasma interactions that are intrinsic to the lower thermosphere-ionosphere (LTI). | <b>SO1 Collisional Electrodynamics (J):</b><br>Determine how collisions between neutral and charged species affect the electrodynamics of the LTI. | SO 1.1: Determine how electric currents flow and close in the LTI, and thereby couple to the magnetospheric electrodynamics.                | Altitudinally-resolved 3D current densities (J) and current densities (J <sub>  </sub> )                                               | B, E, Ni, Ne, Un, Vi                     | Spatial scales from 100m to 1000s kilometers; Temporal scale of < 1 sec to 1 hour                    | Latitudes: high-latitude (FAC and auroral electrojets); Altitudes: 110-300 km (FAC), 100-130 km (auroral electrojets) | Modeling of 3D currents on small-scales (<10 km) and meso-scales (10s-100s km) will provide supportive information for local current closure. | ISR plasma density, electron and ion temperature, plasma velocity; Coherent scatter radar plasma velocity; FPI neutral wind; Ground-level magnetic field perturbation; High-altitude magnetic field perturbation, plasma drift velocity & density; Conductivity from sounding rocket and active radio experiment. |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 1.2: Understand how the various LTI properties and processes determine the Hall conductivities; ion-neutral and Pedersen conductivities. | Altitudinally-resolved Pedersen and Hall conductivities; ion-neutral collision frequency                                               | B, Ni, Ne                                | Spatial scales down to 1 km; Temporal scales to < 1 sec                                              | Latitudes: all; Altitudes: 110-200 km (Pedersen conductivity); 100-160 km (Hall conductivity)                         | Modeling will be required to estimate complete altitudinally-resolved conductivity profiles as well as ion-neutral collision frequency.       | Conductivity from sounding rocket and active radio experiment                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 1.3: Determine the effect of the neutral winds on the LTI electrodynamics.                                                               | Altitudinally-resolved neutral wind-driven currents                                                                                    | B, E, Ni, Ne, Un, Vi                     | Spatial scales down to 1 km; Temporal scales to < 1 sec                                              | Latitudes: all; Altitudes: 100-300 km                                                                                 | Modeling of global current systems (e.g., Sq) will support the interpretation of neutral wind-driven current observations.                    | Same as SO1.1                                                                                                                                                                                                                                                                                                     |
| To understand how the transition from Earth's atmosphere to space is governed by the fundamental neutral-plasma interactions that are intrinsic to the lower thermosphere-ionosphere (LTI). | <b>SO2 Collisional Energetics (J,E):</b><br>Determine how collisions between neutral and charged species affect the energetics of the LTI.         | SO 2.1: Determine how Joule (frictional) heating rate, and electromagnetic heating depends on scale size, altitude and neutral winds.       | Joule heating rate (J/E), frictional heating rate, and Ohmic (convective) heating rate; ion-neutral collision frequency; Poynting Flux | B, E, Ni, Ne, Un, Vi                     | Spatial scales from 100m to 1000 kilometers; Temporal scale of 0.05 sec to 1 hour                    | Latitudes: high-latitude; Altitudes: 110-200 km                                                                       | Modeling will be required to estimate ion-neutral collision frequency.                                                                        | Same as SO1.1                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 2.2: Determine how energy from energetic precipitating particles (EPP) directly heats the LTI.                                           | Heating rate associated with EPP; E-region plasma structure                                                                            | EPP differential energy flux, Ni, Ni, Ne | Spatial scales down to 1 km to 100s kilometers; Temporal scales to < 1 sec                           | Latitudes: high-latitude; Altitudes: 100-200 km                                                                       | Model estimates of heating rate and impact ionization rate associated with EPP.                                                               | ASI auroral images; high-altitude EPP                                                                                                                                                                                                                                                                             |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 2.3: Determine how plasma-neutral collisions cause chemical changes that affect the energetics of the LTI.                               | Radiative cooling rates (e.g., NO IR cooling rate); Composition change                                                                 | [NO] [O], Tn, Nn, Ni, Ne                 | Spatial scales down to a few kilometers; Temporal scales of a few seconds to hours.                  | Latitudes: high-latitude; Altitudes: 100-200 km                                                                       | Modeling of radiative transfer and chemical processes to support interpretation of IR cooling observations.                                   | FPI neutral temperature; space-based IR fluxes                                                                                                                                                                                                                                                                    |
| To understand how the transition from Earth's atmosphere to space is governed by the fundamental neutral-plasma interactions that are intrinsic to the lower thermosphere-ionosphere (LTI). | <b>SO3 Collisional Dynamics (JxB):</b><br>Determine how collisions between neutral and charged species affect the neutral dynamics of the LTI.     | SO 3.1: Determine how winds are accelerated by plasma motions via ion-neutral collisions.                                                   | Ampere force, collision force, and mechanical work rate; ion-neutral collision frequency                                               | B, E, Ni, Ne, Un, Vi                     | Spatial scales down to 10 kilometers; Temporal scales down to a few seconds                          | Latitudes: high-latitude and mid-latitude; Altitudes: 100-190km                                                       | Modeling of neutral wind acceleration (e.g., Walterscheid et al. 1985) will support the interpretation of neutral wind observations.          | Same as SO1.1                                                                                                                                                                                                                                                                                                     |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 3.2: Discover how the exchange of momentum is manifest across scales by means of lower atmospheric forcing in the LTI.                   | Gravity wave momentum and heat flux                                                                                                    | Un, Tn                                   | Spatial scales from 1-3 km to 1000s kilometers; Temporal scale from a few seconds to 100s of minutes | Latitudes: all; Altitudes: 100-300km                                                                                  | Modeling of gravity wave (e.g., Vadas and Liu, 2013) will support the interpretation of gravity momentum and heat flux observations.          | Ground-based lidar airglow imaging                                                                                                                                                                                                                                                                                |
|                                                                                                                                                                                             |                                                                                                                                                    | SO 3.3: Determine how collisional processes drive vertical transport and cause composition changes.                                         | Neutral wind divergence, eddy mixing rate, molecular diffusion rate, Composition changes; ion-neutral collision frequency              | Un, Tn, Nn, Ni                           | Spatial scales from 10s to 1000s kilometers; Temporal scale from a few seconds to 1 hour             | Latitudes: high-latitude; Altitudes: 100-300km                                                                        | Modeling of vertical composition transport processes will support the interpretation of causes of observed composition changes.               | FPI neutral wind and temperature; Conductivity from sounding rocket and active radio experiment (e.g., HAARP)                                                                                                                                                                                                     |

Table B.1. Science strategy matrix (SSM) for dedicated LTI mission concept.